

Aşağıdaki doküman **Customer Operating System / Amiral Gemisi CRM Platformu — Uçtan Uca Teknik Blueprint v1** olarak kullanılabilir.

1. Ürün Hedefi

Bu sistem klasik CRM değil; aşağıdaki platformların birleşimi olacak:

CRM

- + Sales Pipeline
- + Quote / Order
- + Inventory
- + Finance Lite
- + Omnichannel Inbox
- + Support / Ticket
- + Workflow Engine
- + B2B / Dealer Portal
- + AI Assistant
- + API / Integration Hub
- + SaaS Admin Panel

Ana hedef akış:

Lead

- Customer
- Opportunity
- Quote
- Order
- Shipment
- Invoice
- Payment
- Support
- Retention

2. Hedef Mimari

Ana yaklaşım

Modular Monolith + Event-Driven Architecture

İlk aşamada mikroservis yapılmayacak. 50 kişilik ekip modül bazlı çalışacak ama deployment tek backend olarak başlayacak.

İleride ayrılabilir servisler:

notification-service

file-service

integration-service

ai-service

reporting-service

workflow-service

3. Teknoloji Stack

Backend

NestJS

TypeScript

PostgreSQL

Prisma veya TypeORM

Redis

RabbitMQ

BullMQ

Frontend

Next.js

React

TypeScript

TanStack Query

Zustand veya Redux Toolkit

TailwindCSS

Mobil

Flutter

Storage

MinIO / S3

Search

PostgreSQL Full Text Search başlangıç
OpenSearch / Elasticsearch ileri aşama

AI

AI Gateway
LLM Provider Adapter
Vector DB: pgvector başlangıç

DevOps

Docker
Kubernetes
Helm
GitHub Actions / GitLab CI
Terraform
Prometheus
Grafana
Loki
Sentry

4. Sistem Katmanları

Client Layer

- Web App
- Mobile App
- Customer Portal
- Dealer Portal
- Admin Panel

API Layer

- REST API
- Webhook API
- Public API
- Internal API

Application Layer

- IAM
- CRM
- Lead

- └ Sales
- └ Quote
- └ Order
- └ Inventory
- └ Finance
- └ Support
- └ Marketing
- └ Workflow
- └ Communication
- └ AI
- └ Reporting

Data Layer

- └ PostgreSQL
- └ Redis
- └ Object Storage
- └ Search Index
- └ Event Store

Infrastructure Layer

- └ Kubernetes
- └ CI/CD
- └ Monitoring
- └ Logging
- └ Backup / DR

5. Multi-Tenant Model

Ana karar:

Shared Database + tenant_id + PostgreSQL RLS

Tüm iş tablolarında zorunlu alanlar:

id uuid primary key
tenant_id uuid not null
created_at timestampz not null
created_by uuid
updated_at timestampz
updated_by uuid
deleted_at timestampz

```
version int default 1
```

RLS standardı:

```
ALTER TABLE customers ENABLE ROW LEVEL SECURITY;
```

```
CREATE POLICY tenant_isolation_customers  
ON customers  
USING (tenant_id = current_setting('app.tenant_id')::uuid);
```

Backend her request başında:

```
SET app.tenant_id = '<tenant_uuid>';  
SET app.user_id = '<user_uuid>';
```

6. Domain Modülleri

CORE

```
tenants  
tenant_settings  
plans  
subscriptions  
feature_flags  
usage_quotas  
audit_logs  
system_settings
```

IAM

```
users  
roles  
permissions  
role_permissions  
user_roles  
teams  
team_members  
departments  
branches  
dealers  
sessions
```

devices
login_history

CRM

customers
customer_contacts
customer_addresses
customer_segments
customer_tags
customer_notes
customer_files
customer_timeline
customer_scores
customer_risk_scores
customer_lifetime_values

LEAD

leads
lead_sources
lead_tags
lead_scores
lead_assignments
lead_activities
lead_conversion_logs
lead_loss_reasons

SALES

pipelines
pipeline_stages
opportunities
opportunity_products
opportunity_activities
opportunity_stage_history
opportunity_loss_reasons
competitors

QUOTE

quotes
quote_versions

quote_items
quote_discounts
quote_approvals
quote_files
quote_view_logs
quote_signatures
quote_status_history

ORDER

orders
order_items
order_status_history
order_reservations
order_shipments
order_deliveries
order_returns

INVENTORY

products
product_variants
product_categories
product_brands
product_collections
product_prices
warehouses
warehouse_locations
stocks
stock_movements
stock_reservations
stock_counts
stock_adjustments
stock_transfers

FINANCE

accounts
account_transactions
payments
payment_methods
installments
credit_limits

risk_limits
invoices
invoice_items
bank_accounts
bank_transactions

SUPPORT

tickets
ticket_categories
ticket_priorities
ticket_statuses
ticket_messages
ticket_files
ticket_sla_rules
ticket_satisfaction_surveys
knowledge_base_articles

COMMUNICATION

channels
channel_accounts
conversations
conversation_participants
messages
message_attachments
email_logs
sms_logs
whatsapp_logs
call_logs
meeting_logs

WORKFLOW

workflows
workflow_versions
workflow_triggers
workflow_nodes
workflow_edges
workflow_runs
workflow_logs
approval_requests
approval_steps

approval_actions

AI

ai_agents

ai_prompts

ai_conversations

ai_summaries

ai_predictions

ai_embeddings

ai_documents

ai_usage_logs

ai_feedback

INTEGRATION

api_keys

oauth_clients

webhooks

webhook_events

webhook_logs

connectors

connector_accounts

integration_runs

integration_errors

7. Ana İş Akışları

Lead → Customer → Opportunity

LeadCreated

→ LeadAssigned

→ LeadQualified

→ CustomerCreated

→ OpportunityCreated

→ LeadConverted

Opportunity → Quote

OpportunityCreated

→ ProductsAdded

→ QuoteCreated

- QuoteSent
- QuoteViewed
- QuoteApproved / QuoteRejected

Quote → Order

- QuoteApproved
- OrderCreated
- StockReserved
- PaymentPlanCreated
- ShipmentPlanned

Order → Inventory

- OrderCreated
- StockReservationCreated
- ShipmentCreated
- StockMovementOut
- DeliveryCompleted

Payment

- InvoiceCreated
- PaymentReceived
- AccountTransactionCreated
- BalanceUpdated

8. Event Catalog v1

TenantCreated
UserInvited
UserLoggedIn
RoleChanged

CustomerCreated
CustomerUpdated
CustomerMerged
CustomerDeleted

LeadCreated
LeadAssigned

LeadQualified
LeadConverted
LeadLost

OpportunityCreated
OpportunityStageChanged
OpportunityWon
OpportunityLost

QuoteCreated
QuoteSent
QuoteViewed
QuoteApproved
QuoteRejected
QuoteExpired

OrderCreated
OrderConfirmed
OrderCancelled
OrderShipped
OrderDelivered
OrderReturned

ProductCreated
StockChanged
StockReserved
StockReleased
CriticalStockReached

PaymentCreated
PaymentReceived
PaymentOverdue

TicketCreated
TicketAssigned
TicketResolved
TicketClosed

MessageReceived
MessageSent
ConversationAssigned

WorkflowStarted
WorkflowCompleted
WorkflowFailed

AIActionRequested
AIActionCompleted
AIUsageLimitExceeded

9. API Standardı

Base path:

/api/v1

Örnek endpoint yapısı:

```
GET    /customers
POST   /customers
GET    /customers/{id}
PATCH /customers/{id}
DELETE /customers/{id}

POST   /leads/{id}/convert
POST   /quotes/{id}/send
POST   /quotes/{id}/approve
POST   /quotes/{id}/convert-to-order
POST   /orders/{id}/reserve-stock
POST   /payments
```

Standart response:

```
{
  "data": {},
  "meta": {
    "requestId": "uuid",
    "timestamp": "2026-06-19T12:00:00Z"
  }
}
```

Hata formatı:

```
{
  "error": {
    "code": "CUSTOMER_NOT_FOUND",
    "message": "Customer not found",
    "details": {}
  }
}
```

10. Yetkilendirme Modeli

Hibrit model:

RBAC + Permission + Tenant Scope + Branch Scope + Team Scope

Hazır roller:

Super Admin
Tenant Admin
Sales Manager
Sales Representative
Finance User
Warehouse User
Support Agent
Marketing User
Dealer User
Customer Portal User

Permission örnekleri:

customer.read
customer.create
customer.update
customer.delete
customer.export

lead.read
lead.assign
lead.convert

quote.create
quote.send

quote.approve
quote.discount.approve

order.create
order.cancel

payment.read
payment.create

inventory.read
inventory.adjust

workflow.manage
ai.use
admin.manage

11. Ekip Dağılımı — 50 Kişi

Yönetim ve Ürün — 6 kişi

1 Product Director
2 Product Manager
2 Business Analyst
1 Scrum Master / Delivery Manager

Mimari ve Teknik Liderlik — 4 kişi

1 Solution Architect
1 Backend Lead
1 Frontend Lead
1 DevOps Lead

Backend — 14 kişi

Squad A: CORE + IAM — 3 kişi
Squad B: CRM + Lead — 3 kişi
Squad C: Sales + Quote + Order — 3 kişi
Squad D: Inventory + Finance — 3 kişi
Squad E: Workflow + Integration — 2 kişi

Frontend — 10 kişi

Admin Panel – 2 kişi

CRM/Sales UI – 3 kişi

Inventory/Finance UI – 2 kişi

Portal UI – 1 kişi

Design System – 2 kişi

Mobile — 4 kişi

Flutter Core – 2 kişi

Field Sales – 1 kişi

Offline Sync – 1 kişi

QA — 5 kişi

Manual QA – 2 kişi

Automation QA – 2 kişi

Performance/Security QA – 1 kişi

DevOps / Platform — 4 kişi

Infrastructure

CI/CD

Observability

Security/Backup

AI / Data — 3 kişi

AI Engineer

Data Engineer

Analytics Engineer

12. Squad Sahipliği

Squad CORE/IAM

Sorumluluk:

Tenant

Auth

Users

Roles

Permissions
Sessions
Audit
Feature Flags
Subscription

Squad CRM/Lead

Sorumluluk:

Customer
Contact
Address
Timeline
Lead
Lead Source
Lead Assignment
Lead Conversion

Squad Sales/Quote/Order

Sorumluluk:

Pipeline
Opportunity
Quote
Quote PDF
Approval
Order
Order Status

Squad Inventory/Finance

Sorumluluk:

Product
Variant
Warehouse
Stock
Stock Movement
Account
Payment
Invoice

Squad Communication/Support

Sorumluluk:

Inbox

WhatsApp

Email

SMS

Calls

Tickets

SLA

Knowledge Base

Squad Workflow/Integration

Sorumluluk:

Workflow Builder

Triggers

Actions

Approvals

Webhooks

API Keys

Connectors

Squad AI/Data

Sorumluluk:

AI Gateway

Summaries

Lead Scoring

Quote Assistant

Search

Embeddings

Analytics

13. İlk 90 Gün Teslimat Planı

Hafta 1–2: Foundation

Çıktılar:

Repo setup
Coding standards
Docker compose
PostgreSQL
Redis
RabbitMQ
MinIO
CI pipeline
Base NestJS app
Base Next.js app
Auth skeleton
Tenant context

Hafta 3–4: IAM + Tenant

Tenant CRUD
User CRUD
Role CRUD
Permission system
Login / Refresh Token
Audit Log
RLS proof of concept

Hafta 5–6: CRM Core

Customer CRUD
Contact CRUD
Address CRUD
Customer Timeline
Files
Tags
Notes
Customer List UI
Customer Detail UI

Hafta 7–8: Lead

Lead CRUD
Lead Source
Lead Assignment
Lead Status
Lead Conversion

Lead UI

Hafta 9–10: Sales

Pipeline CRUD

Stage CRUD

Opportunity CRUD

Drag-drop pipeline

Opportunity detail

Stage history

Hafta 11–12: Quote + Dashboard

Quote CRUD

Quote Items

Discount

Tax

PDF generation

Quote send

Basic dashboard

Notifications

90 gün sonunda çalışacak akış:

Login

→ Customer Create

→ Lead Create

→ Lead Convert

→ Opportunity Create

→ Quote Create

→ PDF Send

→ Dashboard View

14. İlk 6 Ay Roadmap

Ay 1–3

IAM

CRM

Lead

Sales

Quote
Task
Dashboard

Ay 4

Order
Product
Basic Inventory
Stock Movement

Ay 5

Finance Lite
Account Balance
Payments
Installments
Receivables

Ay 6

Workflow v1
Notification Engine
Email
WhatsApp v1
Customer Portal v1

15. İlk 12 Ay Roadmap

Q1

MVP CRM + Sales + Quote

Q2

Order + Inventory + Finance Lite + Workflow

Q3

Omnichannel + Support + Dealer Portal + API

Q4

AI Assistant + Advanced Reporting + Mobile + Enterprise Security

16. Altyapı Ortamları

local
development
staging
production
demo
sandbox

Her ortamda:

PostgreSQL
Redis
RabbitMQ
MinIO
API
Web
Worker
Scheduler
Monitoring Agent

Production minimum:

3 API pods
2 Worker pods
1 Scheduler pod
PostgreSQL primary + replica
Redis HA
RabbitMQ cluster
Object storage
Load balancer
WAF

17. CI/CD Pipeline

Her PR için:

- lint
- type-check
- unit test
- migration check
- build
- docker image build
- security scan

Main branch merge:

- deploy dev
- run smoke tests

Release branch:

- deploy staging
- run integration tests
- run e2e tests
- manual approval
- deploy production

Production deploy:

- blue/green veya rolling deployment
- automatic rollback
- migration backup
- health check

18. Observability

Zorunlu metrikler:

- request_count
- request_duration
- error_rate
- db_query_duration
- queue_depth
- worker_failures

login_failures
tenant_usage
ai_usage
storage_usage

Log standardı:

```
{  
  "requestId": "uuid",  
  "tenantId": "uuid",  
  "userId": "uuid",  
  "module": "crm",  
  "action": "customer.create",  
  "level": "info",  
  "message": "Customer created"  
}
```

19. Güvenlik Blueprint

Zorunlu:

RLS
RBAC
2FA
JWT rotation
Refresh token revocation
Audit log
IP whitelist
Rate limit
Webhook signature
File upload validation
Virus scan
Encryption at rest
Encryption in transit
Secret manager
Backup encryption

KVKK/GDPR için:

consent records
data export request

data deletion request
retention policies
data masking
audit trail

20. Veritabanı Öncelikli Migration Seti

Migration 001 — Core

tenants
tenant_settings
plans
subscriptions
feature_flags
audit_logs

Migration 002 — IAM

users
roles
permissions
role_permissions
user_roles
sessions
devices

Migration 003 — CRM

customers
customer_contacts
customer_addresses
customer_notes
customer_tags
customer_files
customer_timeline

Migration 004 — Lead

leads
lead_sources
lead_activities
lead_assignments

lead_conversion_logs

Migration 005 — Sales

pipelines
pipeline_stages
opportunities
opportunity_products
opportunity_stage_history

Migration 006 — Quote

quotes
quote_versions
quote_items
quote_approvals
quote_files
quote_status_history

Migration 007 — Task/Notification

tasks
notifications
reminders

Migration 008 — Product/Inventory

products
product_variants
categories
warehouses
stocks
stock_movements
stock_reservations

Migration 009 — Finance

accounts
account_transactions
payments
payment_methods
installments
invoices

invoice_items

Migration 010 — Workflow

workflows

workflow_versions

workflow_nodes

workflow_edges

workflow_runs

workflow_logs

approval_requests

approval_steps

21. Frontend Ana Ekran Haritasi

/auth/login

/auth/forgot-password

/app/dashboard

/app/customers

/app/customers/:id

/app/customers/new

/app/leads

/app/leads/:id

/app/pipeline

/app/opportunities/:id

/app/quotes

/app/quotes/:id

/app/quotes/:id/preview

/app/orders

/app/orders/:id

/app/products

/app/inventory

/app/tasks
/app/calendar

/app/inbox
/app/tickets

/app/reports

/app/settings/users
/app/settings/roles
/app/settings/permissions
/app/settings/tenant
/app/settings/integrations

22. Backend Klasör Yapısı

```
src/  
├── app.module.ts  
├── main.ts  
├── common/  
│   ├── guards/  
│   ├── decorators/  
│   ├── filters/  
│   ├── interceptors/  
│   ├── pipes/  
│   └── utils/  
├── config/  
├── database/  
│   ├── migrations/  
│   ├── seeds/  
│   └── prisma-or-typeorm/  
├── modules/  
│   ├── core/  
│   ├── iam/  
│   ├── crm/  
│   └── lead/
```

```
|
| └─ sales/
| └─ quote/
| └─ order/
| └─ inventory/
| └─ finance/
| └─ support/
| └─ communication/
| └─ workflow/
| └─ notification/
| └─ integration/
| └─ reporting/
| └─ ai/
|
└─ workers/
└─ schedulers/
└─ events/
```

23. Frontend Klasör Yapısı

```
src/
└─ app/
└─ components/
    └─ ui/
    └─ forms/
    └─ tables/
    └─ layout/
    └─ charts/
└─ features/
    └─ auth/
    └─ customers/
    └─ leads/
    └─ opportunities/
    └─ quotes/
    └─ orders/
    └─ products/
    └─ tasks/
    └─ dashboard/
    └─ settings/
```

- └─ hooks/
- └─ lib/
- └─ services/
- └─ stores/
- └─ types/

24. Definition of Done

Her feature için zorunlu:

- API tamam
- DB migration tamam
- RLS policy tamam
- Permission kontrolü tamam
- Audit log tamam
- Unit test tamam
- Integration test tamam
- Frontend ekran tamam
- Validation tamam
- Error handling tamam
- Loading/empty/error states tamam
- Log/metric tamam
- Documentation tamam

25. Kritik Riskler

Risk 1: Kapsam patlaması

Çözüm:

- MVP freeze
- Change control board
- Icebox backlog

Risk 2: Tenant izolasyonu hatası

Çözüm:

- RLS mandatory
- tenant_id mandatory
- automated tests

security review

Risk 3: 50 kişilik ekipte çakışma

Çözüm:

module ownership
API contract
weekly architecture review
code ownership

Risk 4: Yanlış veri modeli

Çözüm:

ERD review board
migration review
data dictionary approval

Risk 5: Performans

Çözüm:

index strategy
pagination mandatory
query review
read replicas
partitioning for logs/messages/events

26. İlk Sprint Backlog

Sprint 1

CORE-001 Repository setup
CORE-002 Docker compose
CORE-003 PostgreSQL setup
CORE-004 Redis setup
CORE-005 RabbitMQ setup
CORE-006 MinIO setup
CORE-007 Base NestJS app
CORE-008 Base Next.js app
CORE-009 Auth module skeleton

CORE-010 Tenant context middleware

Sprint 2

IAM-001 Tenant table
IAM-002 User table
IAM-003 Role table
IAM-004 Permission table
IAM-005 Login API
IAM-006 Refresh token API
IAM-007 User management UI
IAM-008 Role management UI
IAM-009 RLS POC
IAM-010 Audit log base

Sprint 3

CRM-001 Customer table
CRM-002 Contact table
CRM-003 Address table
CRM-004 Customer CRUD API
CRM-005 Customer list UI
CRM-006 Customer detail UI
CRM-007 Customer create/edit UI
CRM-008 Activity base
CRM-009 File metadata
CRM-010 Tags

27. MVP Kabul Kriteri

MVP şu senaryo çalışınca kabul edilir:

1. Tenant oluşturulur.
2. Admin kullanıcı giriş yapar.
3. Kullanıcı ve rol tanımlanır.
4. Müşteri oluşturulur.
5. Lead oluşturulur.
6. Lead fırsata çevrilir.
7. Pipeline'da fırsat taşınır.
8. Teklif oluşturulur.
9. Teklif PDF üretilir.

10. Teklif gönderildi olarak işaretlenir.
11. Dashboard'da lead, fırsat ve teklif sayıları görünür.
12. Tüm işlemler audit log'a düşer.
13. Tenant A, Tenant B verisini göremez.

28. Nihai Teslimat Sırası

1. ERD v1
2. Data Dictionary v1
3. PostgreSQL Migration v1
4. OpenAPI v1
5. Backend Skeleton
6. Frontend Skeleton
7. IAM
8. CRM
9. Lead
10. Sales
11. Quote
12. Dashboard
13. Pilot
14. Order
15. Inventory
16. Finance
17. Workflow
18. Omnichannel
19. Support
20. AI

29. Yönetici Özeti

Bu proje için doğru başlangıç:

50 kişilik ekip doğrudan kod yazmayacak.

İlk 2 hafta:

ERD

Data Dictionary

OpenAPI

Architecture Contracts

Repository Setup

CI/CD

Sonra ekipler paralel bölünecek:

CORE/IAM

CRM/Lead

Sales/Quote

Inventory/Finance

Frontend

Mobile

DevOps

QA

AI/Data

İlk büyük hedef:

90 gün içinde çalışan CRM + Lead + Sales + Quote MVP

İlk ticari hedef:

6 ay içinde Order + Inventory + Finance Lite ile satılabilir SaaS

Uzun vadeli hedef:

12 ay içinde Omnichannel + Workflow + Portal + AI destekli Customer Operating System

Aşağıda **CRM OS MASTER ERD v1** başlangıç modeli var. Bu, 50 kişilik ekibin modül modül bölünebilmesi için ana veri omurgasıdır.

CRM OS MASTER ERD v1

1. CORE / SaaS

tenants

├─ tenant_settings

├─ tenant_domains

├─ tenant_modules

├─ tenant_feature_flags

├─ tenant_usage_quotas

├─ subscriptions

- └ subscription_items
- └ plans
- └ plan_features
- └ audit_logs

Ana ilişki:

tenants 1 — N users
tenants 1 — N customers
tenants 1 — N leads
tenants 1 — N products
tenants 1 — N workflows

2. IAM / Kullanıcı ve Yetki

users

- └ user_profiles
- └ user_roles
- └ user_permissions
- └ user_sessions
- └ user_devices
- └ user_login_history
- └ user_password_history
- └ user_2fa_methods
- └ user_api_tokens

roles

- └ role_permissions
- └ role_scopes

permissions

Organizasyon:

branches

- └ branch_users
- └ branch_warehouses
- └ branch_targets

departments

- └ department_users

teams

- | team_members
- | team_targets

dealers

- | dealer_users
- | dealer_price_lists
- | dealer_targets
- | dealer_regions

Ana ilişkiler:

users N — N roles

roles N — N permissions

users N — N teams

branches 1 — N users

dealers 1 — N dealer_users

3. CRM / Customer 360

customers

- | customer_contacts
- | customer_addresses
- | customer_tags
- | customer_segments
- | customer_notes
- | customer_files
- | customer_locations
- | customer_relationships
- | customer_scores
- | customer_risk_scores
- | customer_lifetime_values
- | customer_preferences
- | customer_consent
- | customer_merge_logs
- | customer_timeline_events

Ana ilişkiler:

customers 1 — N customer_contacts
customers 1 — N customer_addresses
customers 1 — N opportunities
customers 1 — N quotes
customers 1 — N orders
customers 1 — N tickets
customers 1 — N payments
customers 1 — N conversations

4. LEAD

leads
├ lead_sources
├ lead_tags
├ lead_scores
├ lead_assignments
├ lead_activities
├ lead_notes
├ lead_files
├ lead_conversion_logs
├ lead_loss_reasons
└ lead_enrichment_data

Ana akış:

leads 0..1 — 1 customers
leads 0..1 — 1 opportunities

Dönüşüm:

Lead
→ Customer
→ Opportunity

5. SALES / Opportunity & Pipeline

pipelines
├ pipeline_stages
├ pipeline_stage_rules
└ pipeline_stage_automations

opportunities

- | opportunity_products
- | opportunity_contacts
- | opportunity_activities
- | opportunity_notes
- | opportunity_files
- | opportunity_competitors
- | opportunity_stage_history
- | opportunity_loss_reasons
- | opportunity_quotes
- | opportunity_forecasts

Ana ilişkiler:

pipelines 1 — N pipeline_stages
pipeline_stages 1 — N opportunities
customers 1 — N opportunities
opportunities 1 — N quotes

6. QUOTE / Teklif

quotes

- | quote_versions
- | quote_items
- | quote_item_discounts
- | quote_taxes
- | quote_approvals
- | quote_approval_steps
- | quote_files
- | quote_view_logs
- | quote_signatures
- | quote_status_history
- | quote_payment_terms
- | quote_delivery_terms
- | quote_loss_reasons

Ana ilişkiler:

customers 1 — N quotes
opportunities 1 — N quotes

quotes 1 — N quote_items
quotes 1 — N quote_versions
quotes 0..1 — 1 orders

7. ORDER / Sipariş

orders

- | order_items
- | order_status_history
- | order_reservations
- | order_shipments
- | order_shipment_items
- | order_deliveries
- | order_delivery_confirmations
- | order_returns
- | order_return_items
- | order_payments
- | order_files
- | order_notes

Ana ilişkiler:

customers 1 — N orders
quotes 0..1 — 1 orders
orders 1 — N order_items
orders 1 — N shipments
orders 1 — N payments

8. PRODUCT / Catalog

products

- | product_variants
- | product_categories
- | product_brands
- | product_collections
- | product_images
- | product_files
- | product_attributes
- | product_attribute_values
- | product_barcode

- |─ product_supplier_links
- |─ product_prices
- |─ product_price_lists
- |─ product_related_items
- └─ product_ai_descriptions

Ana ilişkiler:

products 1 — N product_variants
products N — 1 product_categories
products N — 1 product_brands
product_variants 1 — N stocks
product_variants 1 — N quote_items
product_variants 1 — N order_items

9. INVENTORY / Stok

warehouses

- |─ warehouse_locations
- |─ warehouse_users

stocks

- |─ stock_movements
- |─ stock_reservations
- |─ stock_counts
- |─ stock_count_items
- |─ stock_adjustments
- |─ stock_transfers
- |─ stock_transfer_items
- |─ stock_alerts
- └─ stock_lots

Ana ilişkiler:

warehouses 1 — N stocks
product_variants 1 — N stocks
stocks 1 — N stock_movements
orders 1 — N stock_reservations

10. PROCUREMENT / Satın Alma

suppliers

- └ supplier_contacts
- └ supplier_addresses
- └ supplier_price_lists
- └ supplier_products
- └ supplier_performance_scores

purchase_requests

- └ purchase_request_items
- └ purchase_request_approvals

purchase_orders

- └ purchase_order_items
- └ purchase_order_status_history
- └ purchase_receipts
- └ purchase_receipt_items
- └ purchase_files

Ana ilişkiler:

suppliers 1 — N purchase_orders

purchase_orders 1 — N purchase_order_items

purchase_receipts 1 — N stock_movements

11. FINANCE / Cari, Tahsilat, Fatura

accounts

- └ account_transactions
- └ account_balances
- └ credit_limits
- └ risk_limits

payments

- └ payment_allocations
- └ payment_methods
- └ payment_receipts

installments

- └ installment_payments

invoices

- └ invoice_items
- └ invoice_taxes
- └ invoice_files

bank_accounts

- └ bank_transactions
- └ bank_reconciliation_logs

Ana ilişkiler:

customers 1 — 1 accounts

accounts 1 — N account_transactions

orders 1 — N invoices

invoices 1 — N payments

payments N — N invoices via payment_allocations

12. TASK / Activity

tasks

- └ task_assignees
- └ task_comments
- └ task_files
- └ task_checklist_items
- └ task_reminders
- └ task_recurrence_rules

activities

- └ activity_participants
- └ activity_files
- └ activity_comments

Evrensel bağlantı:

tasks.entity_type + tasks.entity_id

activities.entity_type + activities.entity_id

Bağlanabilir varlıklar:

customer

lead

opportunity
quote
order
ticket
project
invoice

13. COMMUNICATION / Omnichannel

channels

- └ channel_accounts
- └ channel_settings

conversations

- └ conversation_participants
- └ conversation_assignments
- └ conversation_tags
- └ conversation_status_history

messages

- └ message_attachments
- └ message_delivery_logs
- └ message_ai_summaries

email_logs

sms_logs

whatsapp_logs

call_logs

meeting_logs

Ana ilişkiler:

customers 1 — N conversations

conversations 1 — N messages

channels 1 — N channel_accounts

channel_accounts 1 — N conversations

14. SUPPORT / Ticket

tickets

- | ticket_categories
- | ticket_priorities
- | ticket_statuses
- | ticket_messages
- | ticket_files
- | ticket_assignments
- | ticket_sla_logs
- | ticket_satisfaction_surveys
- | ticket_root_causes
- | ticket_resolution_notes

knowledge_base_articles

- | knowledge_base_categories
- | knowledge_base_feedback

Ana ilişkiler:

customers 1 — N tickets

tickets 1 — N ticket_messages

tickets 1 — N ticket_files

tickets N — 1 users as assigned_user

15. MARKETING

campaigns

- | campaign_segments
- | campaign_channels
- | campaign_messages
- | campaign_recipients
- | campaign_events
- | campaign_conversions
- | campaign_ab_tests

journeys

- | journey_steps
- | journey_runs
- | journey_run_events

Ana ilişkiler:

campaigns N — N customer_segments
campaigns 1 — N campaign_recipients
journeys 1 — N journey_steps
customers 1 — N journey_runs

16. PROJECT / Uygulama ve Montaj

projects

- └ project_rooms
- └ project_measurements
- └ project_products
- └ project_tasks
- └ project_files
- └ project_photos
- └ project_costs
- └ project_payments
- └ project_installation_teams
- └ project_checklists
- └ project_close_reports

Ana ilişkiler:

customers 1 — N projects
projects 1 — N project_rooms
projects 1 — N project_tasks
projects 1 — N project_products

17. DEALER / B2B Portal

dealer_portal_users
dealer_catalogs
dealer_product_visibility
dealer_price_rules
dealer_orders
dealer_order_items
dealer_campaigns
dealer_balances
dealer_targets

dealer_commissions
dealer_regions

Ana ilişkiler:

dealers 1 — N dealer_users
dealers 1 — N dealer_orders
dealers 1 — N dealer_price_rules

18. WORKFLOW / Automation

workflows

- | workflow_versions
- | workflow_triggers
- | workflow_nodes
- | workflow_edges
- | workflow_conditions
- | workflow_actions
- | workflow_runs
- | workflow_run_steps
- | workflow_logs
- | workflow_errors

Approval engine:

approval_requests

- | approval_steps
- | approval_actions
- | approval_delegations
- | approval_logs

Ana ilişkiler:

workflows 1 — N workflow_versions
workflow_versions 1 — N workflow_nodes
workflow_runs 1 — N workflow_run_steps
approval_requests 1 — N approval_steps

19. AI Platform

ai_agents

- └ ai_agent_tools
- └ ai_agent_permissions
- └ ai_agent_runs

ai_prompts

- └ ai_prompt_versions

ai_conversations

- └ ai_conversation_messages

ai_documents

- └ ai_document_chunks
- └ ai_embeddings

ai_predictions

ai_summaries

ai_classifications

ai_recommendations

ai_usage_logs

ai_feedback

Ana ilişkiler:

ai_agents 1 — N ai_agent_runs

ai_documents 1 — N ai_document_chunks

ai_document_chunks 1 — 1 ai_embeddings

users 1 — N ai_conversations

20. REPORTING / Analytics

dashboards

- └ dashboard_widgets
- └ dashboard_filters
- └ dashboard_permissions

reports

- └ report_columns
- └ report_filters
- └ report_schedules

|— report_exports

metrics

metric_snapshots

kpi_targets

Ana ilişkiler:

dashboards 1 — N dashboard_widgets

reports 1 — N report_filters

users N — N dashboards via dashboard_permissions

21. INTEGRATION / API & Marketplace

api_keys

oauth_clients

oauth_tokens

webhooks

webhook_events

webhook_logs

connectors

connector_accounts

connector_mappings

integration_runs

integration_errors

marketplace_apps

marketplace_app_installs

Ana ilişkiler:

tenants 1 — N api_keys

tenants 1 — N connector_accounts

webhooks 1 — N webhook_logs

marketplace_apps 1 — N marketplace_app_installs

22. NOTIFICATION

notifications

|— notification_recipients

|— notification_channels

- |─ notification_templates
- |─ notification_delivery_logs
- |─ notification_preferences
- |─ notification_rules

Kanallar:

- in_app
- email
- sms
- whatsapp
- push
- webhook

23. LOW-CODE / Customization

- custom_modules
 - |─ custom_module_fields
 - |─ custom_module_records
 - |─ custom_module_record_values

- custom_fields
 - |─ custom_field_options
 - |─ custom_field_values

- custom_views
- custom_forms
- custom_pages
- custom_dashboards
- custom_validations
- custom_business_rules

Ana ilişkiler:

- custom_fields polymorphic entity_type/entity_id
- custom_module_records 1 — N custom_module_record_values

Evrensel Platform Tabloları

Bu tablolar tüm modüllere hizmet eder.

files

file_versions

file_permissions

comments

mentions

tags

entity_tags

timeline_events

domain_events

search_index

saved_searches

import_jobs

import_job_rows

export_jobs

data_quality_rules

data_quality_issues

merge_requests

merge_logs

Polymorphic bağlantı standardı:

entity_type

entity_id

Kullanılacağı tablolar:

files

comments

tags

timeline_events

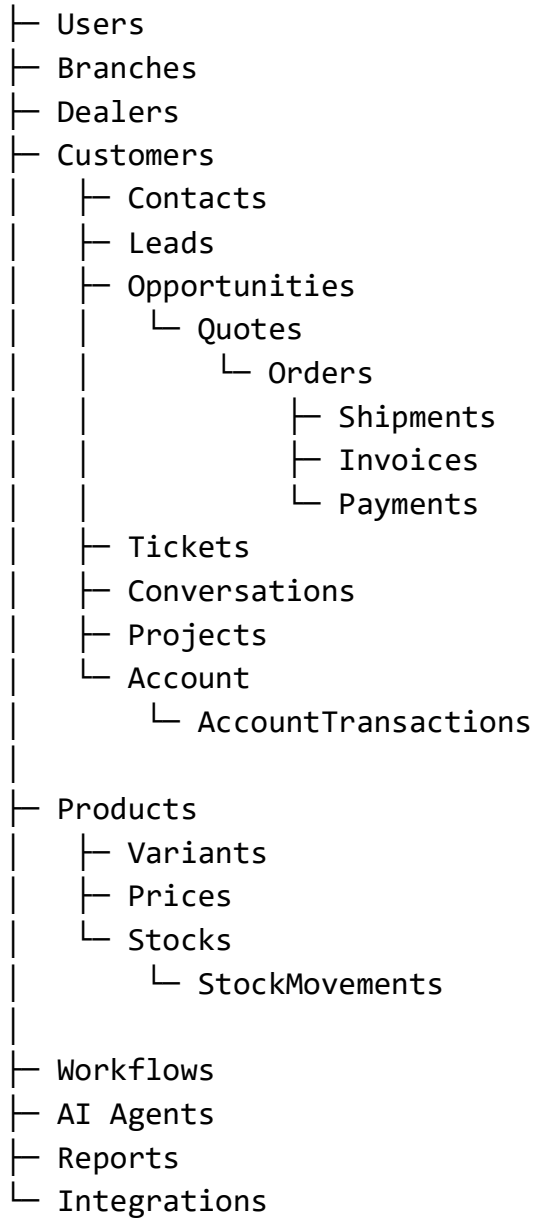
activities

tasks

approval_requests
workflow_runs
ai_summaries

Ana Veri Akış ERD Özeti

Tenant



Minimum Ana Tablo Sayısı

CORE	10
IAM	25
CRM	20
LEAD	10
SALES	15
QUOTE	15
ORDER	15
PRODUCT	15
INVENTORY	18
PROCUREMENT	15
FINANCE	20
TASK/ACTIVITY	12
COMMUNICATION	18
SUPPORT	15
MARKETING	15
PROJECT	12
DEALER	12
WORKFLOW	15
AI	18
REPORTING	12
INTEGRATION	14
NOTIFICATION	8
LOW-CODE	12
PLATFORM COMMON	18
TOTAL	~350 tablo

İlk Geliştirilecek ERD Slice

50 kişilik ekip için önce çekirdek slice:

CORE
IAM
CRM
LEAD
SALES
QUOTE
TASK
NOTIFICATION

AUDIT

İlk migration paketi:

001_core
002_iam
003_crm
004_lead
005_sales
006_quote
007_task_activity
008_notification
009_audit_event

Kritik Foreign Key Kuralları

Her tabloda:

tenant_id → tenants.id
created_by → users.id
updated_by → users.id
deleted_by → users.id

İş tablolarında:

customer_id → customers.id
assigned_user_id → users.id
branch_id → branches.id
dealer_id → dealers.id

Satış zinciri:

lead_id → leads.id
opportunity_id → opportunities.id
quote_id → quotes.id
order_id → orders.id
invoice_id → invoices.id
payment_id → payments.id

Ürün zinciri:

product_id → products.id
product_variant_id → product_variants.id
warehouse_id → warehouses.id
stock_id → stocks.id

PostgreSQL Tasarım Standardı

Her iş tablosu:

id uuid primary key
tenant_id uuid not null references tenants(id)

created_at timestamptz not null default now()
created_by uuid references users(id)

updated_at timestamptz
updated_by uuid references users(id)

deleted_at timestamptz
deleted_by uuid references users(id)

version integer not null default 1

Zorunlu indeks:

```
CREATE INDEX idx_<table>_tenant_id ON <table>(tenant_id);  
CREATE INDEX idx_<table>_deleted_at ON <table>(deleted_at);
```

Tenant + durum:

```
CREATE INDEX idx_<table>_tenant_status  
ON <table>(tenant_id, status);
```

RLS Standardı

```
ALTER TABLE <table_name> ENABLE ROW LEVEL SECURITY;
```

```
CREATE POLICY tenant_isolation_<table_name>  
ON <table_name>
```

```
USING (  
    tenant_id = current_setting('app.tenant_id')::uuid  
);
```

Backend her request başında:

```
SET app.tenant_id = '<tenant_id>';  
SET app.user_id = '<user_id>';
```

Büyük Tablolar İçin Partition Adayları

audit_logs
domain_events
timeline_events
messages
notifications
activity_logs
workflow_logs
ai_usage_logs
api_logs

Partition stratejisi:

tenant_id hash partition
+
created_at monthly partition

Master ERD v1 Sonucu

Bu model:

CRM
ERP Lite
Omnichannel
B2B Portal
Workflow Engine
AI Platform
Low-Code Customization

Enterprise SaaS Admin

tamamını taşıyabilecek ana veri omurgasıdır.

Bir sonraki teknik çıktı:

CRM OS Data Dictionary v1

olmalı. Bu dokümanda her tablo için:

field_name
data_type
nullable
default
foreign_key
index
description
business_rule

tanımlanır.

Aşağıda **CRM OS Data Dictionary v1** veriyorum. Bu sürüm, 50 kişilik ekibin aynı veri standardıyla çalışması için **çekirdek + enterprise omurga** sözlüğüdür.

CRM OS Data Dictionary v1

0. Global Field Standard

Tüm ana iş tablolarında bulunacak ortak alanlar:

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Primary key
tenant_id	uuid	Evet	Tenant izolasyonu
created_at	timestampz	Evet	Oluşturma zamanı
created_by	uuid	Hayır	Oluşturan kullanıcı
updated_at	timestampz	Hayır	Güncelleme zamanı

updated_by	uuid	Hayır	Güncelleyen kullanıcı
deleted_at	timestampz	Hayır	Soft delete
deleted_by	uuid	Hayır	Silen kullanıcı
version	int	Evet	Optimistic locking

1. CORE

tenants

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Tenant ID
name	varchar(255)	Evet	Firma adı
slug	varchar(100)	Evet	URL/tenant kodu
plan_id	uuid	Hayır	Paket
status	varchar(30)	Evet	active, suspended, cancelled
default_currency	varchar(3)	Evet	TRY, USD, EUR
timezone	varchar(80)	Evet	Europe/Istanbul
locale	varchar(10)	Evet	tr-TR
created_at	timestampz	Evet	Oluşturma

tenant_settings

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	PK
tenant_id	uuid	Evet	Tenant
key	varchar(150)	Evet	Ayar anahtarı
value	jsonb	Hayır	Ayar değeri
is_encrypted	boolean	Evet	Şifreli mi

plans

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Plan ID
code	varchar(50)	Evet	starter, pro, enterprise
name	varchar(100)	Evet	Plan adı
monthly_price	numeric(18,2)	Hayır	Aylık fiyat
yearly_price	numeric(18,2)	Hayır	Yıllık fiyat
is_active	boolean	Evet	Aktif mi

2. IAM

users

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	User ID
tenant_id	uuid	Evet	Tenant
email	varchar(255)	Evet	E-posta
phone	varchar(50)	Hayır	Telefon
password_hash	text	Hayır	Şifre hash
first_name	varchar(100)	Evet	Ad
last_name	varchar(100)	Evet	Soyad
avatar_url	text	Hayır	Profil resmi
status	varchar(30)	Evet	invited, active, blocked
last_login_at	timestamp	Hayır	Son giriş
branch_id	uuid	Hayır	Şube
department_id	uuid	Hayır	Departman
manager_id	uuid	Hayır	Üst yönetici

roles

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Role ID
tenant_id	uuid	Evet	Tenant
code	varchar(80)	Evet	sales_manager
name	varchar(120)	Evet	Rol adı
is_system	boolean	Evet	Sistem rolü mü

permissions

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Permission ID
code	varchar(150)	Evet	customer.read
module	varchar(80)	Evet	crm, sales, finance
description	text	Hayır	Açıklama

user_roles

Alan	Tip	Zorunlu	Açıklama
user_id	uuid	Evet	User FK
role_id	uuid	Evet	Role FK

role_permissions

Alan	Tip	Zorunlu	Açıklama
role_id	uuid	Evet	Role FK

permission_id	uuid	Evet	Permission FK
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3. ORGANIZATION

branches

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Şube ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Şube adı
code	varchar(50)	Hayır	Şube kodu
address	text	Hayır	Adres
city	varchar(100)	Hayır	Şehir
status	varchar(30)	Evet	active/passive

departments

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Departman ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Departman adı
parent_id	uuid	Hayır	Üst departman

teams

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Team ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Takım adı
leader_id	uuid	Hayır	Takım lideri

4. CRM / Customer 360

customers

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Customer ID
tenant_id	uuid	Evet	Tenant
customer_type	varchar(30)	Evet	individual, corporate
code	varchar(80)	Hayır	Müşteri kodu
company_name	varchar(255)	Hayır	Firma adı
first_name	varchar(100)	Hayır	Bireysel ad
last_name	varchar(100)	Hayır	Bireysel soyad

tax_number	varchar(50)	Hayır	Vergi no
tax_office	varchar(100)	Hayır	Vergi dairesi
email	varchar(255)	Hayır	Ana e-posta
phone	varchar(50)	Hayır	Ana telefon
source	varchar(100)	Hayır	Kaynak
segment_id	uuid	Hayır	Segment
status	varchar(30)	Evet	new, active, passive, risk, vip
assigned_user_id	uuid	Hayır	Müşteri temsilcisi
branch_id	uuid	Hayır	Şube
dealer_id	uuid	Hayır	Bayi
credit_limit	numeric(18,2)	Hayır	Kredi limiti
risk_score	numeric(5,2)	Hayır	Risk skoru
lifetime_value	numeric(18,2)	Hayır	Yaşam boyu değer

customer_contacts

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Contact ID
tenant_id	uuid	Evet	Tenant
customer_id	uuid	Evet	Customer
full_name	varchar(255)	Evet	Yetkili kişi
title	varchar(120)	Hayır	Ünvan
email	varchar(255)	Hayır	E-posta
phone	varchar(50)	Hayır	Telefon
mobile	varchar(50)	Hayır	Cep
is_primary	boolean	Evet	Ana kişi
is_decision_maker	boolean	Evet	Karar verici mi

customer_addresses

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Address ID
tenant_id	uuid	Evet	Tenant
customer_id	uuid	Evet	Customer
address_type	varchar(30)	Evet	billing, shipping, office
country	varchar(100)	Hayır	Ülke
city	varchar(100)	Hayır	Şehir
district	varchar(100)	Hayır	İlçe
address_line	text	Evet	Açık adres
postal_code	varchar(20)	Hayır	Posta kodu
latitude	numeric(10,7)	Hayır	Enlem

longitude	numeric(10,7)	Hayır	Boylam
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customer_segments

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Segment ID
tenant_id	uuid	Evet	Tenant
name	varchar(120)	Evet	Segment adı
description	text	Hayır	Açıklama

5. LEAD

leads

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Lead ID
tenant_id	uuid	Evet	Tenant
full_name	varchar(255)	Hayır	Kişi adı
company_name	varchar(255)	Hayır	Firma
email	varchar(255)	Hayır	E-posta
phone	varchar(50)	Hayır	Telefon
source_id	uuid	Hayır	Lead kaynağı
status	varchar(40)	Evet	new, contacted, qualified, lost, converted
score	int	Evet	Lead skoru
temperature	varchar(20)	Hayır	cold, warm, hot
assigned_user_id	uuid	Hayır	Temsilci
customer_id	uuid	Hayır	Dönüşen müşteri
opportunity_id	uuid	Hayır	Dönüşen fırsat
converted_at	timestampz	Hayır	Dönüşüm zamanı
lost_reason_id	uuid	Hayır	Kayıp nedeni

lead_sources

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Source ID
tenant_id	uuid	Evet	Tenant
name	varchar(120)	Evet	Kaynak adı
channel	varchar(80)	Hayır	web, whatsapp, instagram, fair
is_active	boolean	Evet	Aktif mi

lead_conversion_logs

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Log ID
tenant_id	uuid	Evet	Tenant
lead_id	uuid	Evet	Lead
customer_id	uuid	Hayır	Oluşan müşteri
opportunity_id	uuid	Hayır	Oluşan fırsat
converted_by	uuid	Evet	Kullanıcı
converted_at	timestampz	Evet	Tarih

6. SALES

pipelines

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Pipeline ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Pipeline adı
module	varchar(50)	Hayır	sales, project, dealer
is_default	boolean	Evet	Varsayılan mı

pipeline_stages

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Stage ID
tenant_id	uuid	Evet	Tenant
pipeline_id	uuid	Evet	Pipeline
name	varchar(120)	Evet	Aşama adı
probability	numeric(5,2)	Evet	Olasılık
sort_order	int	Evet	Sıra
stage_type	varchar(30)	Evet	open, won, lost

opportunities

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Opportunity ID
tenant_id	uuid	Evet	Tenant
customer_id	uuid	Evet	Müşteri
lead_id	uuid	Hayır	Kaynak lead
pipeline_id	uuid	Evet	Pipeline
stage_id	uuid	Evet	Aşama
title	varchar(255)	Evet	Fırsat adı

amount	numeric(18,2)	Evet	Beklenen tutar
currency	varchar(3)	Evet	Para birimi
probability	numeric(5,2)	Hayır	Kazanma olasılığı
expected_close_date	date	Hayır	Tahmini kapanış
status	varchar(30)	Evet	open, won, lost
assigned_user_id	uuid	Hayır	Temsilci
lost_reason_id	uuid	Hayır	Kayıp nedeni
won_at	timestamp tz	Hayır	Kazanma zamanı
lost_at	timestamp tz	Hayır	Kaybetme zamanı

opportunity_products

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	PK
tenant_id	uuid	Evet	Tenant
opportunity_id	uuid	Evet	Fırsat
product_id	uuid	Hayır	Ürün
product_variant_id	uuid	Hayır	Varyant
quantity	numeric(18,3)	Evet	Miktar
estimated_unit_price	numeric(18,2)	Hayır	Tahmini fiyat

7. QUOTE

quotes

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Quote ID
tenant_id	uuid	Evet	Tenant
quote_no	varchar(80)	Evet	Teklif no
customer_id	uuid	Evet	Müşteri
opportunity_id	uuid	Hayır	Fırsat
status	varchar(40)	Evet	draft, sent, viewed, approved, rejected, expired
currency	varchar(3)	Evet	Para birimi
subtotal_amount	numeric(18,2)	Evet	Ara toplam
discount_amount	numeric(18,2)	Evet	İndirim
tax_amount	numeric(18,2)	Evet	Vergi
total_amount	numeric(18,2)	Evet	Genel toplam
cost_amount	numeric(18,2)	Hayır	Maliyet
profit_amount	numeric(18,2)	Hayır	Kâr
margin_percent	numeric(6,2)	Hayır	Marj

valid_until	date	Hayır	Geçerlilik
sent_at	timestampz	Hayır	Gönderim
approved_at	timestampz	Hayır	Onay
rejected_at	timestampz	Hayır	Red

quote_items

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Item ID
tenant_id	uuid	Evet	Tenant
quote_id	uuid	Evet	Teklif
product_id	uuid	Hayır	Ürün
product_variant_id	uuid	Hayır	Varyant
description	text	Hayır	Açıklama
quantity	numeric(18,3)	Evet	Miktar
unit	varchar(30)	Hayır	Birim
unit_price	numeric(18,2)	Evet	Birim fiyat
discount_amount	numeric(18,2)	Evet	İndirim
tax_rate	numeric(5,2)	Evet	KDV oranı
line_total	numeric(18,2)	Evet	Satır toplamı

quote_approvals

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Approval ID
tenant_id	uuid	Evet	Tenant
quote_id	uuid	Evet	Teklif
requested_by	uuid	Evet	Talep eden
status	varchar(30)	Evet	pending, approved, rejected
reason	text	Hayır	Onay gerekçesi
requested_at	timestampz	Evet	Talep tarihi

8. ORDER

orders

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Order ID
tenant_id	uuid	Evet	Tenant
order_no	varchar(80)	Evet	Sipariş no
customer_id	uuid	Evet	Müşteri
quote_id	uuid	Hayır	Kaynak teklif

			draft, confirmed, preparing, shipped, delivered, cancelled
status	varchar(40)	Evet	
currency	varchar(3)	Evet	Para birimi
subtotal_amount	numeric(18,2)	Evet	Ara toplam
tax_amount	numeric(18,2)	Evet	Vergi
total_amount	numeric(18,2)	Evet	Genel toplam
expected_delivery_date	date	Hayır	Termin
confirmed_at	timestampz	Hayır	Onay zamanı
delivered_at	timestampz	Hayır	Teslim zamanı

order_items

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Item ID
tenant_id	uuid	Evet	Tenant
order_id	uuid	Evet	Sipariş
product_id	uuid	Hayır	Ürün
product_variant_id	uuid	Hayır	Varyant
quantity	numeric(18,3)	Evet	Miktar
unit_price	numeric(18,2)	Evet	Birim fiyat
tax_rate	numeric(5,2)	Evet	KDV
line_total	numeric(18,2)	Evet	Satır toplamı
reserved_quantity	numeric(18,3)	Evet	Rezerve miktar
shipped_quantity	numeric(18,3)	Evet	Sevk edilen

9. PRODUCT / CATALOG

products

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Product ID
tenant_id	uuid	Evet	Tenant
sku	varchar(100)	Evet	SKU
name	varchar(255)	Evet	Ürün adı
description	text	Hayır	Açıklama
category_id	uuid	Hayır	Kategori
brand_id	uuid	Hayır	Marka
collection_id	uuid	Hayır	Koleksiyon

product_type	varchar(50)	Evet	wallpaper, fabric, service
unit	varchar(30)	Evet	pcs, roll, m2
is_stock_tracked	boolean	Evet	Stok takip edilsin mi
status	varchar(30)	Evet	active, passive

product_variants

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Variant ID
tenant_id	uuid	Evet	Tenant
product_id	uuid	Evet	Ürün
sku	varchar(100)	Evet	Varyant SKU
barcode	varchar(100)	Hayır	Barkod
color	varchar(100)	Hayır	Renk
pattern	varchar(100)	Hayır	Desen
size	varchar(100)	Hayır	Ölçü
roll_width	numeric(10,2)	Hayır	Rulo eni
roll_length	numeric(10,2)	Hayır	Rulo boyu
status	varchar(30)	Evet	active/passive

product_prices

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Price ID
tenant_id	uuid	Evet	Tenant
product_variant_id	uuid	Evet	Varyant
price_list_id	uuid	Evet	Fiyat listesi
currency	varchar(3)	Evet	Para birimi
price	numeric(18,2)	Evet	Satış fiyatı
cost_price	numeric(18,2)	Hayır	Maliyet
valid_from	date	Hayır	Başlangıç
valid_to	date	Hayır	Bitiş

10. INVENTORY

warehouses

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Warehouse ID
tenant_id	uuid	Evet	Tenant
branch_id	uuid	Hayır	Şube

name	varchar(150)	Evet	Depo adı
code	varchar(50)	Hayır	Depo kodu
status	varchar(30)	Evet	active/passive

stocks

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Stock ID
tenant_id	uuid	Evet	Tenant
warehouse_id	uuid	Evet	Depo
product_variant_id	uuid	Evet	Varyant
quantity_on_hand	numeric(18,3)	Evet	Fiziki stok
quantity_reserved	numeric(18,3)	Evet	Rezerve
quantity_available	numeric(18,3)	Evet	Satılabilir
critical_level	numeric(18,3)	Hayır	Kritik seviye

stock_movements

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Movement ID
tenant_id	uuid	Evet	Tenant
warehouse_id	uuid	Evet	Depo
product_variant_id	uuid	Evet	Varyant
movement_type	varchar(30)	Evet	in, out, transfer, reserve, release, adjust
quantity	numeric(18,3)	Evet	Miktar
reference_type	varchar(50)	Hayır	order, purchase, count
reference_id	uuid	Hayır	İlgili kayıt
movement_at	timestampz	Evet	Hareket zamanı

11. FINANCE

accounts

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Account ID
tenant_id	uuid	Evet	Tenant
customer_id	uuid	Hayır	Müşteri cari
supplier_id	uuid	Hayır	Tedarikçi cari
account_code	varchar(80)	Evet	Cari kod

account_type	varchar(30)	Evet	customer, supplier, dealer
currency	varchar(3)	Evet	Para birimi
balance	numeric(18,2)	Evet	Güncel bakiye
credit_limit	numeric(18,2)	Hayır	Kredi limiti
status	varchar(30)	Evet	active/passive

account_transactions

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Transaction ID
tenant_id	uuid	Evet	Tenant
account_id	uuid	Evet	Cari hesap
transaction_type	varchar(40)	Evet	debit, credit, payment, invoice, refund
debit_amount	numeric(18,2)	Evet	Borç
credit_amount	numeric(18,2)	Evet	Alacak
currency	varchar(3)	Evet	Para birimi
reference_type	varchar(50)	Hayır	invoice, payment, order
reference_id	uuid	Hayır	İlgili kayıt
transaction_date	date	Evet	İşlem tarihi

payments

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Payment ID
tenant_id	uuid	Evet	Tenant
account_id	uuid	Evet	Cari
customer_id	uuid	Hayır	Müşteri
payment_method_id	uuid	Evet	Ödeme yöntemi
amount	numeric(18,2)	Evet	Tutar
currency	varchar(3)	Evet	Para birimi
payment_date	date	Evet	Ödeme tarihi
status	varchar(30)	Evet	pending, completed, cancelled

12. TASK / ACTIVITY

tasks

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Task ID
tenant_id	uuid	Evet	Tenant
title	varchar(255)	Evet	Başlık
description	text	Hayır	Açıklama
entity_type	varchar(50)	Hayır	customer, lead, quote
entity_id	uuid	Hayır	İlgili kayıt
assigned_user_id	uuid	Hayır	Atanan kişi
priority	varchar(20)	Evet	low, normal, high, urgent
status	varchar(30)	Evet	todo, doing, done, cancelled
due_at	timestampz	Hayır	Son tarih
completed_at	timestampz	Hayır	Tamamlanma

activities

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Activity ID
tenant_id	uuid	Evet	Tenant
entity_type	varchar(50)	Evet	customer, lead, opportunity
entity_id	uuid	Evet	İlgili kayıt
activity_type	varchar(50)	Evet	call, meeting, email, whatsapp, note
subject	varchar(255)	Hayır	Konu
content	text	Hayır	İçerik
activity_at	timestampz	Evet	Aktivite zamanı
created_by	uuid	Hayır	Oluşturan

13. COMMUNICATION

conversations

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Conversation ID
tenant_id	uuid	Evet	Tenant
customer_id	uuid	Hayır	Müşteri

channel_id	uuid	Evet	Kanal
channel_account_id	uuid	Evet	Bağlı hesap
subject	varchar(255)	Hayır	Konu
status	varchar(30)	Evet	open, pending, resolved
assigned_user_id	uuid	Hayır	Atanan temsilci
last_message_at	timestampz	Hayır	Son mesaj zamanı

messages

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Message ID
tenant_id	uuid	Evet	Tenant
conversation_id	uuid	Evet	Konuşma
sender_type	varchar(30)	Evet	customer, user, system, ai
sender_id	uuid	Hayır	Gönderen
message_type	varchar(30)	Evet	text, image, file, audio
body	text	Hayır	Mesaj içeriği
direction	varchar(20)	Evet	inbound, outbound
status	varchar(30)	Evet	sent, delivered, read, failed
sent_at	timestampz	Hayır	Gönderim

14. SUPPORT

tickets

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Ticket ID
tenant_id	uuid	Evet	Tenant
ticket_no	varchar(80)	Evet	Ticket no
customer_id	uuid	Hayır	Müşteri
subject	varchar(255)	Evet	Konu
description	text	Hayır	Açıklama
category_id	uuid	Hayır	Kategori
priority	varchar(20)	Evet	low, normal, high, urgent
status	varchar(30)	Evet	open, pending, resolved, closed
assigned_user_id	uuid	Hayır	Temsilci

sla_due_at	timestamp	Hayır	SLA son tarihi
resolved_at	timestamp	Hayır	Çözüm

15. WORKFLOW

workflows

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Workflow ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Workflow adı
module	varchar(80)	Evet	crm, sales, finance
status	varchar(30)	Evet	draft, active, paused
trigger_type	varchar(80)	Hayır	event, schedule, manual

workflow_nodes

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Node ID
tenant_id	uuid	Evet	Tenant
workflow_id	uuid	Evet	Workflow
node_type	varchar(50)	Evet	trigger, condition, action, delay
config	jsonb	Evet	Node ayarları
position_x	int	Hayır	Canvas X
position_y	int	Hayır	Canvas Y

workflow_runs

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Run ID
tenant_id	uuid	Evet	Tenant
workflow_id	uuid	Evet	Workflow
status	varchar(30)	Evet	running, completed, failed
entity_type	varchar(50)	Hayır	Tetikleyen varlık
entity_id	uuid	Hayır	Kayıt
started_at	timestamp	Evet	Başlangıç
finished_at	timestamp	Hayır	Bitiş

16. AI

ai_agents

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Agent ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Agent adı
agent_type	varchar(80)	Evet	sales, support, finance
instructions	text	Hayır	Sistem talimatı
status	varchar(30)	Evet	active, inactive

ai_usage_logs

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Usage ID
tenant_id	uuid	Evet	Tenant
user_id	uuid	Hayır	Kullanıcı
agent_id	uuid	Hayır	Agent
action_type	varchar(80)	Evet	summarize, classify, generate
input_tokens	int	Hayır	Girdi token
output_tokens	int	Hayır	Çıktı token
cost_amount	numeric(18,6)	Hayır	Maliyet
created_at	timestampz	Evet	Zaman

17. REPORTING

dashboards

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Dashboard ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Dashboard adı
owner_id	uuid	Hayır	Sahip
visibility	varchar(30)	Evet	private, team, tenant
layout_config	jsonb	Hayır	Yerleşim

dashboard_widgets

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Widget ID
tenant_id	uuid	Evet	Tenant

dashboard_id	uuid	Evet	Dashboard
widget_type	varchar(80)	Evet	chart, metric, table
title	varchar(150)	Evet	Başlık
query_config	jsonb	Evet	Veri sorgusu
layout	jsonb	Evet	Pozisyon

18. INTEGRATION

api_keys

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	API Key ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Anahtar adı
key_hash	text	Evet	Hashlenmiş key
scopes	text[]	Evet	Yetki kapsamı
status	varchar(30)	Evet	active, revoked
last_used_at	timestampz	Hayır	Son kullanım

webhooks

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Webhook ID
tenant_id	uuid	Evet	Tenant
name	varchar(150)	Evet	Webhook adı
url	text	Evet	Hedef URL
events	text[]	Evet	Dinlenen eventler
secret	text	Evet	İmza secret
status	varchar(30)	Evet	active, inactive

19. COMMON PLATFORM

files

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	File ID
tenant_id	uuid	Evet	Tenant
entity_type	varchar(50)	Hayır	Bağlı varlık tipi
entity_id	uuid	Hayır	Bağlı kayıt
file_name	varchar(255)	Evet	Dosya adı
mime_type	varchar(120)	Hayır	MIME
size_bytes	bigint	Hayır	Boyut

storage_provider	varchar(50)	Evet	s3, minio
storage_path	text	Evet	Dosya yolu
uploaded_by	uuid	Hayır	Yükleyen

comments

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Comment ID
tenant_id	uuid	Evet	Tenant
entity_type	varchar(50)	Evet	Varlık tipi
entity_id	uuid	Evet	Varlık ID
body	text	Evet	Yorum
created_by	uuid	Evet	Kullanıcı

tags

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Tag ID
tenant_id	uuid	Evet	Tenant
name	varchar(80)	Evet	Etiket
color	varchar(20)	Hayır	Renk
module	varchar(80)	Hayır	Modül

entity_tags

Alan	Tip	Zorunlu	Açıklama
tenant_id	uuid	Evet	Tenant
tag_id	uuid	Evet	Tag
entity_type	varchar(50)	Evet	Varlık tipi
entity_id	uuid	Evet	Kayıt ID

timeline_events

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Event ID
tenant_id	uuid	Evet	Tenant
entity_type	varchar(50)	Evet	customer, lead, quote
entity_id	uuid	Evet	Kayıt
event_type	varchar(100)	Evet	QuoteSent, PaymentReceived
title	varchar(255)	Evet	Görünen başlık
payload	jsonb	Hayır	Detay veri
occurred_at	timestamp	Evet	Olay zamanı

domain_events

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Event ID
tenant_id	uuid	Evet	Tenant
event_type	varchar(100)	Evet	LeadCreated
aggregate_type	varchar(80)	Evet	lead, quote, order
aggregate_id	uuid	Evet	Kayıt ID
payload	jsonb	Evet	Event verisi
published_at	timestampz	Hayır	Yayın zamanı
created_at	timestampz	Evet	Oluşturma

audit_logs

Alan	Tip	Zorunlu	Açıklama
id	uuid	Evet	Audit ID
tenant_id	uuid	Evet	Tenant
user_id	uuid	Hayır	Kullanıcı
entity_type	varchar(80)	Evet	Tablo/varlık
entity_id	uuid	Hayır	Kayıt
action	varchar(50)	Evet	create, update, delete, login
old_data	jsonb	Hayır	Eski veri
new_data	jsonb	Hayır	Yeni veri
ip_address	inet	Hayır	IP
user_agent	text	Hayır	Cihaz
created_at	timestampz	Evet	Zaman

20. Index Standardı

Her tenant tablosunda:

```
CREATE INDEX idx_<table>_tenant_id ON <table>(tenant_id);  
CREATE INDEX idx_<table>_tenant_deleted ON <table>(tenant_id,  
deleted_at);
```

Durum bazlı tablolarda:

```
CREATE INDEX idx_<table>_tenant_status ON <table>(tenant_id,  
status);
```

Sahip/temsilci bazlı tablolarda:

```
CREATE INDEX idx_<table>_assigned_user ON <table>(tenant_id,  
assigned_user_id);
```

Zaman bazlı büyük tablolarda:

```
CREATE INDEX idx_<table>_tenant_created_at ON <table>(tenant_id,  
created_at DESC);
```

21. Naming Standard

Tablo: snake_case plural

Alan: snake_case

PK: id

FK: <entity>_id

Status: varchar(30/40)

Amount: numeric(18,2)

Quantity: numeric(18,3)

DateTime: timestamptz

Date: date

JSON: jsonb

22. Data Dictionary v1 Sonucu

Bu sözlük:

CORE

IAM

CRM

LEAD

SALES

QUOTE

ORDER

PRODUCT

INVENTORY

FINANCE

TASK

COMMUNICATION

SUPPORT

WORKFLOW

AI
REPORTING
INTEGRATION
COMMON

modülleri için başlangıç veri standardını tanımlar.

Sonraki çıktı doğrudan:

CRM OS PostgreSQL Schema v1

olmalı.

Aşağıda **CRM OS OpenAPI Contract v1** çekirdek API sözleşmesi var.

openapi: 3.1.0

info:

title: CRM OS API

version: 1.0.0

description: Multi-tenant SaaS CRM OS API

servers:

- url: <https://api.crm-os.com/api/v1>

security:

- bearerAuth: []

tags:

- name: Auth
- name: Tenants
- name: Users
- name: Roles
- name: Customers
- name: Leads
- name: Pipelines
- name: Opportunities
- name: Quotes
- name: Tasks
- name: Activities
- name: Files

- name: Notifications
- name: Audit

components:

securitySchemes:

bearerAuth:

type: http

scheme: bearer

bearerFormat: JWT

parameters:

TenantId:

name: X-Tenant-Id

in: header

required: true

schema:

type: string

format: uuid

Page:

name: page

in: query

schema:

type: integer

default: 1

PageSize:

name: pageSize

in: query

schema:

type: integer

default: 25

schemas:

ApiResponse:

type: object

properties:

data:

type: object

meta:

type: object

ErrorResponse:

- type: object
- properties:
 - error:
 - type: object
 - properties:
 - code:
 - type: string
 - message:
 - type: string
 - details:
 - type: object

LoginRequest:

- type: object
- required: [email, password]
- properties:
 - email:
 - type: string
 - password:
 - type: string

LoginResponse:

- type: object
- properties:
 - accessToken:
 - type: string
 - refreshToken:
 - type: string
 - user:
 - \$ref: "#/components/schemas/User"

User:

- type: object
- properties:
 - id:
 - type: string
 - format: uuid
 - tenantId:
 - type: string
 - format: uuid
 - email:

```
    type: string
  firstName:
    type: string
  lastName:
    type: string
  status:
    type: string
```

Customer:

```
  type: object
  properties:
    id:
      type: string
      format: uuid
    customerType:
      type: string
      enum: [individual, corporate]
    companyName:
      type: string
    firstName:
      type: string
    lastName:
      type: string
    email:
      type: string
    phone:
      type: string
    status:
      type: string
    assignedUserId:
      type: string
      format: uuid
```

CreateCustomerRequest:

```
  type: object
  required: [customerType]
  properties:
    customerType:
      type: string
      enum: [individual, corporate]
    companyName:
      type: string
```

```
  firstName:
    type: string
  lastName:
    type: string
  email:
    type: string
  phone:
    type: string
  assignedUserId:
    type: string
    format: uuid
```

```
Lead:
  type: object
  properties:
    id:
      type: string
      format: uuid
    fullName:
      type: string
    companyName:
      type: string
    email:
      type: string
    phone:
      type: string
    status:
      type: string
    score:
      type: integer
    assignedUserId:
      type: string
      format: uuid
```

```
CreateLeadRequest:
  type: object
  properties:
    fullName:
      type: string
    companyName:
      type: string
    email:
```


type: string
phone:
type: string
sourceId:
type: string
format: uuid

ConvertLeadRequest:

type: object
properties:
createCustomer:
type: boolean
customerId:
type: string
format: uuid
createOpportunity:
type: boolean
opportunityTitle:
type: string
amount:
type: number

Opportunity:

type: object
properties:
id:
type: string
format: uuid
customerId:
type: string
format: uuid
pipelineId:
type: string
format: uuid
stageId:
type: string
format: uuid
title:
type: string
amount:
type: number
status:

type: string

Quote:

type: object

properties:

id:

type: string

format: uuid

quoteNo:

type: string

customerId:

type: string

format: uuid

opportunityId:

type: string

format: uuid

status:

type: string

totalAmount:

type: number

paths:

/auth/login:

post:

tags: [Auth]

summary: Login

security: []

requestBody:

required: true

content:

application/json:

schema:

\$ref: "#/components/schemas/LoginRequest"

responses:

"200":

description: Login success

content:

application/json:

schema:

\$ref: "#/components/schemas/LoginResponse"

/auth/refresh:

```
post:
  tags: [Auth]
  summary: Refresh access token
  responses:
    "200":
      description: Token refreshed

/auth/logout:
  post:
    tags: [Auth]
    summary: Logout
    responses:
      "204":
        description: Logged out

/users:
  get:
    tags: [Users]
    summary: List users
    parameters:
      - $ref: "#/components/parameters/TenantId"
      - $ref: "#/components/parameters/Page"
      - $ref: "#/components/parameters/PageSize"
    responses:
      "200":
        description: Users list

  post:
    tags: [Users]
    summary: Create user
    parameters:
      - $ref: "#/components/parameters/TenantId"
    responses:
      "201":
        description: User created

/users/{id}:
  get:
    tags: [Users]
    summary: Get user
    parameters:
      - $ref: "#/components/parameters/TenantId"
```

- name: id
 - in: path
 - required: true
 - schema:
 - type: string
 - format: uuid

responses:

"200":

description: User detail

patch:

tags: [Users]

summary: Update user

parameters:

- \$ref: "#/components/parameters/TenantId"
- name: id
 - in: path
 - required: true
 - schema:
 - type: string
 - format: uuid

responses:

"200":

description: User updated

delete:

tags: [Users]

summary: Soft delete user

parameters:

- \$ref: "#/components/parameters/TenantId"
- name: id
 - in: path
 - required: true
 - schema:
 - type: string
 - format: uuid

responses:

"204":

description: Deleted

/customers:

get:

```
tags: [Customers]
summary: List customers
parameters:
  - $ref: "#/components/parameters/TenantId"
  - $ref: "#/components/parameters/Page"
  - $ref: "#/components/parameters/PageSize"
  - name: status
    in: query
    schema:
      type: string
  - name: q
    in: query
    schema:
      type: string
responses:
  "200":
    description: Customer list
```

```
post:
  tags: [Customers]
  summary: Create customer
  parameters:
    - $ref: "#/components/parameters/TenantId"
  requestBody:
    required: true
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/CreateCustomerRequest"
  responses:
    "201":
      description: Customer created
```

```
/customers/{id}:
get:
  tags: [Customers]
  summary: Get customer detail
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
```

```
    schema:
      type: string
      format: uuid
  responses:
    "200":
      description: Customer detail
```

```
patch:
  tags: [Customers]
  summary: Update customer
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Customer updated
```

```
delete:
  tags: [Customers]
  summary: Soft delete customer
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "204":
      description: Deleted
```

```
/customers/{id}/timeline:
  get:
    tags: [Customers]
    summary: Get customer timeline
    parameters:
```

- \$ref: "#/components/parameters/TenantId"
- name: id
 - in: path
 - required: true
 - schema:
 - type: string
 - format: uuid

responses:

"200":

description: Customer timeline

/customers/{id}/contacts:

get:

tags: [Customers]

summary: List customer contacts

parameters:

- \$ref: "#/components/parameters/TenantId"
- name: id
 - in: path
 - required: true
 - schema:
 - type: string
 - format: uuid

responses:

"200":

description: Contact list

post:

tags: [Customers]

summary: Create customer contact

parameters:

- \$ref: "#/components/parameters/TenantId"
- name: id
 - in: path
 - required: true
 - schema:
 - type: string
 - format: uuid

responses:

"201":

description: Contact created

```
/leads:
  get:
    tags: [Leads]
    summary: List leads
    parameters:
      - $ref: "#/components/parameters/TenantId"
      - $ref: "#/components/parameters/Page"
      - $ref: "#/components/parameters/PageSize"
    responses:
      "200":
        description: Lead list

  post:
    tags: [Leads]
    summary: Create lead
    parameters:
      - $ref: "#/components/parameters/TenantId"
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/CreateLeadRequest"
    responses:
      "201":
        description: Lead created

/leads/{id}:
  get:
    tags: [Leads]
    summary: Get lead
    parameters:
      - $ref: "#/components/parameters/TenantId"
      - name: id
        in: path
        required: true
        schema:
          type: string
          format: uuid
    responses:
      "200":
        description: Lead detail
```



```
patch:
  tags: [Leads]
  summary: Update lead
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Lead updated
```

```
/leads/{id}/convert:
  post:
    tags: [Leads]
    summary: Convert lead to customer and opportunity
    parameters:
      - $ref: "#/components/parameters/TenantId"
      - name: id
        in: path
        required: true
        schema:
          type: string
          format: uuid
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/ConvertLeadRequest"
    responses:
      "200":
        description: Lead converted
```

```
/pipelines:
  get:
    tags: [Pipelines]
    summary: List pipelines
```

```
parameters:
  - $ref: "#/components/parameters/TenantId"
responses:
  "200":
    description: Pipeline list
```

```
post:
  tags: [Pipelines]
  summary: Create pipeline
  parameters:
    - $ref: "#/components/parameters/TenantId"
  responses:
    "201":
      description: Pipeline created
```

```
/pipelines/{id}/stages:
get:
  tags: [Pipelines]
  summary: List pipeline stages
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Stage list
```

```
post:
  tags: [Pipelines]
  summary: Create pipeline stage
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
```

```
responses:
  "201":
    description: Stage created
```

/opportunities:

```
get:
  tags: [Opportunities]
  summary: List opportunities
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - $ref: "#/components/parameters/Page"
    - $ref: "#/components/parameters/PageSize"
  responses:
    "200":
      description: Opportunity list
```

```
post:
  tags: [Opportunities]
  summary: Create opportunity
  parameters:
    - $ref: "#/components/parameters/TenantId"
  responses:
    "201":
      description: Opportunity created
```

/opportunities/{id}:

```
get:
  tags: [Opportunities]
  summary: Get opportunity
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Opportunity detail
```

```
patch:
```

```
tags: [Opportunities]
summary: Update opportunity
parameters:
  - $ref: "#/components/parameters/TenantId"
  - name: id
    in: path
    required: true
    schema:
      type: string
      format: uuid
responses:
  "200":
    description: Opportunity updated
```

/opportunities/{id}/move-stage:

```
post:
  tags: [Opportunities]
  summary: Move opportunity to another stage
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Stage changed
```

/opportunities/{id}/win:

```
post:
  tags: [Opportunities]
  summary: Mark opportunity as won
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
```

```
responses:
  "200":
    description: Opportunity won
```

/opportunities/{id}/lose:

```
post:
  tags: [Opportunities]
  summary: Mark opportunity as lost
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Opportunity lost
```

/quotes:

```
get:
  tags: [Quotes]
  summary: List quotes
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - $ref: "#/components/parameters/Page"
    - $ref: "#/components/parameters/PageSize"
  responses:
    "200":
      description: Quote list
```

```
post:
  tags: [Quotes]
  summary: Create quote
  parameters:
    - $ref: "#/components/parameters/TenantId"
  responses:
    "201":
      description: Quote created
```

/quotes/{id}:

```
get:
  tags: [Quotes]
  summary: Get quote
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Quote detail
```

```
patch:
  tags: [Quotes]
  summary: Update quote
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Quote updated
```

```
/quotes/{id}/items:
  post:
    tags: [Quotes]
    summary: Add quote item
    parameters:
      - $ref: "#/components/parameters/TenantId"
      - name: id
        in: path
        required: true
        schema:
          type: string
          format: uuid
```

```
responses:
  "201":
    description: Quote item added
```

/quotes/{id}/send:

```
post:
  tags: [Quotes]
  summary: Send quote
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Quote sent
```

/quotes/{id}/approve:

```
post:
  tags: [Quotes]
  summary: Approve quote
  parameters:
    - $ref: "#/components/parameters/TenantId"
    - name: id
      in: path
      required: true
      schema:
        type: string
        format: uuid
  responses:
    "200":
      description: Quote approved
```

/quotes/{id}/reject:

```
post:
  tags: [Quotes]
  summary: Reject quote
  parameters:
    - $ref: "#/components/parameters/TenantId"
```

- name: id
- in: path
- required: true
- schema:
 - type: string
 - format: uuid

responses:

"200":

description: Quote rejected

/quotes/{id}/pdf:

get:

tags: [Quotes]

summary: Generate or download quote PDF

parameters:

- \$ref: "#/components/parameters/TenantId"
- name: id
- in: path
- required: true
- schema:
 - type: string
 - format: uuid

responses:

"200":

description: Quote PDF

/tasks:

get:

tags: [Tasks]

summary: List tasks

parameters:

- \$ref: "#/components/parameters/TenantId"
- \$ref: "#/components/parameters/Page"
- \$ref: "#/components/parameters/PageSize"

responses:

"200":

description: Task list

post:

tags: [Tasks]

summary: Create task

parameters:

- \$ref: "#/components/parameters/TenantId"

responses:

"201":

description: Task created

/tasks/{id}/complete:

post:

tags: [Tasks]

summary: Complete task

parameters:

- \$ref: "#/components/parameters/TenantId"
- name: id
- in: path
- required: true
- schema:
- type: string
- format: uuid

responses:

"200":

description: Task completed

/activities:

get:

tags: [Activities]

summary: List activities

parameters:

- \$ref: "#/components/parameters/TenantId"
- name: entityType
- in: query
- schema:
- type: string
- name: entityId
- in: query
- schema:
- type: string
- format: uuid

responses:

"200":

description: Activity list

post:

tags: [Activities]

summary: Create activity
parameters:
- \$ref: "#/components/parameters/TenantId"
responses:
"201":
description: Activity created

/files:
post:
tags: [Files]
summary: Upload file metadata
parameters:
- \$ref: "#/components/parameters/TenantId"
responses:
"201":
description: File registered

/notifications:
get:
tags: [Notifications]
summary: List notifications
parameters:
- \$ref: "#/components/parameters/TenantId"
responses:
"200":
description: Notification list

/notifications/{id}/read:
post:
tags: [Notifications]
summary: Mark notification as read
parameters:
- \$ref: "#/components/parameters/TenantId"
- name: id
in: path
required: true
schema:
type: string
format: uuid
responses:
"200":
description: Notification read

```
/audit-logs:
  get:
    tags: [Audit]
    summary: List audit logs
    parameters:
      - $ref: "#/components/parameters/TenantId"
      - $ref: "#/components/parameters/Page"
      - $ref: "#/components/parameters/PageSize"
    responses:
      "200":
        description: Audit logs
```

Bu v1 sözleşmesi ilk MVP akışını kapsar:

Login
→ Customer
→ Lead
→ Lead Convert
→ Opportunity
→ Pipeline Stage Move
→ Quote
→ Quote PDF
→ Task
→ Activity
→ Notification
→ Audit

Sonraki çıktı: **CRM OS NestJS Architecture v1.**

CRM OS NestJS Architecture v1

1. Mimari Karar

İlk sürüm için doğru yaklaşım:

Modular Monolith

+

Domain-Based Modules

+

Event-Driven Internal Architecture

+
PostgreSQL RLS
+
Queue Workers

İlk gün mikroservis yok. Ama modüller ileride ayrılabilir şekilde tasarlanacak.

2. Ana Repository Yapısı

crm-os-backend/

```
|— apps/
|   |— api/
|   |— worker/
|   |— scheduler/
|
|— libs/
|   |— common/
|   |— config/
|   |— database/
|   |— auth/
|   |— events/
|   |— logger/
|   |— queue/
|   |— storage/
|   |— security/
|
|— modules/
|   |— core/
|   |— iam/
|   |— crm/
|   |— lead/
|   |— sales/
|   |— quote/
|   |— order/
|   |— product/
|   |— inventory/
|   |— finance/
|   |— task/
|   |— communication/
|   |— support/
```

```
|   |— workflow/
|   |— notification/
|   |— reporting/
|   |— integration/
|   |— ai/
|
|— prisma/
|   |— schema.prisma
|   |— migrations/
|   |— seed.ts
|
|— docker/
|— helm/
|— scripts/
|— test/
|— package.json
|— tsconfig.json
|— docker-compose.yml
```

3. App Ayrımı

apps/api

REST API burada çalışır.

```
apps/api/
|— src/
|   |— main.ts
|   |— app.module.ts
|   |— swagger.ts
|   |— health.controller.ts
```

apps/worker

Queue işleri.

```
apps/worker/
|— src/
|   |— main.ts
|   |— worker.module.ts
|   |— processors/
```

İşler:

email gönderimi
pdf üretimi
webhook dispatch
ai job
workflow run
notification delivery
import/export

`apps/scheduler`

Cron / zamanlı işler.

`apps/scheduler/`

```
|— src/  
|   |— main.ts  
|   |— scheduler.module.ts  
|   |— jobs/
```

İşler:

expired quotes
overdue tasks
payment reminders
subscription checks
stock alerts

4. Lib Yapısı

`libs/common`

`libs/common/`

```
|— dto/  
|— enums/  
|— exceptions/  
|— filters/  
|— guards/  
|— interceptors/  
|— decorators/  
|— pipes/  
|— pagination/
```

└─ utils/

libs/config

libs/config/

- └─ app.config.ts
- └─ database.config.ts
- └─ redis.config.ts
- └─ jwt.config.ts
- └─ storage.config.ts
- └─ queue.config.ts
- └─ ai.config.ts

libs/database

libs/database/

- └─ database.module.ts
- └─ prisma.service.ts
- └─ transaction.service.ts
- └─ tenant-context.service.ts
- └─ rls.service.ts
- └─ repositories/

libs/auth

libs/auth/

- └─ auth.module.ts
- └─ auth.service.ts
- └─ jwt.strategy.ts
- └─ refresh-token.strategy.ts
- └─ guards/
 - └─ jwt-auth.guard.ts
 - └─ permission.guard.ts
 - └─ tenant.guard.ts
- └─ decorators/
 - └─ current-user.decorator.ts
 - └─ tenant-id.decorator.ts
 - └─ require-permission.decorator.ts

5. Modül Standardı

Her domain modülü aynı formatta olacak.

Örnek: CRM modülü

```
modules/crm/
├── crm.module.ts
├── customers/
│   ├── customers.controller.ts
│   ├── customers.service.ts
│   ├── customers.repository.ts
│   ├── dto/
│   │   ├── create-customer.dto.ts
│   │   ├── update-customer.dto.ts
│   │   ├── customer-filter.dto.ts
│   │   └── customer-response.dto.ts
│   ├── mappers/
│   │   └── customer.mapper.ts
│   ├── policies/
│   │   └── customer.policy.ts
│   └── events/
│       ├── customer-created.event.ts
│       └── customer-updated.event.ts
├── contacts/
├── addresses/
├── timeline/
└── segments/
```

6. Ana Modüller

core

tenants

tenant_settings

plans

subscriptions

feature_flags

audit_logs

iam

- users
- roles
- permissions
- sessions
- teams
- branches
- departments

crm

- customers
- contacts
- addresses
- segments
- customer_360
- timeline
- tags
- files
- comments

lead

- leads
- lead_sources
- lead_assignment
- lead_conversion
- lead_scoring

sales

- pipelines
- pipeline_stages
- opportunities
- stage_history
- forecast

quote

- quotes
- quote_items
- quote_pdf
- quote_approval

quote_status

task

tasks

activities

reminders

calendar

notification

notifications

templates

delivery

preferences

7. Request Lifecycle

Her API request şu sıradan geçer:

HTTP Request

↓

CorrelationId Middleware

↓

Tenant Resolver

↓

JWT Auth Guard

↓

Permission Guard

↓

Validation Pipe

↓

Controller

↓

Service

↓

Repository

↓

Prisma Transaction

↓

PostgreSQL RLS Context

↓

Response Interceptor

8. Tenant Context

Her request'te header:

X-Tenant-Id: uuid

Authorization: Bearer token

Backend içinde:

```
export interface RequestContext {  
    tenantId: string;  
    userId: string;  
    requestId: string;  
    permissions: string[];  
}
```

RLS için transaction başında:

```
SET LOCAL app.tenant_id = '<tenant_id>';
```

```
SET LOCAL app.user_id = '<user_id>';
```

9. Controller Standardı

```
@Controller('customers')
```

```
@UseGuards(JwtAuthGuard, TenantGuard, PermissionGuard)
```

```
export class CustomersController {  
    constructor(private readonly service: CustomersService) {}
```

```
    @Get()
```

```
    @RequirePermission('customer.read')
```

```
    list(@Query() query: CustomerFilterDto) {  
        return this.service.list(query);  
    }
```

```
    @Post()
```

```
    @RequirePermission('customer.create')
```

```
    create(@Body() dto: CreateCustomerDto) {
```

```

        return this.service.create(dto);
    }

    @Get('/:id')
    @RequirePermission('customer.read')
    get(@Param('id') id: string) {
        return this.service.getById(id);
    }

    @Patch('/:id')
    @RequirePermission('customer.update')
    update(@Param('id') id: string, @Body() dto: UpdateCustomerDto) {
        return this.service.update(id, dto);
    }

    @Delete('/:id')
    @RequirePermission('customer.delete')
    delete(@Param('id') id: string) {
        return this.service.softDelete(id);
    }
}

```

10. Service Standard

```

@Injectable()
export class CustomersService {
    constructor(
        private readonly repo: CustomersRepository,
        private readonly events: DomainEventBus,
        private readonly audit: AuditService,
    ) {}

    async create(dto: CreateCustomerDto) {
        const customer = await this.repo.create(dto);

        await this.events.publish({
            type: 'CustomerCreated',
            aggregateType: 'customer',
            aggregateId: customer.id,
            payload: customer,

```

```

    });

    return customer;
  }
}

```

11. Repository Standardı

```

@Injectable()
export class CustomersRepository {
  constructor(private readonly prisma: PrismaService) {}

  async findMany(filter: CustomerFilterDto) {
    return this.prisma.customer.findMany({
      where: {
        deletedAt: null,
        status: filter.status,
      },
      orderBy: {
        createdAt: 'desc',
      },
      take: filter.pageSize,
      skip: (filter.page - 1) * filter.pageSize,
    });
  }
}

```

Not: tenant_id kodda filtrelenmez; PostgreSQL RLS zorunlu uygular.

12. Event Bus

Internal event yapısı:

domain_events tablosu

+

RabbitMQ

+

NestJS EventEmitter

Event publish akışı:

```
Service
↓
domain_events insert
↓
transaction commit
↓
outbox worker
↓
RabbitMQ publish
↓
handlers
```

13. Outbox Pattern

Tablo:

domain_events

Worker:

Unpublished events çekilir
RabbitMQ'ya publish edilir
published_at güncellenir

Bu sayede event kaybı önlenir.

14. Event Handler Örneği

```
@EventHandler(CustomerCreatedEvent)
export class CustomerCreatedHandler {
  constructor(
    private readonly timeline: TimelineService,
    private readonly notification: NotificationService,
  ) {}

  async handle(event: CustomerCreatedEvent) {
    await this.timeline.create({
      entityType: 'customer',
```

```
        entityId: event.customerId,  
        eventType: 'CustomerCreated',  
        title: 'Customer created',  
        payload: event.payload,  
    });  
}  
}
```

15. Queue Yapısı

Queue isimleri:

```
email.queue  
sms.queue  
whatsapp.queue  
pdf.queue  
webhook.queue  
workflow.queue  
ai.queue  
import.queue  
export.queue  
notification.queue
```

Processor örnekleri:

```
QuotePdfProcessor  
EmailDeliveryProcessor  
WebhookDispatchProcessor  
WorkflowRunProcessor  
AiSummaryProcessor
```

16. Permission Guard

Permission decorator:

```
@RequirePermission('quote.approve')
```

Guard çalışma mantığı:

JWT içindeki user id
↓
Redis permission cache
↓
yoksa DB
↓
permission doğrulama

Cache key:

permissions:{tenantId}:{userId}

17. Audit Standardı

Her kritik işlemde:

create
update
delete
approve
reject
login
export
permission_change

audit yazılır.

Audit interceptor opsiyonel:

```
@AuditAction('customer.create')
```

18. DTO Standardı

```
export class CreateCustomerDto {  
  @IsEnum(['individual', 'corporate'])  
  customerType: 'individual' | 'corporate';  
  
  @IsOptional()  
  @IsString()  
  companyName?: string;
```



```
@IsOptional()
@IsEmail()
email?: string;

@IsOptional()
@IsString()
phone?: string;
}
```

19. Response Standardı

Başarılı response:

```
{
  "data": {},
  "meta": {
    "requestId": "uuid",
    "timestamp": "2026-06-20T12:00:00Z"
  }
}
```

Liste response:

```
{
  "data": [],
  "pagination": {
    "page": 1,
    "pageSize": 25,
    "total": 240
  },
  "meta": {
    "requestId": "uuid"
  }
}
```

Hata response:

```
{
  "error": {
    "code": "CUSTOMER_NOT_FOUND",
  }
}
```

```
    "message": "Customer not found",
    "details": {}
  },
  "meta": {
    "requestId": "uuid"
  }
}
```

20. Error Code Standard

AUTH_INVALID_CREDENTIALS
AUTH_TOKEN_EXPIRED
TENANT_NOT_FOUND
PERMISSION_DENIED
CUSTOMER_NOT_FOUND
LEAD_ALREADY_CONVERTED
OPPORTUNITY_NOT_FOUND
QUOTE_NOT_FOUND
QUOTE_ALREADY_APPROVED
VALIDATION_ERROR
RATE_LIMIT_EXCEEDED

21. Logging Standard

```
{
  "level": "info",
  "requestId": "uuid",
  "tenantId": "uuid",
  "userId": "uuid",
  "module": "crm",
  "action": "customer.create",
  "message": "Customer created",
  "durationMs": 42
}
```

22. Security Middleware

Zorunlu:

- helmet
- cors allowlist
- rate limit
- request size limit
- file upload validation
- JWT verification
- tenant verification
- permission guard

23. API Versioning

- /api/v1
- /api/v2

NestJS:

```
app.enableVersioning({  
  type: VersioningType.URI,  
});
```

24. Swagger

Swagger build:

/api/docs

Sadece staging/dev açık.

Production'da ya kapalı ya da IP whitelist.

25. Test Yapısı

- unit tests
- integration tests
- e2e tests
- contract tests

security tests

Klasör:

test/

├─ unit/

├─ integration/

├─ e2e/

├─ fixtures/

└─ factories/

26. Module Ownership

50 kişilik ekip için sahiplik:

core/iam	Squad Core
crm/lead	Squad CRM
sales/quote	Squad Sales
order/inventory	Squad Ops
finance	Squad Finance
communication	Squad Communication
workflow	Squad Automation
ai/reporting	Squad AI/Data
integration	Squad Platform

27. İlk Sprint Teknik Çıktısı

Repo oluştur

Docker Compose

NestJS app

PostgreSQL bağlantısı

Prisma setup

RLS context proof

Auth skeleton

Tenant middleware

Permission guard

Customer CRUD vertical slice

28. Vertical Slice v1

İlk gerçek uçtan uca kesit:

Login

↓

Tenant Context

↓

Customer Create API

↓

RLS ile DB insert

↓

Audit Log

↓

Domain Event

↓

Timeline Event

↓

Notification

↓

Customer List UI

Bu çalışıyorsa mimari doğru kurulmuş demektir.

29. Gelecekte Mikroservice Ayrılacak Modüller

notification-service

file-service

ai-service

workflow-service

integration-service

reporting-service

communication-service

Ama ilk 12 ay:

Modular Monolith

kalmalı.

30. Sonraki Çıktı

Bundan sonra sıradaki doküman:

CRM OS Frontend Architecture v1

olmalı.

CRM OS Design System v1

1. Tasarım Yönü

Dark Titanium Enterprise

Referans hissiyat:

Linear

+ Attio

+ Stripe Dashboard

+ Salesforce Lightning

+ Bloomberg Terminal

Amaç:

Premium

Kompakt

Kurumsal

Veri yoğun

Hızlı kullanılabilir

2. Ana Tema

Dark Theme Default

```
:root {  
  --bg: #0A0D14;  
  --bg-soft: #0F1420;  
  
  --surface: #111827;
```

```
--surface-2: #161F31;
--surface-3: #1D2638;

--border: #273246;
--border-soft: #1E293B;

--text: #F9FAFB;
--text-soft: #CBD5E1;
--text-muted: #94A3B8;
--text-disabled: #64748B;

--primary: #4F8CFF;
--primary-hover: #6EA2FF;
--primary-muted: rgba(79, 140, 255, 0.14);

--success: #22C55E;
--warning: #F59E0B;
--danger: #EF4444;
--info: #38BDF8;
}
```

3. Renk Kullanımı

Kullanım kuralı

%80 nötr koyu ton
%15 gri/mavi yüzey
%5 accent renk

Yanlış:

Her kart neon
Her buton gradient
Her panel cam

Doğru:

Accent sadece aksiyon, durum ve vurgu için

4. Typography

Font:

Inter

Alternatif:

SF Pro

Font Scale

Kullanım	Boyut	Weight
Caption	11px	500
Table Text	13px	500
Body	14px	400
Section Title	15px	600
Page Title	22px	700
Hero Metric	32px	700

5. Spacing

Base grid:

4px

Primary spacing:

8px grid

Standartlar:

Page padding: 24px

Card padding: 16px

Table row height: 44px

Form row height: 40px

Sidebar width: 240px

Topbar height: 64px

6. Radius

Input: 8px

Button: 8px

Card: 12px

Modal: 16px

Drawer: 16px

Yasak:

30px+

aşırı yuvarlak SaaS oyuncak görünümü

7. Shadow / Glow

Varsayılan:

```
--shadow-sm: 0 1px 2px rgba(0,0,0,0.24);
```

```
--shadow-md: 0 8px 24px rgba(0,0,0,0.28);
```

```
--shadow-lg: 0 20px 60px rgba(0,0,0,0.36);
```

Glow sadece:

Aktif menü

Primary CTA

AI panel

Canlı durum

için kullanılacak.

8. Layout

App Shell

Sidebar 240px	Topbar
	Content

Content

Max width: 1600px
Page padding: 24px
Grid gap: 16px

9. Sidebar

Genişlik:

240px

Bölümler:

HOME
CRM
SALES
OPERATIONS
FINANCE
COMMUNICATION
SUPPORT
AUTOMATION
ANALYTICS
ADMIN

Aktif item:

background: rgba(79, 140, 255, 0.12);
border: 1px solid rgba(79, 140, 255, 0.28);
color: #F9FAFB;

10. Topbar

İçerik:

Page title
Breadcrumb
Global search
Notifications
AI Ask

User menu

Yükseklik:

64px

11. Button Sistemi

Primary

background: #4F8CFF;

color: #FFFFFF;

height: 36px;

border-radius: 8px;

Secondary

background: #161F31;

border: 1px solid #273246;

color: #F9FAFB;

Ghost

background: transparent;

color: #94A3B8;

Danger

background: rgba(239,68,68,0.14);

color: #EF4444;

12. Form Elements

Input

height: 40px;

background: #0F1420;

border: 1px solid #273246;

border-radius: 8px;

color: #F9FAFB;

States:

default

hover

focus

error

disabled

Focus:

border-color: #4F8CFF;

box-shadow: 0 0 0 3px rgba(79,140,255,0.16);

13. Card

background: #111827;

border: 1px solid #273246;

border-radius: 12px;

padding: 16px;

Kart türleri:

MetricCard

TableCard

ChartCard

AIInsightCard

TimelineCard

SummaryCard

14. Table / DataGrid

CRM'in en kritik bileşeni.

Row height

44px compact

52px comfortable

Özellikler

Server pagination
Sorting
Filtering
Column visibility
Saved views
Bulk select
Inline status
Sticky header
Resizable columns

Status Cell

Badge kullanılır
Renk az kullanılır

15. Badge

Boyut:

height: 22px
padding: 0 8px
font-size: 11px
radius: 999px

Tipler:

neutral
success
warning
danger
info
primary

16. Kanban

Kullanım:

Pipeline
Tasks

Tickets
Projects
Approvals

Column

background: #0F1420;
border: 1px solid #273246;
border-radius: 12px;
padding: 12px;

Card

background: #111827;
border: 1px solid #273246;
border-radius: 10px;
padding: 12px;

17. Timeline

Event tipleri:

call
email
whatsapp
meeting
quote
order
payment
ticket
system
ai

Timeline kompakt olmalı:

Saat
Icon
Başlık
Açıklama
Kullanıcı

18. Drawer

Detay ekranlarında modal yerine drawer kullanılacak.

Kullanım:

Opportunity Inspector

Customer Quick View

Quote Preview

Task Detail

Genişlik:

480px compact

640px detail

840px large

19. Modal

Sadece kısa işlemler için:

Confirm delete

Approve quote

Convert lead

Assign user

20. Toast

Konum:

top-right

Tipler:

success

error

warning

info

21. Command Menu

Kısayol:

Cmd/Ctrl + K

Aksiyonlar:

Müşteri ara

Lead oluştur

Teklif oluştur

Görev ekle

Sayfaya git

AI'a sor

22. AI UI Pattern

AI panel farklı ama abartısız görünmeli.

```
background: linear-gradient(  
  180deg,  
  rgba(79,140,255,0.12),  
  rgba(17,24,39,1)  
);  
border: 1px solid rgba(79,140,255,0.28);
```

AI bileşenleri:

AI Copilot Panel

AI Suggestion Card

AI Summary Box

AI Confidence Indicator

Ask CRM Input

23. Dashboard Components

MetricCard

Label

Value

Delta

Mini trend

ChartCard

Title

Filter

Chart

Legend

InsightCard

AI insight

Risk

Recommended action

CTA

24. Customer 360 Components

CustomerHeader

CustomerKpiStrip

SmartActionBar

CustomerTabs

CustomerTimeline

CustomerContactCard

CustomerRiskPanel

CustomerFinanceSummary

25. Quote Builder Components

QuoteHeader

ProductSelector

QuoteItemTable

PricingPanel

MarginPanel
ApprovalStatus
PdfPreview

26. Pipeline Components

PipelineToolbar
PipelineColumn
DealCard
DealInspectorDrawer
StageSummary
WinProbabilityBadge
NextActionPanel

27. Inbox Components

ConversationList
ChatThread
MessageBubble
Composer
CustomerContextPanel
AISuggestedReply
SLAIndicator

28. Responsive Rules

Desktop first

Ana hedef:

1440px+

Breakpoints:

sm: 640
md: 768
lg: 1024
xl: 1280

2xl: 1536

Mobile:

Saha satış için ayrı mobil deneyim

Web app mobile'da sadece temel responsive

29. Accessibility

Zorunlu:

AA contrast

Keyboard navigation

Focus ring

ARIA labels

Modal focus trap

Screen reader labels

30. Design Tokens Dosyası

```
export const colors = {  
  background: "#0A0D14",  
  surface: "#111827",  
  surface2: "#161F31",  
  border: "#273246",  
  text: "#F9FAFB",  
  muted: "#94A3B8",  
  primary: "#4F8CFF",  
  success: "#22C55E",  
  warning: "#F59E0B",  
  danger: "#EF4444",  
};
```

```
export const radii = {  
  input: "8px",  
  button: "8px",  
  card: "12px",  
  modal: "16px",  
};
```

```
export const spacing = {
  xs: "4px",
  sm: "8px",
  md: "16px",
  lg: "24px",
  xl: "32px",
};
```

31. Tailwind Token Örneği

```
theme: {
  extend: {
    colors: {
      bg: "#0A0D14",
      surface: "#111827",
      surface2: "#161F31",
      border: "#273246",
      muted: "#94A3B8",
      primary: "#4F8CFF",
      success: "#22C55E",
      warning: "#F59E0B",
      danger: "#EF4444",
    },
    borderRadius: {
      sm: "8px",
      md: "12px",
      lg: "16px",
    }
  }
}
```

32. UI Kalite Kriteri

Bir ekran kabul edilmeden önce:

Grid hizalı mı?

Tablo kompakt mı?

Primary action net mi?

Boşluklar dengeli mi?
Renk fazla mı?
Klavye ile kullanılabiliyor mu?
Loading/empty/error var mı?
Permission state var mı?
Mobil kırılım bozuluyor mu?

33. Design System Sonucu

Bu sistemin UI dili:

Kurumsal
Karanlık
Kompakt
Hızlı
Az efektli
Yüksek veri yoğunluklu
Premium SaaS

olmalı.

Sonraki çıktı:

CRM OS DevOps & Infrastructure Blueprint v1

CRM OS DevOps & Infrastructure Blueprint v1

1. Hedef Mimari

CRM OS için altyapı hedefi:

Multi-tenant SaaS
High availability
Zero-downtime deployment
Observable system
Secure by default
Backup / disaster recovery ready
Scalable modular monolith

İlk mimari:

Kubernetes

- + PostgreSQL
- + Redis
- + RabbitMQ
- + MinIO / S3
- + Next.js Web
- + NestJS API
- + Worker
- + Scheduler
- + Monitoring Stack

2. Ortamlar

local
development
staging
production
demo
sandbox

Ortam Amaçları

Ortam	Amaç
local	Geliştirici makinesi
development	Sürekli entegrasyon
staging	Production öncesi test
production	Canlı sistem
demo	Satış/pilot demo
sandbox	Kurumsal müşteri test tenantları

3. Local Development

services:

api:

image: crm-os-api:local

web:

image: crm-os-web:local

postgres:

image: postgres:16

redis:

image: redis:7
rabbitmq:
image: rabbitmq:3-management
minio:
image: minio/minio

Local servisler:

api : <http://localhost:3000>
web : <http://localhost:3001>
postgres : localhost:5432
redis : localhost:6379
rabbitmq ui : <http://localhost:15672>
minio ui : <http://localhost:9001>

4. Production Kubernetes Bileşenleri

namespace: crm-os-prod

Deploymentlar:

api
web
worker
scheduler
notification-worker
integration-worker
ai-worker

Stateful servisler:

postgres
redis
rabbitmq
minio

Platform servisleri:

ingress-nginx
cert-manager

external-dns
sealed-secrets / external-secrets
prometheus
grafana
loki
tempo
sentry

5. Kubernetes Namespace Yapısı

crm-os-dev
crm-os-staging
crm-os-prod
crm-os-demo
monitoring
ingress
security

Her namespace için:

ResourceQuota
LimitRange
NetworkPolicy
ServiceAccount
Secrets
ConfigMaps

6. Deployment Topolojisi

Internet
↓
WAF / CDN
↓
Load Balancer
↓
Ingress Controller
↓
Web / API
↓

API

- └ PostgreSQL
- └ Redis
- └ RabbitMQ
- └ MinIO/S3
- └ Workers

7. Production Minimum Pod Plan

API

replicas: 3
cpu request: 500m
memory request: 1Gi
cpu limit: 2
memory limit: 2Gi

Web

replicas: 2
cpu request: 300m
memory request: 512Mi
cpu limit: 1
memory limit: 1Gi

Worker

replicas: 2
cpu request: 500m
memory request: 1Gi
cpu limit: 2
memory limit: 2Gi

Scheduler

replicas: 1
leader election: true

8. Database Architecture

Başlangıç

PostgreSQL Primary

+

Daily backup

+

Point-in-time recovery

Production

PostgreSQL Primary

+

Read Replica

+

WAL archiving

+

PITR

+

Automated backup

Ölçekleme

PgBouncer

Read replicas

Partitioning

Materialized views

Partition adayları:

audit_logs

domain_events

timeline_events

messages

notifications

ai_usage_logs

workflow_logs

api_logs

9. PostgreSQL Standardı

PostgreSQL 16

UTF-8

timezone UTC

uuid primary keys

RLS enabled

pgcrypto

pg_stat_statements

Zorunlu ayarlar:

statement_timeout

idle_in_transaction_session_timeout

log_min_duration_statement

shared_preload_libraries = pg_stat_statements

10. PgBouncer

Kullanım:

API → PgBouncer → PostgreSQL

Worker → PgBouncer → PostgreSQL

Pool mode:

transaction

Dikkat:

PostgreSQL RLS için SET LOCAL app.tenant_id transaction içinde çalışmalı

11. Redis

Kullanım alanları:

session cache

permission cache

rate limit

temporary locks
dashboard cache
query cache

Key standardı:

tenant:{tenantId}:permissions:{userId}
tenant:{tenantId}:rate-limit:{key}
tenant:{tenantId}:dashboard:{dashboardId}
tenant:{tenantId}:lock:{resource}

TTL zorunlu.

12. RabbitMQ

Queue yapısı:

email.queue
sms.queue
whatsapp.queue
notification.queue
pdf.queue
webhook.queue
workflow.queue
ai.queue
import.queue
export.queue
audit.queue

Dead letter queue:

email.dlq
webhook.dlq
workflow.dlq
ai.dlq

Retry standardı:

1. deneme: hemen
2. deneme: 1 dakika
3. deneme: 5 dakika
4. deneme: 30 dakika

5. deneme: DLQ

13. Object Storage

Başlangıç:

MinIO

Production alternatif:

AWS S3

Azure Blob

GCP Cloud Storage

Bucket yapısı:

crm-os-files

crm-os-public

crm-os-backups

crm-os-exports

Path standardı:

tenant/{tenantId}/{module}/{entityId}/{fileId}/{filename}

Örnek:

tenant/uuid/customer/uuid/file/contract.pdf

14. Secrets Management

Local:

.env

Development/Staging:

Kubernetes Secret

Production:

External Secrets

+ AWS Secrets Manager / Vault / Doppler

Secret türleri:

DATABASE_URL

REDIS_URL

RABBITMQ_URL

JWT_SECRET

REFRESH_TOKEN_SECRET

S3_SECRET_KEY

SMTP_PASSWORD

WHATSAPP_API_TOKEN

AI_PROVIDER_KEY

15. CI/CD Pipeline

Pull Request

install

lint

type-check

unit-test

build

docker-build

security-scan

migration-check

Merge to develop

deploy development

run smoke tests

Release branch

deploy staging

run integration tests

run e2e tests

run load smoke

manual approval

Production

backup database
run migration
deploy rolling/blue-green
health check
run smoke tests
monitor error rate
auto rollback if unhealthy

16. Git Branching

main
develop
release/*
hotfix/*
feature/*
bugfix/*

Kural:

main = production
develop = development
release/* = staging

17. Container Strategy

Image isimleri:

registry/crm-os-api:<git_sha>
registry/crm-os-web:<git_sha>
registry/crm-os-worker:<git_sha>
registry/crm-os-scheduler:<git_sha>

Dockerfile kuralları:

multi-stage build
non-root user
small base image
healthcheck
no secrets in image

18. Helm Chart Yapısı

```
helm/
├── crm-os/
│   ├── Chart.yaml
│   ├── values.yaml
│   ├── values-dev.yaml
│   ├── values-staging.yaml
│   ├── values-prod.yaml
│   └── templates/
│       ├── api-deployment.yaml
│       ├── web-deployment.yaml
│       ├── worker-deployment.yaml
│       ├── scheduler-deployment.yaml
│       ├── service.yaml
│       ├── ingress.yaml
│       ├── configmap.yaml
│       ├── hpa.yaml
│       └── pdb.yaml
```

19. Ingress

Örnek domainler:

app.crm-os.com
api.crm-os.com
admin.crm-os.com
portal.crm-os.com
dealer.crm-os.com

TLS:

cert-manager
Let's Encrypt

Ingress özellikleri:

rate limit
body size limit
timeout config
CORS
security headers

20. Horizontal Pod Autoscaler

API için:

minReplicas: 3
maxReplicas: 20
targetCPUUtilization: 65
targetMemoryUtilization: 75

Worker için:

queue depth bazlı autoscaling
KEDA önerilir

21. Observability Stack

Metrics

Prometheus
Grafana

Logs

Loki
Promtail / Fluent Bit

Tracing

OpenTelemetry

Tempo

Errors

Sentry

22. Temel Metrikler

API:

http_requests_total
http_request_duration_seconds
http_error_rate
active_users
tenant_request_count

Database:

db_query_duration
db_connections
slow_queries
deadlocks
replication_lag

Queue:

queue_depth
queue_processing_duration
queue_failed_jobs
dlq_count

Business:

lead_created_count
quote_sent_count
order_created_count
payment_received_count
workflow_run_count

ai_usage_count

23. Logging Standardı

JSON log:

```
{
  "timestamp": "2026-06-20T12:00:00Z",
  "level": "info",
  "requestId": "uuid",
  "tenantId": "uuid",
  "userId": "uuid",
  "module": "crm",
  "action": "customer.create",
  "message": "Customer created",
  "durationMs": 42
}
```

Yasak:

password

token

credit card

secret

personal sensitive data

24. Alerting

Kritik alarmlar:

API error rate > 2%

P95 latency > 800ms

DB CPU > 80%

DB connection saturation

Queue DLQ > 0

Redis unavailable

RabbitMQ unavailable

Disk usage > 80%

Backup failed

Certificate expiring < 14 days

25. Backup Strategy

PostgreSQL

daily full backup

continuous WAL archiving

PITR enabled

retention 30 days

monthly archive 12 months

Object Storage

versioning enabled

lifecycle rules

cross-region replication optional

Config / Secrets

encrypted backup

restricted access

26. Disaster Recovery

RPO

15 minutes

RTO

4 hours

Recovery test:

monthly restore test

quarterly DR drill

27. Security Baseline

Zorunlu:

TLS everywhere

WAF

rate limiting

RBAC

RLS

2FA

secret rotation

image scanning

dependency scanning

SAST

DAST

audit logs

network policies

least privilege IAM

28. Network Policy

Varsayılan:

deny all

İzinler:

web → api

api → postgres

api → redis

api → rabbitmq

api → minio

worker → postgres

worker → rabbitmq

worker → minio

29. Rate Limiting

Global:

1000 req / minute / tenant

Auth:

5 login attempts / 5 minutes / IP

Public API:

plan-based quota

Webhook:

signature verification mandatory

30. Migration Strategy

Kurallar:

migration backward compatible olmalı

long lock yaratmamalı

production deploy öncesi staging test edilmeli

rollback planı olmalı

data migration ayrı job olmalı

Sıra:

1. add nullable column

2. deploy code

3. backfill

4. set not null

5. remove old field later

31. Feature Flags

Kullanım:

new module rollout

tenant-based enablement

beta customer access

dark launch

kill switch

Feature flag örneği:

```
tenant:{tenantId}:feature:ai_copilot  
tenant:{tenantId}:feature:workflow_builder  
tenant:{tenantId}:feature:dealer_portal
```

32. Tenant Provisioning

Yeni tenant oluşturma akışı:

```
tenant kaydı  
default settings  
default roles  
default permissions  
default pipeline  
default statuses  
default dashboard  
default feature flags  
demo data optional  
welcome email
```

Provision job idempotent olmalı.

33. Release Strategy

MVP

weekly release

Production stable

bi-weekly release

Enterprise customers

scheduled maintenance window

34. Deployment Strategy

İlk aşama:

Rolling update

İleri aşama:

Blue/Green

Canary

Canary ölçümleri:

error rate

latency

business transaction success

queue failures

35. Monitoring Dashboardları

Grafana dashboardları:

API Overview

Database Overview

Queue Overview

Tenant Usage

Worker Jobs

Error Tracking

Business KPIs

AI Usage

Storage Usage

Security Events

36. Compliance / KVKK

Gerekli teknik imkanlar:

data export

data deletion

consent records

audit trail

data masking
retention policies
access logs
encryption

37. Infrastructure as Code

Terraform modülleri:

network
kubernetes
postgres
redis
rabbitmq
object_storage
dns
cdn
waf
monitoring
secrets

Klasör:

```
infra/  
├── environments/  
│   ├── dev/  
│   ├── staging/  
│   └── prod/  
├── modules/  
│   ├── network/  
│   ├── kubernetes/  
│   ├── postgres/  
│   ├── redis/  
│   ├── rabbitmq/  
│   ├── object-storage/  
│   └── monitoring/  
└── global/
```

38. İlk 30 Gün DevOps Planı

Hafta 1

- repo setup
- docker compose
- base Dockerfile
- CI lint/test/build
- dev environment

Hafta 2

- Kubernetes dev cluster
- Helm chart
- Ingress
- TLS
- PostgreSQL
- Redis
- RabbitMQ
- MinIO

Hafta 3

- staging cluster
- CI/CD deployment
- migration pipeline
- basic monitoring
- central logging

Hafta 4

- production baseline
- backup
- alerting
- security scan
- release process
- rollback process

39. Minimum Production Go-Live Checklist

- TLS active
- WAF active
- DB backup active

PITR tested
RLS tested
Secrets externalized
Monitoring active
Alerts configured
Logs centralized
Error tracking active
Rate limit active
Health checks active
Rollback tested
Smoke tests passing
Security scan clean

40. Sonraki Çıktı

Bundan sonra teknik olarak sıradaki belge:

CRM OS Jira Backlog & Squad Plan v1

olmalı.

CRM OS Jira Backlog & Squad Plan v1

1. Jira Proje Yapısı

Project Key: CRMOS
Project Name: CRM OS Platform
Method: Scrum
Sprint Length: 2 weeks
Release Cadence: 2 weeks
Planning Horizon: 12 months

Issue hierarchy:

Initiative
└ Epic
 └ Story
 └ Sub-task

2. Squad Yapısı

Squad 1 – Platform / Core

Sorumluluk:

Tenant

Plan

Subscription

Feature Flag

Audit

System Settings

Jira prefix:

CORE

Squad 2 – IAM

Auth

Users

Roles

Permissions

Sessions

Branches

Teams

Departments

Prefix:

IAM

Squad 3 – CRM / Customer 360

Customers

Contacts

Addresses

Segments

Timeline

Files

Tags

Comments

Customer 360

Prefix:

CRM

Squad 4 – Lead

Lead Capture

Lead Assignment

Lead Scoring

Lead Conversion

Lead Sources

Prefix:

LEAD

Squad 5 – Sales

Pipeline

Stages

Opportunities

Stage History

Forecast

Prefix:

SALES

Squad 6 – Quote

Quotes

Quote Items

Pricing

Discount

Approval

PDF

Send

Prefix:

QUOTE

Squad 7 – Product / Inventory

Products

Variants

Warehouses

Stocks

Stock Movements

Reservations

Prefix:

INV

Squad 8 – Finance

Accounts

Transactions

Payments

Invoices

Collections

Credit Limits

Prefix:

FIN

Squad 9 – Communication / Support

Inbox

WhatsApp

Email

SMS

Calls

Tickets

SLA

Prefix:

COMM

Squad 10 – Workflow / Integration / AI

Workflows

Approvals

Webhooks
API Keys
Connectors
AI Gateway
AI Assistant

Prefix:

AUTO

Shared Squads

Frontend Platform
DevOps
QA Automation
UX/UI
Data/Analytics

3. Release Plan

Release 0 – Foundation

Süre:

Sprint 1-2

Amaç:

Repo, altyapı, auth, tenant, RLS proof

Release 1 – CRM MVP

Süre:

Sprint 3-6

Amaç:

Customer
Lead
Pipeline
Quote
Task

Dashboard

Release 2 – Commercial MVP

Süre:

Sprint 7-12

Amaç:

Order

Product

Inventory

Finance Lite

Notification

Release 3 – Enterprise CRM

Süre:

Sprint 13-20

Amaç:

Omnichannel

Support

Workflow

Portal

API

Release 4 – AI + Scale

Süre:

Sprint 21-26

Amaç:

AI Assistant

Advanced Reporting

Mobile

Marketplace basics

4. Initiative Listesi

CRMOS-I1 SaaS Foundation
CRMOS-I2 Identity & Access Management
CRMOS-I3 Customer 360
CRMOS-I4 Lead Management
CRMOS-I5 Sales Pipeline
CRMOS-I6 Quote Management
CRMOS-I7 Product & Inventory
CRMOS-I8 Finance Lite
CRMOS-I9 Communication & Support
CRMOS-I10 Workflow & Automation
CRMOS-I11 AI Platform
CRMOS-I12 Reporting & Analytics
CRMOS-I13 DevOps & Observability
CRMOS-I14 Frontend Design System
CRMOS-I15 Mobile CRM

5. Epics

CRMOS-I1 – SaaS Foundation

CORE-E1 Tenant Management
CORE-E2 Plan & Subscription
CORE-E3 Feature Flags
CORE-E4 Audit Log
CORE-E5 Domain Events
CORE-E6 System Settings

CRMOS-I2 – IAM

IAM-E1 Authentication
IAM-E2 User Management
IAM-E3 Role Management
IAM-E4 Permission Management
IAM-E5 Organization Hierarchy
IAM-E6 Session & Device Security

CRMOS-I3 – Customer 360

CRM-E1 Customer Management
CRM-E2 Contact Management

- CRM-E3 Address Management
- CRM-E4 Customer Timeline
- CRM-E5 Tags & Segments
- CRM-E6 Customer 360 Dashboard
- CRM-E7 Files & Comments

CRMOS-I4 – Lead Management

- LEAD-E1 Lead CRUD
- LEAD-E2 Lead Sources
- LEAD-E3 Lead Assignment
- LEAD-E4 Lead Conversion
- LEAD-E5 Lead Scoring
- LEAD-E6 Lead Loss Reasons

CRMOS-I5 – Sales Pipeline

- SALES-E1 Pipeline Management
- SALES-E2 Pipeline Stages
- SALES-E3 Opportunity Management
- SALES-E4 Kanban Pipeline
- SALES-E5 Stage History
- SALES-E6 Win/Loss Flow

CRMOS-I6 – Quote Management

- QUOTE-E1 Quote CRUD
- QUOTE-E2 Quote Items
- QUOTE-E3 Pricing & Discount
- QUOTE-E4 Quote Approval
- QUOTE-E5 Quote PDF
- QUOTE-E6 Quote Send & View Tracking

CRMOS-I7 – Product & Inventory

- INV-E1 Product Catalog
- INV-E2 Product Variants
- INV-E3 Warehouses
- INV-E4 Stock Ledger
- INV-E5 Stock Reservation
- INV-E6 Stock Alerts

CRMOS-I8 – Finance Lite

- FIN-E1 Customer Accounts
- FIN-E2 Account Transactions
- FIN-E3 Payments
- FIN-E4 Installments
- FIN-E5 Receivables
- FIN-E6 Invoice Integration Placeholder

CRMOS-I9 – Communication & Support

- COMM-E1 Omnichannel Inbox
- COMM-E2 Messages
- COMM-E3 WhatsApp Integration
- COMM-E4 Email Integration
- COMM-E5 Ticket Management
- COMM-E6 SLA Management

CRMOS-I10 – Workflow & Automation

- AUTO-E1 Workflow Builder
- AUTO-E2 Workflow Triggers
- AUTO-E3 Workflow Actions
- AUTO-E4 Approval Engine
- AUTO-E5 Webhooks
- AUTO-E6 API Keys

CRMOS-I11 – AI Platform

- AI-E1 AI Gateway
- AI-E2 AI Assistant
- AI-E3 AI Summaries
- AI-E4 Quote Assistant
- AI-E5 Lead Scoring AI
- AI-E6 Usage Quotas

CRMOS-I12 – Reporting

- REP-E1 Dashboard Framework
- REP-E2 Sales Dashboard
- REP-E3 Customer Reports
- REP-E4 Finance Reports
- REP-E5 Custom Report Builder

CRMOS-I13 – DevOps

DEVOPS-E1 Local Dev Environment

DEVOPS-E2 CI/CD

DEVOPS-E3 Kubernetes

DEVOPS-E4 Monitoring

DEVOPS-E5 Logging

DEVOPS-E6 Backup & DR

CRMOS-I14 – Frontend Platform

UI-E1 Design System

UI-E2 App Shell

UI-E3 DataGrid

UI-E4 Kanban

UI-E5 Timeline

UI-E6 Command Menu

6. Sprint Plan – İlk 12 Sprint

Sprint 1 – Foundation Setup

CORE

CORE-001 Create monorepo structure

CORE-002 Setup NestJS API app

CORE-003 Setup Next.js web app

CORE-004 Setup Docker Compose

CORE-005 Setup PostgreSQL

CORE-006 Setup Redis

CORE-007 Setup RabbitMQ

CORE-008 Setup MinIO

DEVOPS

DEVOPS-001 Create CI pipeline

DEVOPS-002 Add lint/typecheck/test pipeline

DEVOPS-003 Create base Dockerfiles

DEVOPS-004 Setup dev environment variables

UI

- UI-001 Setup design tokens
- UI-002 Build App Shell skeleton
- UI-003 Build Sidebar component
- UI-004 Build Topbar component

Sprint 2 – Auth + Tenant + RLS

CORE

- CORE-009 Create tenants table migration
- CORE-010 Implement tenant context middleware
- CORE-011 Implement tenant provisioning service
- CORE-012 Implement audit log base service

IAM

- IAM-001 Implement login API
- IAM-002 Implement refresh token API
- IAM-003 Implement logout API
- IAM-004 Implement users table migration
- IAM-005 Implement roles and permissions migration
- IAM-006 Implement permission guard
- IAM-007 Implement default roles seed

DEVOPS

- DEVOPS-005 Add migration pipeline
- DEVOPS-006 Add RLS test job

UI

- UI-005 Login screen
- UI-006 Auth store
- UI-007 Tenant switcher shell

Sprint 3 – Customer Core

CRM

- CRM-001 Create customers migration
- CRM-002 Create customer_contacts migration
- CRM-003 Create customer_addresses migration
- CRM-004 Implement customer create API
- CRM-005 Implement customer list API
- CRM-006 Implement customer detail API
- CRM-007 Implement customer update API
- CRM-008 Implement customer soft delete API

UI

- UI-008 Build DataGrid base
- UI-009 Customer list page
- UI-010 Customer create form
- UI-011 Customer detail shell

QA

- QA-001 Customer CRUD API tests
- QA-002 Customer UI smoke test

Sprint 4 – Customer 360

CRM

- CRM-009 Implement contacts API
- CRM-010 Implement addresses API
- CRM-011 Implement tags API
- CRM-012 Implement files metadata API
- CRM-013 Implement comments API
- CRM-014 Implement timeline service
- CRM-015 CustomerCreated timeline event
- CRM-016 CustomerUpdated timeline event

UI

- UI-012 Customer header
- UI-013 Customer KPI strip
- UI-014 Customer timeline
- UI-015 Customer contacts tab

UI-016 Customer files tab

Sprint 5 – Lead Management

LEAD

LEAD-001 Create leads migration
LEAD-002 Create lead_sources migration
LEAD-003 Implement lead create API
LEAD-004 Implement lead list API
LEAD-005 Implement lead detail API
LEAD-006 Implement lead update API
LEAD-007 Implement lead assignment API

UI

UI-017 Lead list page
UI-018 Lead detail page
UI-019 Lead create form
UI-020 Lead assignment component

QA

QA-003 Lead CRUD tests

Sprint 6 – Lead Conversion + Pipeline Setup

LEAD

LEAD-008 Implement lead convert API
LEAD-009 Create lead_conversion_logs migration
LEAD-010 Emit LeadConverted event

SALES

SALES-001 Create pipelines migration
SALES-002 Create pipeline_stages migration
SALES-003 Seed default pipeline
SALES-004 Implement pipeline CRUD API
SALES-005 Implement stage CRUD API

UI

UI-021 Lead convert modal

UI-022 Pipeline settings screen

Sprint 7 – Opportunity Management

SALES

SALES-006 Create opportunities migration

SALES-007 Create opportunity_products migration

SALES-008 Implement opportunity create API

SALES-009 Implement opportunity list API

SALES-010 Implement opportunity detail API

SALES-011 Implement opportunity update API

UI

UI-023 Opportunity detail drawer

UI-024 Opportunity create form

UI-025 Opportunity product list

Sprint 8 – Visual Pipeline

SALES

SALES-012 Implement stage move API

SALES-013 Implement stage history

SALES-014 Implement win opportunity API

SALES-015 Implement lose opportunity API

UI

UI-026 Pipeline kanban board

UI-027 Deal card component

UI-028 Drag-drop stage move

UI-029 Deal inspector drawer

QA

QA-004 Pipeline drag-drop e2e

Sprint 9 – Quote Core

QUOTE

- QUOTE-001 Create quotes migration
- QUOTE-002 Create quote_items migration
- QUOTE-003 Implement quote create API
- QUOTE-004 Implement quote list API
- QUOTE-005 Implement quote detail API
- QUOTE-006 Implement quote update API
- QUOTE-007 Implement quote item add API
- QUOTE-008 Implement quote item update API

UI

- UI-030 Quote list page
- UI-031 Quote builder shell
- UI-032 Quote item table
- UI-033 Pricing summary panel

Sprint 10 – Quote PDF + Approval

QUOTE

- QUOTE-009 Implement pricing calculation service
- QUOTE-010 Implement quote approval request
- QUOTE-011 Implement quote approve API
- QUOTE-012 Implement quote reject API
- QUOTE-013 Implement PDF generation job
- QUOTE-014 Implement quote send API

UI

- UI-034 Quote preview panel
- UI-035 Quote approval modal
- UI-036 Quote PDF preview

WORKER

- WORKER-001 PDF queue processor
- WORKER-002 Email queue processor placeholder

Sprint 11 – Tasks + Activities

TASK

TASK-001 Create tasks migration
TASK-002 Create activities migration
TASK-003 Implement task create API
TASK-004 Implement task list API
TASK-005 Implement task complete API
TASK-006 Implement activity create API
TASK-007 Implement activity list API

UI

UI-037 My Work page
UI-038 Task list
UI-039 Task create modal
UI-040 Activity composer

Sprint 12 – Dashboard + Notifications + MVP Hardening

NOTIFICATION

NOTIF-001 Create notifications migration
NOTIF-002 Implement notification create service
NOTIF-003 Implement notification list API
NOTIF-004 Implement mark as read API

REPORTING

REP-001 Implement MVP dashboard metrics API
REP-002 Lead count metric
REP-003 Opportunity value metric
REP-004 Quote count metric
REP-005 Activity count metric

UI

UI-041 Dashboard shell
UI-042 Metric cards
UI-043 Recent activity widget
UI-044 Notification center

UI-045 Command menu v1

QA

QA-005 MVP happy path e2e

QA-006 Tenant isolation test

QA-007 Permission test suite

7. Story Template

As a [role],
I want to [action],
so that [business value].

Acceptance Criteria Template

Given

When

Then

Definition of Ready

Business rule clear

API contract available

Data model available

Design available

Permission defined

Test scenario defined

Definition of Done

Backend implemented

Frontend implemented

DB migration done

Permission check done

Audit log done

Events emitted

Tests passing

Docs updated

Reviewed

Deployed to staging

8. Epic Detayları

CORE-E1 Tenant Management

Stories:

- CORE-101 Create tenant entity
- CORE-102 Create tenant settings
- CORE-103 Tenant provisioning job
- CORE-104 Default roles seed per tenant
- CORE-105 Default pipeline seed per tenant
- CORE-106 Tenant status management
- CORE-107 Tenant usage quotas

IAM-E1 Authentication

Stories:

- IAM-101 Login
- IAM-102 Logout
- IAM-103 Refresh token
- IAM-104 Forgot password
- IAM-105 Reset password
- IAM-106 2FA setup
- IAM-107 Login history
- IAM-108 Session revoke

IAM-E2 User Management

- IAM-201 Create user
- IAM-202 Invite user
- IAM-203 Update user
- IAM-204 Deactivate user
- IAM-205 Assign role
- IAM-206 Assign branch
- IAM-207 Assign manager

CRM-E1 Customer Management

- CRM-101 Create customer
- CRM-102 List customers
- CRM-103 Search customers
- CRM-104 View customer detail

CRM-105 Update customer
CRM-106 Soft delete customer
CRM-107 Restore customer
CRM-108 Customer duplicate detection placeholder

CRM-E4 Customer Timeline

CRM-401 Create timeline event
CRM-402 List timeline
CRM-403 Filter timeline by type
CRM-404 Timeline from domain events
CRM-405 AI summary placeholder

LEAD-E4 Lead Conversion

LEAD-401 Convert lead to customer
LEAD-402 Convert lead to opportunity
LEAD-403 Link existing customer
LEAD-404 Create conversion log
LEAD-405 Prevent double conversion

SALES-E4 Kanban Pipeline

SALES-401 Load pipeline board
SALES-402 Drag opportunity to stage
SALES-403 Validate stage permission
SALES-404 Update stage history
SALES-405 Show stage totals
SALES-406 Open deal inspector

QUOTE-E5 Quote PDF

QUOTE-501 Generate PDF
QUOTE-502 Store PDF file
QUOTE-503 Download PDF
QUOTE-504 Regenerate PDF
QUOTE-505 PDF template v1

9. Cross-Cutting Backlog

Security

SEC-001 Tenant isolation automated test
SEC-002 Permission matrix test
SEC-003 Rate limiting
SEC-004 Audit sensitive actions
SEC-005 File upload validation
SEC-006 JWT rotation

Observability

OBS-001 Structured logging
OBS-002 Request ID middleware
OBS-003 API metrics
OBS-004 DB slow query logging
OBS-005 Queue metrics
OBS-006 Grafana dashboard

Performance

PERF-001 Pagination standard
PERF-002 DataGrid virtualization
PERF-003 Query index review
PERF-004 Dashboard cache
PERF-005 Load test MVP flow

Data

DATA-001 Seed demo tenant
DATA-002 Seed default permissions
DATA-003 Seed default statuses
DATA-004 Seed sample products
DATA-005 Import customers from CSV

10. Squad Capacity Önerisi

2 haftalık sprint için ortalama:

Squad	Kişi	Story Point
Core/IAM	6	60–75
CRM/Lead	7	70–85

Sales/Quote	7	70–85
Inventory/Finance	6	55–70
Communication/Support	5	45–60
Workflow/AI	5	45–60
Frontend Platform	5	45–60
DevOps	4	35–45
QA Automation	5	40–55

11. Dependency Map

DevOps Foundation

↓

Backend Foundation

↓

Auth + Tenant

↓

IAM

↓

CRM

↓

Lead

↓

Sales

↓

Quote

↓

Order

↓

Inventory

↓

Finance

↓

Workflow

↓

AI

UI bağımlılığı:

Design System

↓

App Shell

↓

DataGrid / Form / Drawer

↓

Customer

↓

Lead

↓

Pipeline

↓

Quote

12. MVP Acceptance

MVP kabul senaryosu:

1. Tenant oluşturulur.
2. Admin giriş yapar.
3. Kullanıcı ve rol oluşturur.
4. Müşteri oluşturur.
5. Lead oluşturur.
6. Lead fırsata çevrilir.
7. Pipeline'da fırsat taşınır.
8. Teklif oluşturulur.
9. Teklif PDF üretilir.
10. Teklif gönderildi yapılır.
11. Görev oluşturulur.
12. Dashboard metrikleri görünür.
13. Audit log kayıtları oluşur.
14. Tenant izolasyonu testten geçer.

13. Jira Board Kolonları

Backlog

Ready

In Progress

Code Review

QA

UAT

Done

Blocked

14. Jira Etiketleri

backend
frontend
db
api
security
devops
qa
design
mobile
ai
integration
mvp
enterprise

15. Öncelik Skalası

P0 Production blocker
P1 MVP blocker
P2 Important
P3 Normal
P4 Nice to have

16. Sonraki Çıktı

Bundan sonra sıradaki belge:

CRM OS Testing & QA Strategy v1

CRM OS Testing & QA Strategy v1

1. QA Hedefi

CRM OS için test stratejisinin hedefi:

Multi-tenant veri izolasyonunu garanti etmek

Yetki hatalarını engellemek

Satış akışlarının kırılmamasını sağlamak

Regresyonu azaltmak

50 kişilik ekipte kalite standardını korumak

Production hatalarını erken yakalamak

Ana kalite ilkesi:

Her feature test edilmeden Done değildir.

2. Test Katmanları

Unit Test

↓

Integration Test

↓

API Contract Test

↓

E2E Test

↓

Security Test

↓

Performance Test

↓

UAT

↓

Production Smoke Test

3. Test Piramidi

%50 Unit Test

%25 Integration Test

%15 API / Contract Test

%7 E2E Test

%3 Manual Exploratory Test

E2E az ama kritik olmalı. Her şeyi E2E ile test etmek yavaşlatır.

4. Test Teknolojileri

Backend

Jest

Supertest

Testcontainers

Prisma test DB

PostgreSQL disposable database

Frontend

Vitest

React Testing Library

Playwright

Storybook

Chromatic

API Contract

OpenAPI validation

Schemathesis

Dredd alternatif

Load Test

k6

Artillery alternatif

Security

OWASP ZAP

Snyk / Trivy

npm audit

Semgrep

Monitoring Validation

Grafana dashboards

Synthetic checks

Health probes

5. Test Ortamları

local-test

ci-test

development

staging

uat

production-smoke

Ortam Kullanımı

Ortam	Amaç
local-test	Geliştirici testi
ci-test	PR pipeline
development	Entegrasyon
staging	Release testi
uat	Kullanıcı kabul
production-smoke	Canlı sonrası temel kontrol

6. Test Data Strategy

Her test tenant bazlı çalışmalı.

tenant_a

tenant_b

tenant_enterprise

tenant_demo

Zorunlu test verileri:

admin user

sales manager

sales rep

finance user

warehouse user

support agent
dealer user
customer portal user

Test müşterileri:

corporate customer
individual customer
VIP customer
risk customer
deleted customer

Test satış akışı:

lead
customer
opportunity
quote
order
payment
ticket

7. Multi-Tenant Testleri

En kritik test sınıfı.

Tenant Isolation Test

Senaryo:

Tenant A customer oluşturur.
Tenant B login olur.
Tenant B aynı customer id ile sorgu atar.
Sonuç 404 veya 403 olmalıdır.

Backend test:

customer list
customer detail
lead detail
opportunity detail
quote detail

file access
timeline access
audit log access

Zorunlu:

Her yeni tablo için tenant isolation testi yazılacak.

8. RLS Testleri

PostgreSQL seviyesinde test edilir.

Test:

```
SET app.tenant_id = 'tenant-a';  
SELECT * FROM customers;
```

```
SET app.tenant_id = 'tenant-b';  
SELECT * FROM customers;
```

Beklenen:

Tenant A verisi Tenant B'de görünmez.

RLS test kapsamı:

customers
leads
opportunities
quotes
orders
payments
files
messages
tickets
audit_logs
timeline_events

9. Permission / RBAC Testleri

Her endpoint için:

Authorized Role

Unauthorized Role

No Permission

No Token

Wrong Tenant

Örnek:

quote.approve

Test matrix:

	Rol	Beklenen
Admin		200
Sales Manager		200
Sales Rep		403
Finance		403
Anonymous		401

10. Kritik İş Akışı Testleri

MVP Happy Path

Login

Create Customer

Create Lead

Convert Lead

Create Opportunity

Move Stage

Create Quote

Add Quote Item

Generate PDF

Send Quote

Create Task

Dashboard Metrics

Audit Log Check

Bu test her release öncesi zorunlu.

11. Lead Conversion Testleri

Senaryolar:

Yeni lead → yeni customer + opportunity
Lead → mevcut customer + opportunity
Lead zaten converted ise tekrar convert edilemez
Eksik veri ile convert hata verir
Convert sonrası timeline oluşur
Convert sonrası domain event oluşur

12. Quote Testleri

Senaryolar:

Quote create
Quote item add
Tax calculation
Discount calculation
Margin calculation
Approval required
Approve quote
Reject quote
PDF generate
Send quote
Quote expired

Kritik kural:

Toplam tutar backend tarafından hesaplanır.
Frontend toplamına güvenilmez.

13. Pipeline Testleri

Senaryolar:

Opportunity create
Stage move
Invalid stage move
Permission denied stage move
Won

Lost
Stage history created
Pipeline totals updated

14. Customer 360 Testleri

Senaryolar:

Customer detail loads
Contacts loads
Timeline loads
Quotes loads
Orders loads
Payments loads
Files loads
Activities loads
Permission based tab visibility

15. File Upload Testleri

Senaryolar:

Allowed file upload
Blocked file type
Max size exceeded
Tenant path isolation
File metadata created
File download authorized
Unauthorized file blocked

Güvenlik:

Virus scan entegrasyonu ileri faz
MIME sniffing kontrolü
Path traversal engeli

16. API Contract Testleri

OpenAPI ile:

Request schema validation
Response schema validation
Required field validation
Error response validation
Status code validation

Her endpoint OpenAPI ile uyumlu olmalı.

17. Frontend Component Testleri

Test edilecek temel bileşenler:

Button
Input
Select
DataGrid
Kanban
Timeline
Drawer
Modal
Toast
CommandMenu
QuoteItemTable
CustomerHeader

Örnek testler:

renders correctly
handles loading
handles empty state
handles error state
permission hidden action

18. Frontend E2E Testleri

Playwright senaryoları:

login.spec.ts
customer-crud.spec.ts
lead-conversion.spec.ts
pipeline-drag-drop.spec.ts
quote-builder.spec.ts
permission.spec.ts
tenant-isolation.spec.ts

19. Visual Regression

Storybook + Chromatic.

Kapsam:

Design System components
Customer 360
Pipeline
Quote Builder
Dashboard
Inbox

Amaç:

UI kırılmalarını yakalamak
Dark theme tutarlılığını korumak

20. Performance Testing

API Hedefleri

İşlem	Hedef
Customer list	< 300ms p95
Customer detail	< 400ms p95
Pipeline board	< 500ms p95
Quote detail	< 400ms p95
Dashboard metrics	< 800ms p95

k6 Senaryoları

100 concurrent users
500 concurrent users

1000 concurrent users

Akışlar:

login
list customers
open customer
create lead
convert lead
move opportunity
create quote

21. Load Test Data

Minimum staging load dataset:

10 tenants
1,000 users
100,000 customers
250,000 leads
100,000 opportunities
50,000 quotes
1,000,000 timeline events
500,000 activities

Enterprise test dataset:

100 tenants
10,000 users
1,000,000 customers
5,000,000 timeline events
10,000,000 messages

22. Security Testing

Zorunlu kontroller:

SQL injection
XSS
CSRF

JWT tampering
Broken access control
IDOR
File upload abuse
Rate limit bypass
Tenant escape
Privilege escalation

Özellikle IDOR:

Tenant A quote id
Tenant B token
GET /quotes/{id}

Beklenen:

403/404

23. Regression Suite

Her release öncesi:

Auth
Tenant
RBAC
Customer CRUD
Lead Conversion
Pipeline
Quote
Task
Notification
Audit

Çalışmalı.

24. Smoke Test

Deployment sonrası:

Health check
Login
Tenant context
Customer list
Create customer
Create lead
Create quote
Notification list
Audit log

Süre hedefi:

< 5 dakika

25. UAT Planı

Pilot kullanıcı rolleri:

Sales Rep
Sales Manager
Finance User
Admin

UAT senaryoları:

Müşteri oluştur
Lead ekle
Fırsata çevir
Pipeline'da ilerlet
Teklif oluştur
PDF al
Görev ata
Dashboard kontrol et

UAT kabul:

Kritik hata: 0
Major hata: maksimum 3
Minor hata: kabul edilebilir

26. Bug Severity

Critical

Tenant isolation broken
Login impossible
Data loss
Payment/financial wrong calculation
Production unavailable

High

Quote cannot be created
Lead conversion broken
Pipeline broken
Permission bypass

Medium

UI issue with workaround
Slow page
Incorrect non-financial display

Low

Text typo
Minor alignment
Non-blocking UI issue

27. Bug Lifecycle

New
Triaged
In Progress
Ready for QA
Verified
Closed
Reopened

28. Test Ownership

Backend Developer

- unit test
- integration test
- repository test
- service test

Frontend Developer

- component test
- form validation test
- UI state test

QA Engineer

- e2e
- exploratory
- regression
- test plan
- bug triage

DevOps

- pipeline
- deployment smoke
- monitoring validation
- backup restore test

Security Owner

- SAST
- DAST
- tenant isolation review
- permission review

29. CI Quality Gates

PR geçmeden:

- lint pass
- typecheck pass

unit tests pass
integration tests pass
OpenAPI validation pass
security scan pass
migration check pass

Release öncesi:

e2e pass
regression pass
performance smoke pass
staging smoke pass

Production sonrası:

production smoke pass
error rate normal
latency normal
queue normal

30. Test Coverage Hedefleri

Katman	Hedef
Core services	%85
Auth/IAM	%90
CRM/Lead/Sales	%80
Quote/Finance	%90
UI components	%70
E2E critical flows	%100

31. QA Automation Roadmap

Ay 1

unit test standard
integration test standard
customer CRUD tests
auth tests
tenant isolation tests

Ay 2

lead conversion
pipeline
quote
frontend e2e
visual regression

Ay 3

performance smoke
security baseline
regression suite
contract tests

32. Release Test Checklist

Migrations tested
Rollback tested
API tests passed
E2E passed
Tenant isolation passed
Permission tests passed
Performance smoke passed
Security scan passed
UAT approved
Production smoke plan ready

33. QA Dashboard

Metrikler:

open bugs by severity
test pass rate
automation coverage
flaky tests
release readiness
production defects
mean time to detect

mean time to resolve

34. Flaky Test Policy

Flaky test ignored edilmez.

3 kez flaky olan test karantinaya alınır.

Sorumlu squad 48 saat içinde düzeltir.

35. Final QA Kuralı

Bu sistemde en kritik kalite kuralı:

Tenant isolation, permission ve finansal hesaplama testleri kırmızıysa release yapılamaz.

Sonraki çıktı: **CRM OS Security & Compliance Blueprint v1.**

CRM OS Security & Compliance Blueprint v1

1. Güvenlik Hedefi

CRM OS için güvenlik yaklaşımı:

Secure by Design

Privacy by Design

Tenant Isolation First

Zero Trust Internal Access

Least Privilege

Full Auditability

GDPR tarafında “data protection by design and by default” ilkesi; sistem tasarımı sırasında veri minimizasyonu, pseudonymization ve varsayılan gizlilik kontrollerinin gömülü olmasını gerektirir. KVKK tarafında da kişisel verilerin işlenmesi, aktarılması, silinmesi/yok edilmesi/anonim hale getirilmesi gibi süreçlerin yönetilmesi gerekir. (gdpr-info.eu)

2. En Kritik Güvenlik İlkesi

Tenant A hiçbir koşulda Tenant B verisini göremez.

Bu kural:

API
Database
Search
Cache
File Storage
Logs
AI
Exports
Backups
Webhooks

tüm katmanlarda zorunludur.

3. Security Architecture

User
↓
WAF / CDN
↓
Ingress / Rate Limit
↓
API Gateway
↓
JWT Auth
↓
Tenant Guard
↓
Permission Guard
↓
Service Layer
↓
PostgreSQL RLS
↓
Audit Log

4. Kimlik Doğrulama

Zorunlu:

Email/password

Refresh token rotation

Session revocation

Password reset

2FA

Login history

Device history

Failed login protection

Enterprise:

SSO

SAML

OIDC

LDAP

SCIM

IP whitelist

Device trust

Token standardı:

Access token: 15 dakika

Refresh token: 7-30 gün

Rotation: zorunlu

Revocation list: Redis + DB

5. Yetkilendirme

Model:

RBAC

+

Permission

+

Tenant Scope

+

Branch Scope

+

Team Scope

+

Record Ownership

Örnek izinler:

customer.read

customer.create

customer.update

customer.delete

quote.approve

payment.read

inventory.adjust

admin.manage

Kritik kural:

Frontend yetki sadece görünürlük içindir.

Asıl kontrol backend tarafındadır.

6. Row Level Security

PostgreSQL RLS zorunlu.

```
ALTER TABLE customers ENABLE ROW LEVEL SECURITY;
```

```
CREATE POLICY tenant_isolation_customers
```

```
ON customers
```

```
USING (
```

```
    tenant_id = current_setting('app.tenant_id')::uuid
```

```
);
```

Her request transaction içinde:

```
SET LOCAL app.tenant_id = '<tenant_id>';
```

```
SET LOCAL app.user_id = '<user_id>';
```

RLS testleri release blocker olmalı.

7. Veri Sınıflandırma

Public

Ürün adı
Genel katalog bilgisi
Kamuya açık kampanya

Internal

Pipeline
Görevler
Stok bilgisi
Raporlar

Confidential

Müşteri bilgileri
Cari hesap
Teklif
Sipariş
Ticket
Mesajlar

Restricted

Şifre hash
API key
OAuth token
Banka hareketleri
Kimlik/vergi bilgileri
AI konuşmaları
Audit log

8. Şifreleme

Transit

TLS 1.2+
TLS 1.3 tercih
HSTS

Secure cookies

At Rest

PostgreSQL disk encryption

Object storage encryption

Backup encryption

Secret encryption

Field Level Encryption

Aday alanlar:

api_keys

oauth_tokens

bank_account_no

tax_number

integration_secrets

ai_provider_keys

webhook_secrets

9. Secret Management

Production'da .env kullanılmaz.

Vault

AWS Secrets Manager

Azure Key Vault

GCP Secret Manager

Doppler

Kurallar:

Secret kodda olmaz

Secret loglanmaz

Secret image içine girmez

Secret rotation yapılır

Least privilege uygulanır

10. Password Policy

Minimum 12 karakter

Common password engeli

Password reuse engeli

Bcrypt/Argon2

Failed login lockout

2FA opsiyonu

Enterprise için 2FA zorunlu

Öneri:

Argon2id

11. Session Security

Refresh token DB'de hashlenmiş tutulur

Device ID tutulur

IP/user-agent kaydı tutulur

Logout refresh token revoke eder

Admin tüm oturumları kapatabilir

Riskli girişte uyarı gider

12. API Security

Zorunlu:

JWT validation

Tenant header validation

Permission guard

Rate limiting

Request size limit

Input validation

Output filtering

Idempotency key

Replay protection for webhooks

Error response:

Fazla bilgi sızdırmaz
Stack trace production'da dönmez
Tenant var/yok bilgisi sızdırılmaz

13. Rate Limiting

Login: 5 deneme / 5 dakika / IP
Password reset: 3 istek / saat / email
Public API: plana göre
Webhook: signature + timestamp
AI API: kota bazlı
Export: throttled

14. Dosya Güvenliği

Zorunlu:

MIME validation
Extension validation
File size limit
Virus scan
Object storage private bucket
Signed URL
Tenant path isolation
No direct public upload

Path standardı:

tenant/{tenantId}/{module}/{entityId}/{fileId}/{filename}

15. Webhook Güvenliği

Her webhook:

HMAC signature
Timestamp
Replay window
Secret rotation

Delivery log
Retry policy
DLQ

Header:

X-CRMOS-Signature
X-CRMOS-Timestamp
X-CRMOS-Event

16. Audit Log

Audit zorunlu işlemler:

login
logout
failed_login
password_change
role_change
permission_change
customer_export
customer_delete
quote_approve
payment_create
stock_adjust
api_key_create
integration_connect
data_export
data_delete

Audit log değiştirilemez olmalı.

İleri seviye:

append-only audit table
WORM storage
hash chain

17. Logging Güvenliği

Loglanmayacak veriler:

password
token
refresh token
api key
secret
credit card
full bank account
sensitive personal data
AI raw confidential prompts

Log formatı:

```
{  
  "requestId": "uuid",  
  "tenantId": "uuid",  
  "userId": "uuid",  
  "module": "crm",  
  "action": "customer.update",  
  "status": "success"  
}
```

18. KVKK / GDPR Veri Hakları

Platform şu talepleri desteklemeli:

Aydınlatma metni kaydı
Açık rıza kaydı
Veri erişim talebi
Veri dışa aktarım talebi
Veri düzeltme talebi
Veri silme talebi
İşleme kısıtlama
Saklama süresi takibi
İmha politikası

Varsayılan olarak sadece gerekli kişisel veri işlenmeli ve erişim sınırlı tutulmalıdır; Avrupa Komisyonu’nun “privacy by default”

açıklaması da yalnızca gerekli verinin işlenmesi, kısa saklama süresi ve sınırlı erişim örneklerini verir. ([European Commission](#))

19. Consent Management

Tablolar:

consent_records
consent_versions
consent_sources
consent_withdrawals

Alanlar:

customer_id
consent_type
channel
text_version
given_at
withdrawn_at
source
ip_address

Consent tipleri:

marketing_email
marketing_sms
marketing_whatsapp
data_processing
data_sharing
profiling

20. Data Retention

Her modül için saklama politikası:

customers
leads
messages
tickets
audit_logs

files
ai_logs
exports
backups

Örnek:

Export dosyaları: 7 gün
Geçici import dosyaları: 30 gün
Audit logs: 5-10 yıl
AI raw prompts: mümkünse kısa süre

KVKK tarafında kişisel verilerin silinmesi, yok edilmesi veya anonim hale getirilmesi süreçlerinin politika ve prosedürlerle yönetilmesi gerekir. ([KVKK](#))

21. Data Deletion

Silme türleri:

Soft delete
Hard delete
Anonymization
Pseudonymization
Crypto-shredding

Kritik karar:

Finansal kayıtlar doğrudan silinmez.
Yasal saklama gerekiyorsa maskeleme/anonymization uygulanır.

22. Data Export

Müşteri export formatları:

JSON
CSV
XLSX
PDF

Export güvenliği:

Permission required

Audit log

Async job

Signed URL

Expire after 7 days

Watermark optional

23. Data Masking

UI masking:

Telefon: +90 532 *** ** 12

E-posta: a***@domain.com

Vergi no: *****1234

Banka: TR** **** **** 1234

Kim görebilir?

Role

Permission

Field-level permission

24. Field-Level Security

Özellikle:

customer.tax_number

customer.credit_limit

account.balance

payment.amount

bank_transaction

ai_conversation

audit_logs

Permission örnekleri:

customer.tax_number.read

finance.balance.read

payment.export
audit.read

25. IP Whitelist

Enterprise tenant için:

allowed_ip_ranges
deny_unknown_ip
admin_bypass_policy
temporary_access_window

Tablo:

tenant_ip_rules

26. Device Security

trusted devices
device fingerprint
device revoke
new device alert
suspicious login alert

27. Admin Security

Admin işlemleri:

MFA zorunlu
Audit zorunlu
Session timeout kısa
Sensitive action re-auth
IP whitelist opsiyonel

Sensitive action:

role permission update
api key create

- export
- delete
- integration secret change
- billing change

28. API Key Security

- API key sadece bir kez gösterilir
- DB'de hash tutulur
- Scope zorunlu
- Expiration opsiyonel
- Last used tracking
- IP restriction opsiyonel
- Rotation
- Revoke

29. Integration Security

- OAuth token encryption
- Refresh token encryption
- Connector permission scope
- Webhook signature
- Rate limit
- Integration audit log

30. AI Security

- AI katmanı için kurallar:

- Tenant izolasyonu
- Prompt log masking
- PII redaction
- Provider bazlı data retention kontrolü
- AI action permission
- AI usage quota
- Human approval for risky actions

AI şunları doğrudan yapmamalı:

payment delete
stock adjust
quote approve
customer delete
permission change

Bunlar için approval gerekir.

31. AI Data Boundary

Tenant A verisi Tenant B prompt context'ine giremez.

Vector DB'de:

tenant_id
entity_type
entity_id
permission_scope

zorunlu metadata olmalı.

32. Frontend Security

CSP
XSS protection
No token in localStorage tercihen
HttpOnly refresh cookie
Sanitize rich text
Escape user content
Secure file preview
Permission-aware UI

CSP örnek:

```
default-src 'self';  
script-src 'self';  
img-src 'self' data: https;;
```

```
connect-src 'self' api.crm-os.com;
```

33. Backend Security

- Input validation
- DTO whitelist
- SQL injection protection
- Parameterized queries
- Mass assignment protection
- Output DTO filtering
- CORS allowlist
- Helmet
- Request body limit

34. Database Security

- RLS
- Least privilege DB user
- No superuser for app
- Read replica permissions
- Encrypted backups
- Slow query monitoring
- Audit tables protected

DB role ayrımı:

- app_user
- migration_user
- readonly_user
- backup_user

35. Cache Security

Redis key'leri tenant bazlı.

tenant:{tenantId}:...

Cache'e konmayacak:

- password
- secret
- full token
- sensitive financial data
- raw AI confidential context

36. Backup Security

- Encrypted backups
- Access limited
- Restore audit
- Backup integrity check
- PITR
- Cross-region copy optional

Backup test:

- Monthly restore test
- Quarterly DR drill

37. Incident Response

Seviyeler:

- SEV1: Veri sızıntısı / tenant escape / sistem down
- SEV2: Yetki bypass / kritik modül down
- SEV3: Performans / kısmi hata
- SEV4: minor issue

Akış:

- Detect
- Triage
- Contain
- Eradicate
- Recover
- Notify

Postmortem

38. Security Monitoring

Alarm örnekleri:

failed_login_spike
permission_denied_spike
tenant_cross_access_attempt
mass_export
api_key_abuse
suspicious_admin_action
unusual_ai_usage
large_file_download
stock_adjustment_anomaly

39. Compliance Artifacts

Hazırlanacak dokümanlar:

KVKK aydınlatma metni
Açık rıza metni
Veri saklama ve imha politikası
Çerez politikası
Gizlilik politikası
Veri işleyen sözleşmesi
Alt işleyen listesi
Bilgi güvenliği politikası
Yetki matrisi
Veri envanteri
Risk analizi
İhlal müdahale planı

40. Minimum Security Go-Live Checklist

RLS testleri geçti
RBAC testleri geçti
Tenant isolation geçti

Audit log aktif
2FA hazır
Rate limit aktif
TLS aktif
Secrets dışarıda
Backup aktif
Restore test edildi
File upload kontrol edildi
Webhook signature aktif
Security scan temiz
Admin işlemleri audit'li
Export audit'li
KVKK/GDPR veri talepleri için süreç var

41. Sonraki Çıktı

CRM OS Event Storming & Domain Events v1

CRM OS Event Storming & Domain Events v1

1. Amaç

Bu doküman CRM OS içindeki tüm kritik iş olaylarını, event akışlarını, bounded context'leri ve event-driven mimariyi tanımlar.

Ana hedef:

Her önemli iş aksiyonu event üretir.

Workflow, notification, AI, reporting ve integration katmanları bu eventleri dinler.

2. Bounded Context Haritası

CORE

IAM

CRM

LEAD

SALES

QUOTE
ORDER
PRODUCT
INVENTORY
FINANCE
TASK
COMMUNICATION
SUPPORT
WORKFLOW
NOTIFICATION
AI
REPORTING
INTEGRATION

3. Event Naming Standard

Format:

<Entity><ActionPastTense>

Örnek:

CustomerCreated
LeadConverted
QuoteSent
OrderDelivered
PaymentReceived
TicketResolved
WorkflowFailed

Event payload standardı:

```
{  
  "eventId": "uuid",  
  "tenantId": "uuid",  
  "eventType": "CustomerCreated",  
  "aggregateType": "customer",  
  "aggregateId": "uuid",  
  "occurredAt": "2026-06-20T12:00:00Z",  
  "actor": {  
    "userId": "uuid"
```

```
    },  
    "payload": {}  
}
```

4. Event Store / Outbox Standardı

Tablo:

domain_events

Alanlar:

id
tenant_id
event_type
aggregate_type
aggregate_id
payload
published_at
created_at

Akış:

```
Service transaction  
↓  
Business data write  
↓  
domain_events insert  
↓  
commit  
↓  
outbox worker  
↓  
RabbitMQ publish  
↓  
handlers
```

Kritik kural:

Event, transaction commit edilmeden dış sisteme publish edilmez.

5. CORE Events

TenantCreated
TenantActivated
TenantSuspended
TenantCancelled
TenantSettingsUpdated
TenantPlanChanged
TenantQuotaExceeded
FeatureEnabled
FeatureDisabled
SubscriptionCreated
SubscriptionRenewed
SubscriptionExpired

TenantCreated

Tetikleyen:

Yeni firma/workspace oluşturma

Dinleyenler:

IAM
Workflow
Notification
Reporting

Sonuç:

Default roles oluşturulur
Default permissions bağlanır
Default pipeline oluşturulur
Welcome notification gönderilir

6. IAM Events

UserInvited
UserActivated
UserDeactivated

UserLoggedIn
UserLoginFailed
UserLoggedOut
PasswordChanged
PasswordResetRequested
PasswordResetCompleted
TwoFactorEnabled
TwoFactorDisabled
RoleCreated
RoleUpdated
RoleDeleted
PermissionAssignedToRole
PermissionRevokedFromRole
UserAssignedToRole
UserRemovedFromRole
SessionRevoked
DeviceTrusted
DeviceRevoked

UserAssignedToRole

Dinleyenler:

Audit
Notification
Security

Kritik etki:

Permission cache invalidated
Audit log created

7. CRM Events

CustomerCreated
CustomerUpdated
CustomerDeleted
CustomerRestored
CustomerAssigned
CustomerSegmentChanged
CustomerTagged

CustomerUntagged
CustomerMerged
CustomerRiskScoreUpdated
CustomerLifetimeValueUpdated
CustomerConsentGiven
CustomerConsentWithdrawn
CustomerFileUploaded
CustomerNoteAdded
CustomerTimelineEventCreated
ContactCreated
ContactUpdated
ContactDeleted
AddressCreated
AddressUpdated
AddressDeleted

CustomerCreated

Dinleyenler:

Timeline
Audit
Notification
Workflow
Reporting
AI

Yan etkiler:

Customer timeline event
Default account placeholder
AI duplicate check

8. LEAD Events

LeadCreated
LeadUpdated
LeadAssigned
LeadContacted
LeadQualified
LeadDisqualified

LeadScored
LeadTemperatureChanged
LeadLost
LeadConverted
LeadSourceCreated
LeadEnriched
LeadFileUploaded
LeadNoteAdded

LeadCreated

Dinleyenler:

Workflow
Notification
AI
Reporting

Olası otomasyon:

Lead skoru hesapla
Satışçı ata
WhatsApp karşılama gönder
Takip görevi oluştur

LeadConverted

Payload:

```
{  
  "leadId": "uuid",  
  "customerId": "uuid",  
  "opportunityId": "uuid"  
}
```

Dinleyenler:

CRM
SALES
Timeline
Audit
Reporting
Workflow

9. SALES Events

PipelineCreated
PipelineUpdated
PipelineStageCreated
PipelineStageUpdated
PipelineStageDeleted
OpportunityCreated
OpportunityUpdated
OpportunityAssigned
OpportunityStageChanged
OpportunityValueChanged
OpportunityProductAdded
OpportunityProductRemoved
OpportunityWon
OpportunityLost
OpportunityReopened
OpportunityForecastUpdated
OpportunityCompetitorAdded
OpportunityActivityAdded

OpportunityStageChanged

Payload:

```
{  
  "opportunityId": "uuid",  
  "fromStageId": "uuid",  
  "toStageId": "uuid",  
  "fromStageName": "Discovery",  
  "toStageName": "Proposal"  
}
```

Dinleyenler:

Timeline
Workflow
Reporting
AI
Notification

Otomasyon örneği:

Proposal aşamasına geçince teklif oluşturma görevi aç
Negotiation aşamasında yöneticiye bildirim gönder

10. QUOTE Events

QuoteCreated
QuoteUpdated
QuoteItemAdded
QuoteItemUpdated
QuoteItemRemoved
QuoteCalculated
QuoteDiscountApplied
QuoteApprovalRequested
QuoteApproved
QuoteRejected
QuoteSent
QuoteViewed
QuoteExpired
QuoteAcceptedByCustomer
QuoteRejectedByCustomer
QuotePdfGenerated
QuoteConvertedToOrder
QuoteFileUploaded
QuoteVersionCreated

QuoteSent

Dinleyenler:

Timeline
Notification
Workflow
Reporting
AI

Yan etkiler:

Customer timeline update
Quote reminder workflow scheduled

Sales rep notification

QuoteApproved

Kritik:

Margin kontrolü yapılmış olmalı
Audit zorunlu

Dinleyenler:

Workflow
Notification
Order
Reporting

11. ORDER Events

OrderCreated
OrderUpdated
OrderConfirmed
OrderCancelled
OrderItemAdded
OrderItemUpdated
OrderStatusChanged
OrderReservationRequested
OrderReserved
OrderPartiallyReserved
OrderReservationFailed
OrderShipmentCreated
OrderPartiallyShipped
OrderShipped
OrderDelivered
OrderReturned
OrderReturnRequested
OrderPaymentPlanCreated
OrderFileUploaded

OrderCreated

Dinleyenler:

Inventory
Finance
Notification
Timeline
Reporting
Workflow

Yan etkiler:

Stock reservation request
Account transaction draft
Payment plan creation
Customer notification

12. PRODUCT Events

ProductCreated
ProductUpdated
ProductDeactivated
ProductVariantCreated
ProductVariantUpdated
ProductPriceCreated
ProductPriceUpdated
ProductCategoryCreated
ProductBrandCreated
ProductImageUploaded
ProductFileUploaded
ProductDescriptionGenerated

Dinleyenler:

Inventory
Catalog
AI
Reporting
Search

13. INVENTORY Events

WarehouseCreated
WarehouseUpdated
StockCreated
StockChanged
StockMovementCreated
StockReserved
StockReservationReleased
StockReservationFailed
StockCriticalLevelReached
StockCountStarted
StockCountCompleted
StockAdjusted
StockTransferCreated
StockTransferShipped
StockTransferReceived

StockCriticalLevelReached

Dinleyenler:

Notification
Procurement
Workflow
Reporting
AI

Otomasyon:

Satın alma önerisi oluştur
Depo yöneticisine bildirim gönder

14. FINANCE Events

AccountCreated
AccountBalanceUpdated
AccountTransactionCreated
CreditLimitUpdated
CreditLimitExceeded
PaymentCreated
PaymentReceived

PaymentCancelled
PaymentOverdue
InstallmentCreated
InstallmentDueSoon
InstallmentOverdue
InvoiceCreated
InvoiceSent
InvoiceCancelled
BankTransactionImported
BankTransactionMatched

PaymentReceived

Dinleyenler:

Timeline
Order
Reporting
Notification
Workflow

Yan etkiler:

Customer balance update
Receivable status update
Sales rep notification

15. TASK / ACTIVITY Events

TaskCreated
TaskAssigned
TaskUpdated
TaskCompleted
TaskCancelled
TaskOverdue
TaskReminderTriggered
ActivityCreated
CallLogged
MeetingLogged
VisitLogged
EmailLogged

WhatsAppLogged
NoteAdded

Dinleyenler:

Timeline
Notification
Reporting
AI
Workflow

16. COMMUNICATION Events

ChannelConnected
ChannelDisconnected
ConversationCreated
ConversationAssigned
ConversationStatusChanged
MessageReceived
MessageSent
MessageDelivered
MessageRead
MessageFailed
CallStarted
CallEnded
CallRecorded
CallTranscribed
MeetingCreated
MeetingCompleted
ConversationSummarized

MessageReceived

Dinleyenler:

Inbox
Notification
AI
Workflow
CRM Timeline

AI aksiyon:

Duygu analizi

Özetleme

Müşteri niyeti tahmini

Cevap önerisi

17. SUPPORT Events

TicketCreated

TicketAssigned

TicketUpdated

TicketPriorityChanged

TicketStatusChanged

TicketResolved

TicketClosed

TicketReopened

TicketSlaBreached

TicketMessageAdded

TicketFileUploaded

CustomerSatisfactionSubmitted

KnowledgeBaseArticleCreated

TicketSlaBreached

Dinleyenler:

Notification

Workflow

Reporting

Management Dashboard

18. WORKFLOW Events

WorkflowCreated

WorkflowActivated

WorkflowPaused

WorkflowDisabled

WorkflowTriggered

WorkflowRunStarted

WorkflowRunCompleted
WorkflowRunFailed
WorkflowStepStarted
WorkflowStepCompleted
WorkflowStepFailed
ApprovalRequested
ApprovalStepAssigned
ApprovalApproved
ApprovalRejected
ApprovalExpired

WorkflowRunFailed

Dinleyenler:

Notification

Audit

Admin Dashboard

19. NOTIFICATION Events

NotificationCreated
NotificationDelivered
NotificationRead
NotificationFailed
NotificationPreferenceUpdated

20. AI Events

AIActionRequested
AIActionCompleted
AIActionFailed
AISummaryGenerated
AIClassificationCompleted
AILeadScoreCalculated
AIOpportunityWinProbabilityCalculated
AIQuoteTextGenerated
AIRecommendationCreated
AIUsageLimitExceeded

AIFeedbackSubmitted

AIActionRequested

Kritik kontrollerr:

tenant scope

permission scope

usage quota

PII masking

21. REPORTING Events

DashboardCreated

DashboardUpdated

ReportCreated

ReportScheduled

ReportGenerated

ReportExportRequested

ReportExportCompleted

MetricSnapshotCreated

KpiTargetUpdated

22. INTEGRATION Events

ApiKeyCreated

ApiKeyRevoked

WebhookCreated

WebhookUpdated

WebhookDeleted

WebhookDeliveryRequested

WebhookDelivered

WebhookFailed

ConnectorConnected

ConnectorDisconnected

IntegrationRunStarted

IntegrationRunCompleted

IntegrationRunFailed

ExternalOrderImported

ExternalPaymentImported

23. Event → Handler Matrix

Event	Timeline	Notification	Workflow	AI	Reporting	Integration
CustomerCreated	✓		✓	✓	✓	
LeadCreated	✓	✓	✓	✓	✓	
LeadConverted	✓	✓	✓		✓	
OpportunityStageChanged	✓	✓	✓	✓	✓	
QuoteSent	✓	✓	✓	✓	✓	✓
QuoteApproved	✓	✓	✓		✓	
OrderCreated	✓	✓	✓		✓	✓
PaymentReceived	✓	✓	✓		✓	
TicketCreated	✓	✓	✓	✓	✓	
MessageReceived	✓	✓	✓	✓		

24. Critical End-to-End Event Flow

Lead → Quote Flow

LeadCreated

↓

LeadAssigned

↓

LeadQualified

↓

CustomerCreated

↓

OpportunityCreated

↓

LeadConverted

↓

OpportunityStageChanged

↓

QuoteCreated
↓
QuoteItemAdded
↓
QuoteCalculated
↓
QuoteSent
↓
QuoteViewed
↓
QuoteApproved

25. Quote → Order → Payment Flow

QuoteApproved
↓
QuoteConvertedToOrder
↓
OrderCreated
↓
OrderReservationRequested
↓
StockReserved
↓
OrderConfirmed
↓
InvoiceCreated
↓
PaymentCreated
↓
PaymentReceived
↓
AccountBalanceUpdated

26. Support Flow

MessageReceived
↓
ConversationCreated

↓
TicketCreated
↓
TicketAssigned
↓
TicketStatusChanged
↓
TicketResolved
↓
CustomerSatisfactionSubmitted

27. Workflow Trigger Mapping

Workflow trigger kaynakları:

LeadCreated
LeadAssigned
QuoteSent
QuoteViewed
QuoteExpired
OrderCreated
PaymentOverdue
StockCriticalLevelReached
TicketCreated
TicketSlaBreached
MessageReceived
TaskOverdue

Örnek:

Trigger: QuoteSent
Delay: 3 days
Condition: Quote not viewed
Action: Send reminder
Action: Create task

28. Notification Trigger Mapping

LeadAssigned → Sales rep notification
QuoteApprovalRequested → Manager notification
PaymentOverdue → Finance + Sales notification
StockCriticalLevelReached → Warehouse notification
TicketSlaBreached → Support manager notification
WorkflowRunFailed → Admin notification

29. AI Trigger Mapping

LeadCreated → AI lead score
MessageReceived → sentiment + reply suggestion
CallTranscribed → summary + action extraction
OpportunityStageChanged → win probability update
QuoteCreated → quote text suggestion
TicketCreated → classification + solution suggestion

30. Reporting Trigger Mapping

LeadCreated → lead metrics
LeadConverted → conversion rate
OpportunityStageChanged → pipeline velocity
QuoteSent → quote activity
QuoteApproved → win rate
OrderCreated → sales revenue
PaymentReceived → collection metrics
TicketResolved → support SLA metrics

31. Integration Trigger Mapping

QuoteSent → webhook
OrderCreated → webhook
PaymentReceived → webhook
StockChanged → webhook
CustomerCreated → webhook

TicketCreated → webhook

32. Event Idempotency

Her handler idempotent olmalı.

Handler log tablosu:

event_handler_logs

Alanlar:

event_id

handler_name

status

processed_at

error_message

Kural:

Aynı event aynı handler tarafından ikinci kez işlenirse tekrar yan etki üretmez.

33. Event Versioning

Event tipi versiyonlanabilir:

CustomerCreated.v1

CustomerCreated.v2

Payload içinde:

```
{  
  "schemaVersion": 1  
}
```

Kural:

Breaking change varsa yeni versiyon çıkar.

34. Event Priority

High

PaymentReceived
QuoteApproved
OrderCreated
TicketSlaBreached
SecurityEventDetected

Normal

CustomerCreated
LeadCreated
QuoteSent
MessageReceived

Low

DashboardUpdated
ReportGenerated
AISummaryGenerated

35. Event Retention

domain_events: 12-24 ay
audit_logs: 5-10 yıl
workflow_logs: 12 ay
notification_logs: 6-12 ay
ai_usage_logs: 12 ay
webhook_logs: 3-6 ay

36. Event Security

Event payload içinde:

password yok
token yok
secret yok

full payment sensitive data yok
gereksiz PII yok

Sensitive eventlerde payload minimize edilir.

37. Event Testing

Her domain event için test:

event created
payload valid
outbox persisted
handler processed
idempotency works
tenant isolation respected

38. Event Observability

Metrikler:

events_created_total
events_published_total
events_failed_total
event_publish_latency
handler_duration
handler_failure_count
outbox_lag
dlq_count

39. MVP Event List

MVP'de zorunlu eventler:

TenantCreated
UserInvited
UserLoggedIn
CustomerCreated
CustomerUpdated
LeadCreated

LeadAssigned
LeadConverted
OpportunityCreated
OpportunityStageChanged
QuoteCreated
QuoteSent
QuoteApproved
TaskCreated
TaskCompleted
NotificationCreated
AuditLogCreated

40. Sonraki Çıktı

CRM OS Implementation Roadmap v1

CRM OS Implementation Roadmap v1

1. Ana Hedef

12 ay içinde:

Enterprise-ready CRM OS SaaS platformunun

MVP → Commercial → Enterprise → AI aşamalarını tamamlamak.

Ana ürün akışı:

Tenant

→ User

→ Customer

→ Lead

→ Opportunity

→ Quote

→ Order

→ Inventory

→ Finance

→ Support

→ Workflow

→ AI

2. Yol Haritası Özeti

Faz	Süre	Hedef
Phase 0	2 hafta	Hazırlık, repo, mimari, CI/CD
Phase 1	10 hafta	CRM MVP
Phase 2	12 hafta	Commercial MVP
Phase 3	12 hafta	Enterprise CRM
Phase 4	12 hafta	AI + Analytics + Scale
Toplam:		
48 hafta	≈ 12 ay	

3. Phase 0 – Foundation

Süre

Hafta 1-2

Hedef

Kod tabanı, altyapı ve ekip çalışma düzeni kurulur.

Teslimatlar

Monorepo
Backend skeleton
Frontend skeleton
Docker Compose
PostgreSQL
Redis
RabbitMQ
MinIO
CI/CD pipeline
Design tokens
App shell
OpenAPI skeleton
Migration pipeline
RLS proof of concept

Kabul Kriteri

Developer local ortamı 1 komutla ayağa kalkar.

API health check çalışır.

Web app login shell açılır.

PostgreSQL migration pipeline çalışır.

Tenant RLS POC geçer.

4. Phase 1 – CRM MVP

Süre

Hafta 3-12

Hedef

İlk satılabilir CRM çekirdeği.

Modüller

IAM

Customer

Lead

Pipeline

Opportunity

Quote

Task

Timeline

Notification

Dashboard

Audit

Ana Akış

Login

→ Customer Create

→ Lead Create

→ Lead Convert

→ Opportunity Create

→ Pipeline Move

→ Quote Create

→ Quote PDF

→ Task Create

→ Dashboard View

Teslimatlar

Tenant management
User management
Role/permission
Customer CRUD
Customer 360 shell
Lead CRUD
Lead conversion
Pipeline kanban
Opportunity CRUD
Quote builder v1
Quote PDF v1
Task module
Activity module
Notification center
MVP dashboard
Audit log
Tenant isolation tests

Kabul Kriteri

Pilot müşteri sisteme girip müşteri oluşturabilir.
Lead oluşturup fırsata çevirebilir.
Pipeline'da fırsatı taşıyabilir.
Teklif oluşturup PDF alabilir.
Tüm aksiyonlar audit log'a düşer.
Tenant A, Tenant B verisini göremez.

5. Phase 2 – Commercial MVP

Süre

Hafta 13-24

Hedef

CRM satıştan operasyon ve gelire bağlanır.

Modüller

Order
Product
Inventory
Finance Lite
Payment
Receivable
Workflow v1
Email/WhatsApp basic
Customer Portal v1

Ana Akış

Quote Approved
→ Order Created
→ Stock Reserved
→ Payment Plan
→ Invoice Placeholder
→ Payment Received
→ Balance Updated

Teslimatlar

Order CRUD
Order items
Order status history
Product catalog
Product variants
Warehouses
Stocks
Stock movements
Stock reservations
Customer account
Account transactions
Payment entry
Installments
Receivable dashboard
Workflow trigger/action v1
Email sending
WhatsApp placeholder/API integration shell
Customer portal login
Customer quote view

Kabul Kriteri

Teklif siparişe çevrilebilir.
Sipariş stok rezervasyonu oluşturur.
Tahsilat girilebilir.
Cari bakiye güncellenir.
Müşteri portalından teklif görüntülenir.
Basit workflow çalışır.

6. Phase 3 – Enterprise CRM

Süre

Hafta 25-36

Hedef

Kurumsal müşteriler için gelişmiş CRM kabiliyeti.

Modüller

Omnichannel Inbox
Support/Ticket
SLA
Dealer Portal
Advanced Workflow
Approval Engine
Public API
Webhooks
SSO/SAML hazırlığı
Advanced Audit
Branch/Dealer Access

Ana Akışlar

Message Received
→ Conversation Created
→ Ticket Created
→ SLA Tracked
→ Resolution
Dealer Login
→ Catalog View
→ Dealer Price
→ Dealer Order

→ Balance View

Teslimatlar

Conversation list
Messages
Customer context panel
Ticket CRUD
SLA rules
Ticket assignment
Dealer portal
Dealer catalog
Dealer order
Approval engine
Quote approval rules
Discount approval
Webhook system
API key management
Public API v1
Branch scoped access
Dealer scoped access
Enterprise audit screen

Kabul Kriteri

WhatsApp/e-posta mesajları inbox'a düşer.
Mesajdan ticket oluşturulur.
SLA takip edilir.
Bayi portalından sipariş oluşturulur.
İndirim onay akışı çalışır.
Webhook eventleri gönderilir.
API key ile public API çalışır.

7. Phase 4 – AI + Analytics + Scale

Süre

Hafta 37-48

Hedef

Platform AI destekli ve ölçeklenebilir kurumsal ürüne dönüşür.

Modüller

- AI Gateway
- AI Assistant
- AI Summaries
- AI Lead Scoring
- AI Quote Assistant
- Advanced Reporting
- Custom Dashboards
- Mobile CRM v1
- Performance Scaling
- Marketplace foundation

Teslimatlar

- AI provider abstraction
- Ask CRM
- Conversation summary
- Call/meeting summary placeholder
- Lead scoring AI
- Opportunity win probability
- Quote text generator
- AI usage quota
- AI usage logs
- Custom dashboard builder
- Report export
- Mobile CRM v1
- Offline sync foundation
- Read replica support
- Partitioning for large logs
- Marketplace app model

Kabul Kriteri

- Kullanıcı doğal dilde CRM verisini sorgular.
- AI müşteri görüşmesini özetler.
- Lead skor önerisi üretir.
- Teklif açıklaması oluşturur.
- Dashboard özelleştirilebilir.
- Mobil uygulama temel saha akışını destekler.
- Sistem 100 tenant / yüksek veri hacmi testini geçer.

8. 12 Aylık Sprint Planı

Sprint 1-2

Foundation

Repo

Docker

CI/CD

Backend skeleton

Frontend skeleton

RLS POC

Sprint 3-4

IAM

Tenant

Users

Roles

Permissions

Login

Audit base

Sprint 5-6

Customer

Contacts

Addresses

Timeline

Files

Tags

Customer 360

Sprint 7-8

Lead

Lead Sources

Lead Assignment

Lead Conversion

Pipeline setup

Sprint 9-10

Opportunity

Pipeline Kanban

Stage Move

Win/Loss
Stage History

Sprint 11-12

Quote
Quote Items
Pricing
PDF
Approval
Dashboard
Tasks
Notifications

Sprint 13-14

Order
Order Items
Order Status
Quote to Order

Sprint 15-16

Products
Variants
Warehouses
Stocks
Stock Movements

Sprint 17-18

Finance Lite
Accounts
Transactions
Payments
Installments
Receivables

Sprint 19-20

Workflow v1
Email
WhatsApp shell

Customer Portal

Sprint 21-22

Omnichannel Inbox
Messages
Conversation Context

Sprint 23-24

Support
Tickets
SLA
Knowledge Base shell

Sprint 25-26

Dealer Portal
Dealer Catalog
Dealer Orders
Dealer Balance

Sprint 27-28

Approval Engine
Discount Approval
Purchase Approval shell

Sprint 29-30

Public API
API Keys
Webhooks
Integration logs

Sprint 31-32

Enterprise Security
Branch scope
Dealer scope
Advanced audit
SSO preparation

Sprint 33-34

- AI Gateway
- AI Assistant
- AI usage quota
- AI logs

Sprint 35-36

- AI Summaries
- Lead scoring
- Quote assistant
- Next best action

Sprint 37-38

- Advanced Reporting
- Custom Dashboards
- Report Export

Sprint 39-40

- Mobile CRM v1
- Field sales
- Tasks
- Visits
- Offline foundation

Sprint 41-42

- Performance
- Read replicas
- Partitioning
- Load testing
- Query optimization

Sprint 43-44

- Marketplace foundation
- Connector model
- App installs

Sprint 45-46

Hardening
Security review
QA regression
UAT

Sprint 47-48

Launch preparation
Documentation
Training
Pilot rollout
Commercial launch

9. Workstream Plan

Backend Workstream

Core platform
IAM
CRM
Lead
Sales
Quote
Order
Inventory
Finance
Communication
Workflow
AI
Reporting
Integration

Frontend Workstream

Design system
App shell
DataGrid
Customer 360
Lead UI
Pipeline

Quote builder
Order UI
Inventory UI
Finance UI
Inbox
Portal
Dashboard
AI UI

DevOps Workstream

Docker
CI/CD
Kubernetes
Monitoring
Logging
Backup
Secrets
Security scan
Load test infra
Production readiness

QA Workstream

Test automation
API tests
E2E tests
Tenant isolation tests
Permission tests
Regression suite
Performance tests
UAT

Product / UX Workstream

PRD
User stories
Wireframes
Design system
Prototype
Acceptance criteria
UAT scripts

Training material

10. Critical Path

DevOps Foundation

↓

Backend Foundation

↓

Tenant + IAM

↓

Customer

↓

Lead

↓

Opportunity

↓

Quote

↓

Order

↓

Inventory

↓

Finance

↓

Workflow

↓

Omnichannel

↓

AI

Frontend critical path:

Design System

↓

App Shell

↓

DataGrid/Form/Drawer

↓

Customer

↓

Lead

↓
Pipeline

↓
Quote

↓
Order

11. Parallelization Plan for 50 People

Month 1

Core/IAM
DevOps
Frontend Platform
UX/UI
QA Automation

Month 2

CRM
Lead
Sales
Frontend CRM
QA

Month 3

Quote
Task
Dashboard
Notification
MVP hardening

Month 4-6

Order
Inventory
Finance
Workflow
Portal

Month 7-9

Omnichannel

Support

Dealer

Public API

Approval Engine

Month 10-12

AI

Reporting

Mobile

Performance

Marketplace

Hardening

12. MVP Release Criteria

MVP şu koşullarda çıkar:

Tenant isolation passed

RBAC passed

Customer CRUD stable

Lead conversion stable

Pipeline stable

Quote PDF stable

Audit log active

Dashboard working

E2E happy path passed

P95 API latency acceptable

No critical security issue

Pilot user UAT approved

13. Commercial Release Criteria

Order flow complete

Stock reservation complete

Payment entry complete

Account balance accurate

Workflow v1 stable
Customer portal usable
Backup/restore tested
Production monitoring active

14. Enterprise Release Criteria

Omnichannel usable
Ticket/SLA stable
Dealer portal usable
Approval engine stable
Public API documented
Webhook retry/DLQ working
Branch/dealer scoped access tested
Advanced audit available

15. AI Release Criteria

AI actions permission-aware
Tenant isolation in AI context tested
Usage quota active
AI logs active
PII masking active
Human approval for risky actions

16. Key Risks

Risk: Scope Creep

Mitigation:

MVP freeze
Change control board
Icebox backlog

Risk: Tenant Leakage

Mitigation:

RLS

Automated tenant isolation tests

Security review per module

Risk: 50 Kişilik Ekipte Çakışma

Mitigation:

Module ownership

API contracts

Architecture review board

Code owners

Risk: UI Tutarsızlığı

Mitigation:

Design system

Storybook

Visual regression

Frontend platform squad

Risk: Performans

Mitigation:

Index review

Pagination mandatory

Load test

Read replica

Partitioning

17. Architecture Review Board

Her hafta gözden geçirilecek:

New tables

New APIs

Event changes

Security changes

Performance risk

Cross-module dependencies

Zorunlu onay gerekenler:

- Database schema breaking change
- Permission model change
- Tenant isolation affecting code
- Financial calculation
- AI data access
- Public API change

18. Documentation Deliverables

Her release için:

- ERD update
- Data Dictionary update
- OpenAPI update
- Event catalog update
- Permission matrix update
- User guide
- Admin guide
- Deployment notes
- Release notes
- Known issues

19. Launch Strategy

Pilot 1

- 1 tenant
- 10 users
- CRM + Lead + Quote

Pilot 2

- 3 tenants
- 50 users
- Order + Inventory + Finance

Beta

10 tenants

200 users

Workflow + Portal + Support

Commercial Launch

Public onboarding

Paid plans

Support process

SLA

20. Success Metrics

Product

Time to first customer create < 10 min

Time to first quote < 30 min

Lead to quote conversion tracked

Weekly active users > 70%

Technical

API p95 < 500ms

Uptime > 99.5% MVP

Critical bugs = 0

Tenant isolation incidents = 0

Business

Pilot activation rate

Trial to paid conversion

MRR

Churn

Expansion revenue

21. Final Roadmap Output

Bu roadmap 50 kişilik ekibi aşağıdaki sırayla çalıştırır:

Foundation

- CRM MVP
- Commercial MVP
- Enterprise CRM
- AI + Scale

Sonraki belge:

CRM OS Permission Matrix v1

CRM OS Permission Matrix v1

1. Yetki Modeli

CRM OS yetkilendirme modeli:

RBAC

- + Permission-Based Access
- + Tenant Scope
- + Branch Scope
- + Dealer Scope
- + Team Scope
- + Record Ownership
- + Field-Level Security

Temel kural:

Frontend sadece görünürlüğü yönetir.

Backend her istekte yetkiyi tekrar kontrol eder.

PostgreSQL RLS tenant izolasyonunu zorunlu uygular.

2. Standart Roller

Rol	Açıklama
super_admin	Platform yöneticisi
tenant_admin	Firma yöneticisi
sales_director	Satış direktörü
sales_manager	Satış müdürü
sales_rep	Satış temsilcisi

finance_manager	Finans müdürü
finance_user	Finans kullanıcısı
warehouse_manager	Depo müdürü
warehouse_user	Depo kullanıcısı
support_manager	Destek müdürü
support_agent	Destek temsilcisi
marketing_user	Pazarlama kullanıcısı
dealer_admin	Bayi yöneticisi
dealer_user	Bayi kullanıcısı
customer_portal_user	Müşteri portal kullanıcısı
auditor	Denetçi / salt okunur

3. Scope Seviyeleri

tenant_all	Tenant içindeki tüm kayıtlar
branch	Sadece bağlı şube
team	Sadece bağlı ekip
owned	Sadece kendisine atanmış kayıtlar
dealer	Sadece bağlı bayi
customer_self	Sadece kendi müşteri portal verisi
none	Erişim yok

4. Permission Naming Standard

Format:

<module>.<resource>.<action>

Örnek:

```

crm.customer.read
crm.customer.create
crm.customer.update
crm.customer.delete
sales.quote.approve
finance.payment.create
inventory.stock.adjust
admin.user.manage

```

Action standardı:

read
create
update
delete
restore
assign
approve
reject
export
import
manage

5. CORE / Admin Permissions

Permission	Açıklama
core.tenant.read	Tenant bilgisi görüntüleme
core.tenant.update	Tenant ayarlarını güncelleme
core.plan.read	Plan bilgisi görme
core.subscription.manage	Abonelik yönetimi
core.feature.manage	Feature flag yönetimi
core.audit.read	Audit log görüntüleme
core.system.manage	Sistem ayarları

Rol Matrisi

Permission	super_admin	tenant_admin	auditor
core.tenant.read	✓	✓	✓
core.tenant.update	✓	✓	
core.subscription.manage	✓	✓	
core.feature.manage	✓		
core.audit.read	✓	✓	✓
core.system.manage	✓		

6. IAM Permissions

Permission	Açıklama
iam.user.read	Kullanıcı görüntüleme
iam.user.create	Kullanıcı oluşturma

iam.user.update	Kullanıcı güncelleme
iam.user.delete	Kullanıcı pasifleştirme
iam.user.invite	Kullanıcı davet
iam.role.read	Rol görüntüleme
iam.role.create	Rol oluşturma
iam.role.update	Rol güncelleme
iam.role.delete	Rol silme
iam.permission.read	Yetki görüntüleme
iam.permission.assign	Yetki atama
iam.session.revoke	Oturum sonlandırma

Rol Matrisi

Permission	super_admin	tenant_admin	sales_manager	auditor
iam.user.read	✓	✓	team	✓
iam.user.create	✓	✓		
iam.user.update	✓	✓	team	
iam.user.delete	✓	✓		
iam.user.invite	✓	✓		
iam.role.read	✓	✓		✓
iam.role.create	✓	✓		
iam.role.update	✓	✓		
iam.permission.assign	✓	✓		
iam.session.revoke	✓	✓		

7. CRM Permissions

Permission	Açıklama
crm.customer.read	Müşteri görüntüleme
crm.customer.create	Müşteri oluşturma
crm.customer.update	Müşteri güncelleme
crm.customer.delete	Müşteri silme
crm.customer.restore	Silinen müşteri geri alma
crm.customer.assign	Temsilci atama
crm.customer.export	Müşteri dışa aktarma
crm.contact.manage	İletişim kişileri yönetimi
crm.address.manage	Adres yönetimi

crm.file.manage	Müşteri dosyaları
crm.timeline.read	Timeline görüntüleme
crm.tag.manage	Etiket yönetimi
crm.segment.manage	Segment yönetimi
crm.customer.merge	Müşteri birleştirme
crm.customer.risk.read	Risk skoru görme
crm.customer.ltv.read	LTV görme

Rol Matrisi

Permission	tenant_admin	sales_director	sales_manager	sales_rep	support_agent	finance_user	dealer_user	auditor
crm.customer.read	tenant_all	tenant_all	team	owned	assigned	read	dealer	tenant_all
crm.customer.create	✓	✓	✓	✓				
crm.customer.update	✓	✓	team	owned	assigned			
crm.customer.delete	✓							
crm.customer.assign	✓	✓	team					
crm.customer.export	✓	✓						✓
crm.contact.manage	✓	✓	team	owned	assigned			
crm.address.manage	✓	✓	team	owned	assigned			
crm.timeline.read	✓	✓	team	owned	assigned	read	dealer	✓
crm.customer.risk.read	✓	✓	team			✓		✓
crm.customer.ltv.read	✓	✓	team			✓		✓

8. LEAD Permissions

Permission	Açıklama
lead.lead.read	Lead görüntüleme
lead.lead.create	Lead oluşturma

lead.lead.update	Lead güncelleme
lead.lead.delete	Lead silme
lead.lead.assign	Lead atama
lead.lead.convert	Lead dönüştürme
lead.lead.export	Lead dışa aktarma
lead.source.manage	Lead kaynakları yönetimi
lead.score.read	Lead skoru görme
lead.score.update	Lead skoru değiştirme

Rol Matrisi

Permission	tenant_admin	sales_director	sales_manager	sales_rep	marketing_user	auditor
lead.lead.read	tenant_all	tenant_all	team	owned	tenant_all	tenant_all
lead.lead.create	✓	✓	✓	✓	✓	
lead.lead.update	✓	✓	team	owned	✓	
lead.lead.delete	✓					
lead.lead.assign	✓	✓	team			
lead.lead.convert	✓	✓	team	owned		
lead.lead.export	✓	✓			✓	✓
lead.source.manage	✓	✓			✓	
lead.score.read	✓	✓	team	owned	✓	✓
lead.score.update	✓	✓				

9. SALES Permissions

Permission	Açıklama
sales.pipeline.read	Pipeline görüntüleme
sales.pipeline.manage	Pipeline yönetimi
sales.opportunity.read	Fırsat görüntüleme
sales.opportunity.create	Fırsat oluşturma
sales.opportunity.update	Fırsat güncelleme

sales.opportunity.delete	Fırsat silme
sales.opportunity.assign	Fırsat atama
sales.opportunity.move_stage	Aşama değiştirme
sales.opportunity.win	Kazanıldı yapma
sales.opportunity.lose	Kaybedildi yapma
sales.forecast.read	Forecast görüntüleme
sales.forecast.manage	Forecast yönetimi

Rol Matrisi

Permission	tenant_admin	sales_director	sales_manager	sales_rep	finance_user	audit_or
sales.pipeline.read	tenant_all	tenant_all	team	owned	read	tenant_all
sales.pipeline.manage	✓	✓				
sales.opportunity.read	tenant_all	tenant_all	team	owned	read	tenant_all
sales.opportunity.create	✓	✓	✓	✓		
sales.opportunity.update	✓	✓	team	owned		
sales.opportunity.delete	✓					
sales.opportunity.assign	✓	✓	team			
sales.opportunity.move_stage	✓	✓	team	owned		
sales.opportunity.win	✓	✓	team	owned		
sales.opportunity.lose	✓	✓	team	owned		
sales.forecast.read	✓	✓	team		✓	✓
sales.forecast.manage	✓	✓				

10. QUOTE Permissions

Permission	Açıklama
sales.quote.read	Teklif görüntüleme
sales.quote.create	Teklif oluşturma

sales.quote.update	Teklif güncelleme
sales.quote.delete	Teklif silme
sales.quote.send	Teklif gönderme
sales.quote.approve	Teklif onaylama
sales.quote.reject	Teklif reddetme
sales.quote.discount.apply	İndirim uygulama
sales.quote.discount.approve	İndirim onaylama
sales.quote.pdf	PDF üretme/görme
sales.quote.export	Teklif dışa aktarma
sales.quote.margin.read	Kâr marjı görme

Rol Matrisi

Permission	tenant_admin	sales_director	sales_manager	sales_rep	finance_user	dealer_user	customer_portal_user	auditor
sales.quote.read	tenant_all	tenant_all	team	owned	read	dealer	own_quotes	tenant_all
sales.quote.create	✓	✓	✓	✓				
sales.quote.update	✓	✓	team	owned_draft				
sales.quote.delete	✓							
sales.quote.send	✓	✓	team	owned				
sales.quote.approve	✓	✓	✓					
sales.quote.reject	✓	✓	✓					
sales.quote.discount.apply	✓	✓	✓	limited				
sales.quote.discount.approve	✓	✓	✓					
sales.quote.pdf	✓	✓	team	owned	read	dealer	own_quotes	✓
sales.quote.margin.read	✓	✓	✓		✓			✓

11. ORDER Permissions

Permission	Açıklama
order.order.read	Sipariş görüntüleme
order.order.create	Sipariş oluşturma
order.order.update	Sipariş güncelleme
order.order.cancel	Sipariş iptal
order.order.confirm	Sipariş onay
order.order.ship	Sevk işlemi
order.order.deliver	Teslimat
order.order.return	İade

Rol Matrisi

Permission	tenant_admin	sales_manager	sales_rep	warehouse_manager	warehouse_user	finance_user	dealer_user	customer_portal_user
order.order.read	tenant_all	team	owned	tenant_all	branch	read	dealer	own_order_s
order.order.create	✓	✓	✓				✓	
order.order.update	✓	team	owned				dealer_draft	
order.order.cancel	✓	✓					dealer_draft	
order.order.confirm	✓	✓						
order.order.ship				✓	✓			
order.order.deliver				✓	✓			
order.order.return	✓	✓		✓		✓		

12. INVENTORY Permissions

Permission	Açıklama
inventory.product.read	Ürün görüntüleme
inventory.product.create	Ürün oluşturma
inventory.product.update	Ürün güncelleme
inventory.product.delete	Ürün silme
inventory.stock.read	Stok görüntüleme
inventory.stock.adjust	Stok düzeltme
inventory.stock.transfer	Depolar arası transfer
inventory.stock.reserve	Stok rezervasyon
inventory.stock.count	Stok sayım

Rol Matrisi

Permission	tenant_admin	warehouse_manager	warehouse_user	sales_manager	sales_rep	dealer_user	auditor
inventory.product.read	✓	✓	✓	✓	✓	dealer_catalog	✓
inventory.product.create	✓	✓					
inventory.product.update	✓	✓					
inventory.product.delete	✓						
inventory.stock.read	✓	✓	branch	✓	available_only	available_only	✓
inventory.stock.adjust	✓	✓					
inventory.stock.transfer	✓	✓	✓				
inventory.stock.reserve	✓	✓	✓	✓	✓		
inventory.stock.count	✓	✓	✓				

13. FINANCE Permissions

Permission	Açıklama
finance.account.read	Cari hesap görüntüleme
finance.account.update	Cari güncelleme

finance.balance.read	Bakiye görüntüleme
finance.transaction.read	Cari hareket okuma
finance.payment.read	Tahsilat görüntüleme
finance.payment.create	Tahsilat oluşturma
finance.payment.cancel	Tahsilat iptal
finance.invoice.read	Fatura görüntüleme
finance.invoice.create	Fatura oluşturma
finance.credit_limit.update	Kredi limiti güncelleme
finance.report.read	Finans raporları

Rol Matrisi

Permission	tenant_admin	finance_manager	finance_user	sales_director	sales_manager	sales_rep	dealer_user	customer_portal_user	auditor
finance.account.read	✓	✓	✓	✓	team	owned_summary	dealer	own	✓
finance.balance.read	✓	✓	✓	✓	team	owned_summary	dealer	own	✓
finance.transaction.read	✓	✓	✓	✓	team		dealer	own	✓
finance.payment.read	✓	✓	✓	✓	team		dealer	own	✓
finance.payment.create	✓	✓	✓						
finance.payment.cancel	✓	✓							
finance.invoice.read	✓	✓	✓	✓	team		dealer	own	✓
finance.invoice.create	✓	✓	✓						
finance.credit_limit.update	✓	✓							

finance.re	✓	✓	✓	✓		✓
port.read						

14. COMMUNICATION Permissions

Permission	Açıklama
comm.inbox.read	Inbox görüntüleme
comm.conversation.read	Konuşma görüntüleme
comm.conversation.assign	Konuşma atama
comm.message.send	Mesaj gönderme
comm.message.delete	Mesaj silme
comm.channel.manage	Kanal yönetimi
comm.call.read	Çağrı görüntüleme
comm.call.recording.read	Çağrı kaydı dinleme

Rol Matrisi

Permission	tenant_a dmin	sales_ma nager	sales_ rep	support_m anager	support_ agent	audi tor
comm.inbox.read	✓	team	owned	✓	assigned	✓
comm.conversati on.read	✓	team	owned	✓	assigned	✓
comm.conversati on.assign	✓	✓		✓		
comm.message.s end	✓	✓	✓	✓	✓	
comm.message. delete	✓			✓		
comm.channel.m anage	✓			✓		
comm.call.read	✓	team	owned	✓	assigned	✓
comm.call.record ing.read	✓	✓		✓		✓

15. SUPPORT Permissions

Permission	Açıklama
support.ticket.read	Ticket görüntüleme
support.ticket.create	Ticket oluşturma
support.ticket.update	Ticket güncelleme
support.ticket.assign	Ticket atama

support.ticket.resolve	Ticket çözme
support.ticket.close	Ticket kapatma
support.sla.manage	SLA yönetimi
support.kb.manage	Bilgi bankası yönetimi

Rol Matrisi

Permission	tenant_admin	support_manager	support_agent	sales_rep	customer_portal_user	auditor
support.ticket.read	✓	✓	assigned	owned_customer	own	✓
support.ticket.create	✓	✓	✓	✓	✓	
support.ticket.update	✓	✓	assigned		own_open	
support.ticket.assign	✓	✓				
support.ticket.resolve	✓	✓	assigned			
support.ticket.close	✓	✓	assigned			
support.sla.manage	✓	✓				
support.kb.manage	✓	✓				

16. WORKFLOW / AUTOMATION Permissions

Permission	Açıklama
workflow.workflow.read	Workflow görüntüleme
workflow.workflow.create	Workflow oluşturma
workflow.workflow.update	Workflow güncelleme
workflow.workflow.delete	Workflow silme
workflow.workflow.activate	Workflow aktifleştirme
workflow.workflow.pause	Workflow durdurma
workflow.run.read	Workflow çalıştırma logları
workflow.approval.read	Onayları görüntüleme
workflow.approval.action	Onaylama/reddetme

Rol Matrisi

Permission	tenant_admin	sales_director	sales_manager	finance_manager	support_manager	auditor
------------	--------------	----------------	---------------	-----------------	-----------------	---------

workflow.workflow.read	✓	✓		✓	✓	✓
workflow.workflow.create	✓					
workflow.workflow.update	✓					
workflow.workflow.delete	✓					
workflow.workflow.activate	✓					
workflow.workflow.pause	✓					
workflow.run.read	✓	✓		✓	✓	✓
workflow.approval.read	✓	✓	✓	✓	✓	✓
workflow.approval.action	✓	✓	✓	✓	✓	

17. AI Permissions

Permission	Açıklama
ai.assistant.use	AI asistan kullanma
ai.summary.generate	Özet üretme
ai.quote.generate	Teklif metni üretme
ai.lead_score.read	AI lead skoru görme
ai.prediction.read	Tahminleri görme
ai.agent.manage	AI agent yönetimi
ai.usage.read	AI kullanım raporu
ai.settings.manage	AI sağlayıcı ayarları

Rol Matrisi

Permission	tenant_admin	sales_director	sales_manager	sales_rep	support_agent	finance_user	auditor
ai.assistant.use	✓	✓	✓	✓	✓	✓	
ai.summary.generate	✓	✓	✓	✓	✓		
ai.quote.generate	✓	✓	✓	✓			

ai.lead_score.read	✓	✓	✓	✓		✓
ai.prediction.read	✓	✓	✓		✓	✓
ai.agent.manage	✓					
ai.usage.read	✓	✓				✓
ai.settings.manage	✓					

18. REPORTING Permissions

Permission	Açıklama
reporting.dashboard.read	Dashboard görüntüleme
reporting.dashboard.create	Dashboard oluşturma
reporting.dashboard.update	Dashboard güncelleme
reporting.report.read	Rapor görüntüleme
reporting.report.create	Rapor oluşturma
reporting.report.export	Rapor dışa aktarma
reporting.executive.read	Yönetici dashboard

Rol Matrisi

Permission	tenant_admin	sales_director	sales_manager	finance_manager	warehouse_manager	support_manager	auditor
reporting.dashboard.read	✓	✓	team	✓	✓	✓	✓
reporting.dashboard.create	✓	✓		✓			
reporting.dashboard.update	✓	✓		✓			
reporting.report.read	✓	✓	team	✓	✓	✓	✓
reporting.report.create	✓	✓		✓			
reporting.report.export	✓	✓		✓			✓

reporting.exe	✓	✓	✓	✓
cutive.read				

19. INTEGRATION Permissions

Permission	Açıklama
integration.api_key.read	API key görüntüleme
integration.api_key.create	API key oluşturma
integration.api_key.revoke	API key iptal
integration.webhook.read	Webhook görüntüleme
integration.webhook.create	Webhook oluşturma
integration.webhook.update	Webhook güncelleme
integration.webhook.delete	Webhook silme
integration.connector.manage	Connector yönetimi
integration.log.read	Entegrasyon logları

Rol Matrisi

Permission	tenant_admin	integration_admin	auditor
integration.api_key.read	✓	✓	✓
integration.api_key.create	✓	✓	
integration.api_key.revoke	✓	✓	
integration.webhook.read	✓	✓	✓
integration.webhook.create	✓	✓	
integration.webhook.update	✓	✓	
integration.webhook.delete	✓	✓	
integration.connector.manage	✓	✓	
integration.log.read	✓	✓	✓

20. Field-Level Permissions

Customer

Field	Permission
tax_number	crm.customer.tax_number.read
credit_limit	finance.credit_limit.read
risk_score	crm.customer.risk.read
lifetime_value	crm.customer.ltv.read

Quote

Field	Permission
cost_amount	sales.quote.cost.read
profit_amount	sales.quote.margin.read
margin_percent	sales.quote.margin.read
discount_amount	sales.quote.discount.read

Finance

Field	Permission
account.balance	finance.balance.read
account_transactions	finance.transaction.read
bank_transactions	finance.bank.read
payment.amount	finance.payment.read

AI

Field	Permission
ai_raw_prompt	ai.debug.read
ai_cost_amount	ai.usage.read
ai_provider_response	ai.debug.read

21. Portal Permissions

Customer Portal

portal.quote.read_own
portal.quote.approve_own
portal.order.read_own
portal.payment_plan.read_own
portal.ticket.create_own
portal.ticket.read_own
portal.file.download_own

Dealer Portal

dealer.catalog.read
dealer.price.read
dealer.order.create
dealer.order.read
dealer.balance.read
dealer.invoice.read
dealer.support.create

22. Sensitive Action Re-Auth

Şu işlemler yeniden şifre/2FA ister:

- role permission update
- api key create
- api key revoke
- customer export
- finance export
- payment cancel
- stock adjust
- tenant settings update
- integration secret update
- delete customer
- delete quote

23. Approval Required Actions

- quote discount above threshold
- credit limit increase
- payment cancel
- stock adjustment above threshold
- purchase order above threshold
- customer delete
- bulk export
- workflow activation
- ai agent publish

24. Permission Cache

Cache key:

permissions:{tenantId}:{userId}

Invalidation eventleri:

- UserAssignedToRole
- UserRemovedFromRole

PermissionAssignedToRole
PermissionRevokedFromRole
RoleUpdated
UserDeactivated

25. Backend Guard Standard

Her endpoint:

JwtAuthGuard
TenantGuard
PermissionGuard
ScopeGuard

Örnek:

```
@RequirePermission('sales.quote.approve')  
@Post('/:id/approve')  
approveQuote() {}
```

26. UI Permission Standard

```
<Can permission="sales.quote.approve">  
  <ApproveQuoteButton />  
</Can>
```

Ama kural:

UI gizleme güvenlik değildir.
Backend kontrol zorunludur.

27. MVP Permission Set

İlk sürüm için minimum izinler:

iam.user.read
iam.user.create
iam.role.read

iam.role.update

crm.customer.read
crm.customer.create
crm.customer.update
crm.customer.delete

lead.lead.read
lead.lead.create
lead.lead.update
lead.lead.assign
lead.lead.convert

sales.pipeline.read
sales.opportunity.read
sales.opportunity.create
sales.opportunity.update
sales.opportunity.move_stage

sales.quote.read
sales.quote.create
sales.quote.update
sales.quote.send
sales.quote.approve
sales.quote.pdf

task.task.read
task.task.create
task.task.update
task.task.complete

reporting.dashboard.read
core.audit.read

28. Go-Live Permission Checklist

Default roles seeded
Default permissions seeded
Role-permission matrix loaded
Tenant admin exists

Super admin separated
Portal roles isolated
Dealer roles isolated
Permission tests passed
Scope tests passed
Field-level masking tested
Audit for permission changes active

Sonraki çıktı: **CRM OS MVP Build Package v1.**

CRM OS MVP Build Package v1

1. MVP Amacı

Bu paket, CRM OS'in ilk çalışan ürün çekirdeğini üretmek için hazırlanmış **uygulama başlangıç paketidir.**

MVP'nin amacı:

Tenant
→ User
→ Customer
→ Lead
→ Opportunity
→ Quote
→ Task
→ Dashboard

akışını uçtan uca çalıştırmak.

2. MVP Kapsamı

Dahil Modüller

CORE
IAM
CRM
LEAD
SALES
QUOTE

TASK
ACTIVITY
NOTIFICATION
AUDIT
DASHBOARD

Dahil Değil

Order
Inventory
Finance
Omnichannel
Support
Workflow Builder
AI Copilot
Dealer Portal
Mobile App
Marketplace
Low-Code

Bunlar MVP sonrası fazlara alınacak.

3. MVP Uçtan Uca Akış

1. Tenant oluşturulur.
2. Admin kullanıcı giriş yapar.
3. Kullanıcı / rol / yetki tanımlanır.
4. Müşteri oluşturulur.
5. Lead oluşturulur.
6. Lead fırsata çevrilir.
7. Pipeline'da fırsat aşama değiştirir.
8. Teklif oluşturulur.
9. Teklife ürün/hizmet kalemi eklenir.
10. Teklif PDF üretilir.
11. Görev oluşturulur.
12. Dashboard metrikleri güncellenir.
13. Audit log oluşur.
14. Timeline event oluşur.

4. MVP Database Scope

Migration Set

001_core
002_iam
003_crm
004_lead
005_sales
006_quote
007_task_activity
008_common
009_notification
010_audit_event

MVP Tabloları

tenants
tenant_settings
plans

branches
departments
users
roles
permissions
user_roles
role_permissions
teams
team_members

customer_segments
customers
customer_contacts
customer_addresses

lead_sources
leads
lead_conversion_logs

pipelines
pipeline_stages
opportunities
opportunity_products

product_categories
product_brands
products
product_variants

quotes
quote_items
quote_approvals

tasks
activities

files
comments
tags
entity_tags
timeline_events
domain_events
audit_logs
notifications

5. MVP API Scope

Auth

POST /api/v1/auth/login
POST /api/v1/auth/refresh
POST /api/v1/auth/logout

Users / Roles

GET /api/v1/users
POST /api/v1/users
GET /api/v1/users/{id}
PATCH /api/v1/users/{id}
DELETE /api/v1/users/{id}

GET /api/v1/roles
POST /api/v1/roles
PATCH /api/v1/roles/{id}

GET /api/v1/permissions

Customers

GET /api/v1/customers

POST /api/v1/customers

GET /api/v1/customers/{id}

PATCH /api/v1/customers/{id}

DELETE /api/v1/customers/{id}

GET /api/v1/customers/{id}/contacts

POST /api/v1/customers/{id}/contacts

GET /api/v1/customers/{id}/timeline

Leads

GET /api/v1/leads

POST /api/v1/leads

GET /api/v1/leads/{id}

PATCH /api/v1/leads/{id}

POST /api/v1/leads/{id}/convert

Pipeline / Opportunity

GET /api/v1/pipelines

POST /api/v1/pipelines

GET /api/v1/pipelines/{id}/stages

POST /api/v1/pipelines/{id}/stages

GET /api/v1/opportunities

POST /api/v1/opportunities

GET /api/v1/opportunities/{id}

PATCH /api/v1/opportunities/{id}

POST /api/v1/opportunities/{id}/move-stage

POST /api/v1/opportunities/{id}/win

POST /api/v1/opportunities/{id}/lose

Quotes

GET /api/v1/quotes

POST /api/v1/quotes

GET /api/v1/quotes/{id}
PATCH /api/v1/quotes/{id}

POST /api/v1/quotes/{id}/items
PATCH /api/v1/quotes/{id}/items/{itemId}
DELETE /api/v1/quotes/{id}/items/{itemId}

POST /api/v1/quotes/{id}/send
POST /api/v1/quotes/{id}/approve
POST /api/v1/quotes/{id}/reject
GET /api/v1/quotes/{id}/pdf

Tasks / Activities

GET /api/v1/tasks
POST /api/v1/tasks
PATCH /api/v1/tasks/{id}
POST /api/v1/tasks/{id}/complete

GET /api/v1/activities
POST /api/v1/activities

Notifications / Audit

GET /api/v1/notifications
POST /api/v1/notifications/{id}/read

GET /api/v1/audit-logs

6. MVP Frontend Scope

Ekranlar

/login

/my-work

/dashboard

/customers

/customers/new

/customers/[id]

/leads
/leads/new
/leads/[id]

/pipeline
/opportunities/[id]

/quotes
/quotes/new
/quotes/[id]
/quotes/[id]/preview

/tasks

/settings/users
/settings/roles
/settings/permissions

Ana Bileşenler

AppShell
Sidebar
Topbar
CommandMenu
NotificationCenter

DataGrid
Form
Drawer
Modal
Tabs
Timeline
Kanban
MetricCard
QuoteItemTable
PricingPanel
PdfPreview

7. MVP Backend Module Scope

apps/api
apps/worker
apps/scheduler

modules/core
modules/iam
modules/crm
modules/lead
modules/sales
modules/quote
modules/task
modules/notification
modules/reporting

Worker Jobs

domain-events-outbox
quote-pdf-generation
notification-delivery
email-placeholder

8. MVP Events

TenantCreated
UserInvited
UserLoggedIn

CustomerCreated
CustomerUpdated

LeadCreated
LeadAssigned
LeadConverted

OpportunityCreated
OpportunityStageChanged
OpportunityWon
OpportunityLost

QuoteCreated
QuoteItemAdded
QuoteCalculated
QuoteSent
QuoteApproved
QuoteRejected
QuotePdfGenerated

TaskCreated
TaskCompleted

NotificationCreated
AuditLogCreated

9. MVP Permission Set

iam.user.read
iam.user.create
iam.user.update
iam.role.read
iam.role.update
iam.permission.read

crm.customer.read
crm.customer.create
crm.customer.update
crm.customer.delete
crm.contact.manage
crm.timeline.read

lead.lead.read
lead.lead.create
lead.lead.update
lead.lead.assign
lead.lead.convert

sales.pipeline.read
sales.pipeline.manage
sales.opportunity.read
sales.opportunity.create

sales.opportunity.update
sales.opportunity.move_stage
sales.opportunity.win
sales.opportunity.lose

sales.quote.read
sales.quote.create
sales.quote.update
sales.quote.send
sales.quote.approve
sales.quote.reject
sales.quote.pdf

task.task.read
task.task.create
task.task.update
task.task.complete

reporting.dashboard.read
core.audit.read

10. MVP Default Roles

tenant_admin

Tüm MVP izinleri.

sales_manager

crm.customer.read
crm.customer.create
crm.customer.update
crm.contact.manage
crm.timeline.read

lead.lead.read
lead.lead.create
lead.lead.update
lead.lead.assign
lead.lead.convert

sales.pipeline.read

sales.opportunity.read
sales.opportunity.create
sales.opportunity.update
sales.opportunity.move_stage
sales.opportunity.win
sales.opportunity.lose

sales.quote.read
sales.quote.create
sales.quote.update
sales.quote.send
sales.quote.approve
sales.quote.reject
sales.quote.pdf

task.task.read
task.task.create
task.task.update
task.task.complete

reporting.dashboard.read

[sales_rep](#)

crm.customer.read
crm.customer.create
crm.customer.update
crm.contact.manage
crm.timeline.read

lead.lead.read
lead.lead.create
lead.lead.update
lead.lead.convert

sales.pipeline.read
sales.opportunity.read
sales.opportunity.create
sales.opportunity.update
sales.opportunity.move_stage

sales.quote.read

sales.quote.create
sales.quote.update
sales.quote.send
sales.quote.pdf

task.task.read
task.task.create
task.task.update
task.task.complete

reporting.dashboard.read

auditor

crm.customer.read
crm.timeline.read
lead.lead.read
sales.pipeline.read
sales.opportunity.read
sales.quote.read
core.audit.read
reporting.dashboard.read

11. MVP Sprint Build Plan

Sprint 1 – Foundation

Repo setup
Docker Compose
PostgreSQL
Redis
RabbitMQ
MinIO
NestJS API skeleton
Next.js Web skeleton
CI pipeline
Design tokens
App shell

Sprint 2 – Auth / Tenant / IAM

- Tenant migration
- User migration
- Role migration
- Permission migration
- Login API
- Refresh API
- Permission guard
- Tenant context
- RLS proof
- Login UI
- User list shell

Sprint 3 – Customer

- Customer migration
- Contact migration
- Address migration
- Customer CRUD API
- Customer list UI
- Customer create form
- Customer detail shell
- Customer contacts

Sprint 4 – Customer 360 / Common

- Timeline events
- Files metadata
- Comments
- Tags
- Customer timeline UI
- Customer KPI strip
- Audit base
- Domain event base

Sprint 5 – Lead

- Lead migration
- Lead source migration
- Lead CRUD API
- Lead list UI
- Lead detail UI

Lead assignment

Sprint 6 – Lead Conversion / Pipeline Setup

Lead conversion API
Conversion logs
Pipeline migration
Stage migration
Default pipeline seed
Pipeline settings UI
Lead convert modal

Sprint 7 – Opportunity

Opportunity migration
Opportunity product migration
Opportunity CRUD API
Opportunity drawer UI
Opportunity detail

Sprint 8 – Pipeline Kanban

Stage move API
Win/Loss API
Stage history
Pipeline kanban UI
Drag-drop
Deal cards
Deal inspector

Sprint 9 – Quote Core

Quote migration
Quote item migration
Quote CRUD API
Quote item APIs
Quote builder UI
Pricing panel
Quote item table

Sprint 10 – Quote PDF / Approval

Quote calculation service
Quote approval API

Quote send API
PDF generation worker
PDF preview
Approval modal

Sprint 11 – Task / Activity

Task migration
Activity migration
Task CRUD API
Task complete API
Activity API
My Work UI
Task modal
Activity composer

Sprint 12 – Dashboard / Notification / Hardening

Notification migration
Notification API
Dashboard metrics API
Dashboard UI
Notification center
Command menu
MVP E2E
Tenant isolation tests
Permission tests
Regression

12. MVP Acceptance Criteria

MVP kabul için aşağıdaki senaryo eksiksiz çalışmalı:

1. Tenant oluşturulur.
2. Admin giriş yapar.
3. Kullanıcı oluşturur.
4. Rol atar.
5. Müşteri oluşturur.
6. Müşteri detayında timeline görür.
7. Lead oluşturur.
8. Lead'i fırsata çevirir.
9. Pipeline'da fırsatı taşır.

10. Fırsatı kazanıldı/kaybedildi yapar.
11. Teklif oluşturur.
12. Teklife kalem ekler.
13. Teklif toplamı backend'de hesaplanır.
14. Teklif PDF üretir.
15. Teklifi gönderildi yapar.
16. Görev oluşturur.
17. Görevi tamamlar.
18. Dashboard metrikleri görünür.
19. Notification oluşur.
20. Audit log oluşur.
21. Tenant izolasyonu testten geçer.
22. Yetkisiz kullanıcı 403 alır.

13. MVP Non-Functional Requirements

Performans

Customer list p95 < 300ms
Customer detail p95 < 400ms
Pipeline board p95 < 500ms
Quote detail p95 < 400ms
Dashboard p95 < 800ms

Güvenlik

JWT auth
Refresh token
Tenant header
PostgreSQL RLS
RBAC
Audit log
Rate limit
CORS allowlist
Input validation

Kullanılabilirlik

Loading state
Empty state
Error state
Toast feedback

Keyboard accessible
Dark theme
Responsive desktop

14. MVP Test Suite

Backend

Auth tests
Tenant context tests
RLS tests
Permission guard tests
Customer CRUD tests
Lead conversion tests
Opportunity stage move tests
Quote calculation tests
Quote approval tests
Task tests
Audit tests

Frontend

Login E2E
Customer create E2E
Lead convert E2E
Pipeline drag-drop E2E
Quote builder E2E
Permission UI tests
Dashboard smoke

Security

Tenant A cannot read Tenant B customer
Tenant A cannot read Tenant B quote
Sales rep cannot approve quote
Anonymous cannot access API
Deleted customer not visible in list

15. Seed Data

Default Tenant

Demo Tenant

Default Users

admin@demo.local

manager@demo.local

rep@demo.local

auditor@demo.local

Default Roles

tenant_admin

sales_manager

sales_rep

auditor

Default Pipeline

New

Discovery

Proposal

Negotiation

Won

Lost

Default Lead Sources

Web Form

WhatsApp

Instagram

Referral

Manual

16. MVP Deployment Package

Servisler

api

web

worker
scheduler
postgres
redis
rabbitmq
minio

Minimum Production

API: 3 replicas
Web: 2 replicas
Worker: 2 replicas
Scheduler: 1 replica
PostgreSQL: primary + backup
Redis: managed or HA
RabbitMQ: managed or 3-node
Object Storage: S3/MinIO

17. MVP Go-Live Checklist

Migrations applied
Default permissions seeded
Default roles seeded
Default pipeline seeded
Admin user created
RLS enabled
Tenant isolation tests passed
Permission tests passed
Audit logs active
PDF worker active
Notification worker active
Backups active
Monitoring active
Smoke tests passed

18. MVP Deliverables

Backend source
Frontend source

- Database migrations
- Seed scripts
- OpenAPI spec
- Design system v1
- Docker Compose
- Helm chart baseline
- CI/CD pipeline
- Test suite
- MVP user guide
- Admin guide
- Release notes

19. MVP Team Assignment

Core/IAM

- Tenant
- Auth
- Users
- Roles
- Permissions
- Audit

CRM/Lead

- Customer
- Contact
- Address
- Timeline
- Lead
- Lead Conversion

Sales/Quote

- Pipeline
- Opportunity
- Quote
- Quote PDF
- Approval

Frontend Platform

App Shell
DataGrid
Forms
Drawer
Kanban
Timeline
Command Menu

DevOps

Docker
CI/CD
Staging
Monitoring
Backup
Release

QA

Test automation
E2E
Tenant isolation
Permission tests
Regression

20. MVP Final Definition

MVP, aşağıdaki zincir canlıda çalışınca tamamdır:

Login
→ Customer
→ Lead
→ Opportunity
→ Quote
→ Task
→ Dashboard

Bu paket tamamlandıktan sonra sıradaki paket:

CRM OS Commercial Build Package v1

olmalıdır.

CRM OS Enterprise Build Package v1

1. Enterprise Package Amacı

Bu paket, Commercial sürümde çalışan zinciri kurumsal seviyeye taşır:

CRM

- Quote
- Order
- Inventory
- Finance
- Portal

Enterprise sürümde buna şunlar eklenir:

Omnichannel

Support / Ticket

SLA

Dealer Portal

Approval Engine

Public API

Webhooks

Advanced Audit

Branch / Dealer Scoped Access

Enterprise Security

2. Enterprise Kapsam

Dahil Modüller

COMMUNICATION

SUPPORT

DEALER PORTAL

APPROVAL ENGINE

PUBLIC API

WEBHOOKS
INTEGRATION LOGS
ADVANCED AUDIT
BRANCH SCOPE
DEALER SCOPE
SSO PREPARATION
ADVANCED REPORTING

Dahil Değil

Full AI Copilot
Marketplace
Low-Code Builder
Full Mobile Offline
Advanced CDP
Full Accounting

3. Ana Enterprise Akışları

Omnichannel → Ticket

Message Received
→ Conversation Created
→ Assigned to Agent
→ Ticket Created
→ SLA Started
→ Ticket Resolved
→ CSAT Sent

Dealer Portal

Dealer Login
→ Catalog View
→ Dealer Price
→ Stock Availability
→ Dealer Order
→ Balance View

Approval Engine

Discount Request
→ Approval Request

- Manager Review
- Approve / Reject
- Quote Updated
- Audit Log

Public API / Webhook

External App

- API Key Auth
- API Request
- Event Created
- Webhook Delivered
- Delivery Log

4. Enterprise Database Scope

Yeni Migration Set

019_communication
020_support_ticket
021_sla
022_dealer_portal
023_approval_engine
024_public_api
025_webhooks
026_integration_logs
027_branch_dealer_scope
028_enterprise_audit
029_sso_preparation
030_reporting_enterprise

Yeni Tablolar

channels

channel_accounts

channel_settings

conversations

conversation_participants

conversation_assignments

conversation_tags

conversation_status_history

messages

message_attachments

message_delivery_logs

message_read_receipts

email_logs

sms_logs

whatsapp_logs

call_logs

meeting_logs

tickets

ticket_categories

ticket_priorities

ticket_statuses

ticket_messages

ticket_files

ticket_assignments

ticket_sla_logs

ticket_satisfaction_surveys

ticket_resolution_notes

sla_policies

sla_policy_rules

sla_timers

dealer_portal_users

dealer_catalogs

dealer_product_visibility

dealer_price_rules

dealer_orders

dealer_order_items

dealer_balances

dealer_targets

dealer_commissions

approval_requests

approval_steps

approval_actions

approval_delegations

approval_logs

api_keys

api_key_scopes

api_usage_logs

webhooks

webhook_events

webhook_logs

webhook_delivery_attempts

connectors

connector_accounts

integration_runs

integration_errors

branch_access_rules

dealer_access_rules

record_access_overrides

sso_providers

sso_sessions

saml_configs

oidc_configs

audit_log_exports

security_events

5. Enterprise API Scope

Communication

GET /api/v1/channels

POST /api/v1/channels

GET /api/v1/channel-accounts

POST /api/v1/channel-accounts

GET /api/v1/conversations

POST /api/v1/conversations

GET /api/v1/conversations/{id}

PATCH /api/v1/conversations/{id}

POST /api/v1/conversations/{id}/assign
POST /api/v1/conversations/{id}/resolve

GET /api/v1/conversations/{id}/messages
POST /api/v1/conversations/{id}/messages
POST /api/v1/messages/{id}/read

Support

GET /api/v1/tickets
POST /api/v1/tickets
GET /api/v1/tickets/{id}
PATCH /api/v1/tickets/{id}
POST /api/v1/tickets/{id}/assign
POST /api/v1/tickets/{id}/resolve
POST /api/v1/tickets/{id}/close
POST /api/v1/tickets/{id}/reopen

GET /api/v1/tickets/{id}/messages
POST /api/v1/tickets/{id}/messages

GET /api/v1/sla-policies
POST /api/v1/sla-policies
PATCH /api/v1/sla-policies/{id}

Dealer Portal

POST /api/v1/dealer-portal/auth/login
GET /api/v1/dealer-portal/me

GET /api/v1/dealer-portal/catalog
GET /api/v1/dealer-portal/products/{id}
GET /api/v1/dealer-portal/stocks
GET /api/v1/dealer-portal/prices

GET /api/v1/dealer-portal/orders
POST /api/v1/dealer-portal/orders
GET /api/v1/dealer-portal/orders/{id}

GET /api/v1/dealer-portal/account
GET /api/v1/dealer-portal/balance

Approval Engine

GET /api/v1/approval-requests
POST /api/v1/approval-requests
GET /api/v1/approval-requests/{id}
POST /api/v1/approval-requests/{id}/approve
POST /api/v1/approval-requests/{id}/reject
POST /api/v1/approval-requests/{id}/delegate

Public API

GET /api/v1/public/customers
POST /api/v1/public/customers

GET /api/v1/public/products
GET /api/v1/public/stocks

GET /api/v1/public/orders
POST /api/v1/public/orders

GET /api/v1/public/quotes
POST /api/v1/public/quotes

Webhooks

GET /api/v1/webhooks
POST /api/v1/webhooks
GET /api/v1/webhooks/{id}
PATCH /api/v1/webhooks/{id}
DELETE /api/v1/webhooks/{id}

GET /api/v1/webhook-logs
POST /api/v1/webhooks/{id}/test

SSO Preparation

GET /api/v1/sso/providers
POST /api/v1/sso/providers
PATCH /api/v1/sso/providers/{id}
POST /api/v1/sso/providers/{id}/test

6. Enterprise Frontend Scope

Yeni Ekranlar

/inbox

/inbox/conversations/[id]

/tickets

/tickets/[id]

/tickets/settings/sla

/dealer-portal/login

/dealer-portal/catalog

/dealer-portal/products/[id]

/dealer-portal/orders

/dealer-portal/orders/[id]

/dealer-portal/account

/approvals

/approvals/[id]

/settings/api-keys

/settings/webhooks

/settings/integrations

/settings/sso

/settings/security

/settings/audit

/reports/enterprise

/reports/sla

/reports/dealer

/reports/api-usage

Yeni Bileşenler

InboxShell

ConversationList

ChatThread

MessageComposer

CustomerContextPanel

ConversationAssignmentPanel

TicketList

TicketDetail

SlaTimer
SlaPolicyBuilder
CsatSurveyPanel

DealerPortalShell
DealerCatalogGrid
DealerPriceBadge
DealerStockAvailability
DealerOrderBuilder
DealerBalanceCard

ApprovalInbox
ApprovalRequestDetail
ApprovalTimeline
ApprovalDecisionModal

ApiKeyManager
WebhookManager
WebhookDeliveryLogs
IntegrationHealthPanel
SsoProviderForm

EnterpriseAuditTable
SecurityEventsTable

7. Backend Module Scope

communication module

channels
channel_accounts
conversations
messages
assignments
delivery_logs
read_receipts

support module

tickets
ticket_messages

ticket_files
ticket_assignment
ticket_status
ticket_sla
csat
knowledge_base_shell

dealer module

dealer_auth
dealer_catalog
dealer_pricing
dealer_orders
dealer_balance
dealer_targets

approval module

approval_requests
approval_steps
approval_actions
approval_rules
delegation

integration module

api_keys
public_api
webhooks
webhook_delivery
connector_shell
integration_logs

security module

sso_preparation
security_events
advanced_audit
ip_rules
session_controls

8. Enterprise Events

ChannelConnected

ChannelDisconnected

ConversationCreated

ConversationAssigned

ConversationStatusChanged

MessageReceived

MessageSent

MessageDelivered

MessageRead

MessageFailed

TicketCreated

TicketAssigned

TicketStatusChanged

TicketResolved

TicketClosed

TicketReopened

TicketSlaBreached

CustomerSatisfactionSubmitted

DealerUserLoggedIn

DealerCatalogViewed

DealerOrderCreated

DealerBalanceViewed

ApprovalRequested

ApprovalStepAssigned

ApprovalApproved

ApprovalRejected

ApprovalDelegated

ApprovalExpired

ApiKeyCreated

ApiKeyRevoked

PublicApiRequestReceived

WebhookCreated

WebhookUpdated

WebhookDeliveryRequested

WebhookDelivered

WebhookFailed

ConnectorConnected

ConnectorDisconnected

IntegrationRunStarted

IntegrationRunCompleted

IntegrationRunFailed

SsoProviderCreated

SsoLoginSucceeded

SsoLoginFailed

SecurityEventDetected

9. Enterprise Permissions

Communication

comm.channel.manage

comm.inbox.read

comm.conversation.read

comm.conversation.assign

comm.message.send

comm.message.delete

comm.call.read

comm.call.recording.read

Support

support.ticket.read

support.ticket.create

support.ticket.update

support.ticket.assign

support.ticket.resolve

support.ticket.close

support.sla.manage

support.csat.read

support.kb.manage

Dealer

dealer.catalog.read
dealer.price.read
dealer.stock.read
dealer.order.create
dealer.order.read
dealer.balance.read
dealer.target.read
dealer.commission.read

Approval

approval.request.read
approval.request.create
approval.request.approve
approval.request.reject
approval.request.delegate
approval.rule.manage

Integration

integration.api_key.read
integration.api_key.create
integration.api_key.revoke
integration.webhook.read
integration.webhook.create
integration.webhook.update
integration.webhook.delete
integration.log.read
integration.connector.manage

Enterprise Security

security.sso.manage
security.ip_rules.manage
security.session.manage
security.event.read
audit.export
audit.advanced_read

10. Enterprise Default Role Updates

tenant_admin

Tüm enterprise izinleri.

support_manager

comm.inbox.read
comm.conversation.read
comm.conversation.assign
comm.message.send

support.ticket.read
support.ticket.create
support.ticket.update
support.ticket.assign
support.ticket.resolve
support.ticket.close
support.sla.manage
support.csat.read

support_agent

comm.inbox.read assigned
comm.conversation.read assigned
comm.message.send

support.ticket.read assigned
support.ticket.create
support.ticket.update assigned
support.ticket.resolve assigned

dealer_admin

dealer.catalog.read
dealer.price.read
dealer.stock.read
dealer.order.create
dealer.order.read
dealer.balance.read
dealer.target.read
dealer.commission.read

integration_admin

integration.api_key.read
integration.api_key.create
integration.api_key.revoke
integration.webhook.read
integration.webhook.create
integration.webhook.update
integration.webhook.delete
integration.log.read
integration.connector.manage

auditor

audit.advanced_read
audit.export
security.event.read
integration.log.read
support.ticket.read
comm.conversation.read readonly

11. Enterprise Sprint Plan

Sprint 21 – Communication Core

COMM-001 channels migration
COMM-002 channel_accounts migration
COMM-003 conversations migration
COMM-004 messages migration
COMM-005 conversation list API
COMM-006 conversation detail API
COMM-007 message send API
COMM-008 Inbox UI shell
COMM-009 Conversation list UI
COMM-010 Chat thread UI

Sprint 22 – Inbox Operations

COMM-011 conversation assignment
COMM-012 conversation status change
COMM-013 message delivery logs
COMM-014 read receipts

COMM-015 customer context panel
COMM-016 AI placeholder panel
COMM-017 inbox permissions
COMM-018 inbox E2E tests

Sprint 23 – Ticket Core

SUP-001 tickets migration
SUP-002 ticket_messages migration
SUP-003 ticket_files migration
SUP-004 ticket CRUD API
SUP-005 ticket assignment API
SUP-006 ticket list UI
SUP-007 ticket detail UI
SUP-008 ticket message composer

Sprint 24 – SLA / Support Management

SUP-009 sla_policies migration
SUP-010 sla timers
SUP-011 SLA breach detection
SUP-012 ticket resolve/close flow
SUP-013 CSAT survey shell
SUP-014 SLA settings UI
SUP-015 support dashboard widgets

Sprint 25 – Dealer Portal Core

DEALER-001 dealer portal users
DEALER-002 dealer catalog visibility
DEALER-003 dealer price rules
DEALER-004 dealer portal auth
DEALER-005 dealer catalog API
DEALER-006 dealer catalog UI
DEALER-007 dealer product detail UI

Sprint 26 – Dealer Orders / Balance

DEALER-008 dealer order create API
DEALER-009 dealer order list/detail
DEALER-010 dealer balance API
DEALER-011 dealer portal order UI
DEALER-012 dealer account UI

DEALER-013 dealer permission tests

Sprint 27 – Approval Engine

APPROVAL-001 approval_requests migration
APPROVAL-002 approval_steps migration
APPROVAL-003 approval_actions migration
APPROVAL-004 approval request API
APPROVAL-005 approve/reject API
APPROVAL-006 approval inbox UI
APPROVAL-007 quote discount approval integration

Sprint 28 – Advanced Approval Rules

APPROVAL-008 approval delegation
APPROVAL-009 approval expiry
APPROVAL-010 approval timeline
APPROVAL-011 credit limit approval integration
APPROVAL-012 stock adjustment approval shell
APPROVAL-013 approval reporting widgets

Sprint 29 – Public API / API Keys

INT-001 api_keys migration
INT-002 api key creation
INT-003 api key hashing
INT-004 scope validation
INT-005 public API auth middleware
INT-006 public customers/products/orders APIs
INT-007 API usage logs
INT-008 API key manager UI

Sprint 30 – Webhooks

INT-009 webhooks migration
INT-010 webhook event subscription
INT-011 webhook HMAC signature
INT-012 webhook delivery worker
INT-013 webhook retry/DLQ
INT-014 webhook logs UI
INT-015 webhook test button

Sprint 31 – Enterprise Security / SSO Prep

- SEC-001 sso_providers migration
- SEC-002 SAML config storage
- SEC-003 OIDC config storage
- SEC-004 SSO provider management UI
- SEC-005 IP allowlist rules
- SEC-006 session management screen
- SEC-007 security events table

Sprint 32 – Enterprise Hardening

- ENT-001 branch scope enforcement
- ENT-002 dealer scope enforcement
- ENT-003 advanced audit filters
- ENT-004 audit export
- ENT-005 enterprise reporting widgets
- ENT-006 permission regression
- ENT-007 security regression
- ENT-008 UAT hardening

12. Communication Business Rules

Conversation

Bir conversation bir customer ile bağlanabilir.
Conversation channel bazlıdır.
Açık conversation aynı kanalda tekrar açılmaz; mevcut conversation güncellenir.
Conversation assigned user veya team alabilir.

Message

Inbound message customer/system kaynaklıdır.
Outbound message user/ai/system kaynaklıdır.
Her message delivery log üretir.
Message tenant ve channel scope dışına çıkamaz.

Assignment

Support manager conversation atayabilir.
Agent sadece kendisine atanmış conversation üzerinde işlem

yapabilir.

13. Ticket Business Rules

Ticket Status

open
pending
waiting_customer
resolved
closed
reopened

SLA

Ticket açılınca SLA timer başlar.
Önceliğe göre first response ve resolution süresi hesaplanır.
SLA breach event üretir.
Resolved olunca timer durur.
Reopened olursa timer devam eder veya yeni timer açılır.

14. Dealer Portal Business Rules

Dealer sadece kendisine açık ürünleri görür.
Dealer sadece bayi fiyatını görür.
Dealer margin/cost göremez.
Dealer kendi siparişlerini görür.
Dealer kendi cari bakiyesini görür.
Dealer başka bayi verisini göremez.

15. Approval Engine Business Rules

Approval Status

pending
approved
rejected
expired

cancelled
delegated

Rules

İndirim eşiği aşılsa approval gerekir.
Credit limit artışı approval gerektirir.
Stock adjustment threshold aşılsa approval gerekir.
Approval tamamlanmadan ilgili işlem finalize edilemez.
Her karar audit log'a düşer.

16. Public API Business Rules

API key hashlenmiş saklanır.
API key sadece bir kez gösterilir.
Scope zorunludur.
Rate limit plan bazlıdır.
Her request api_usage_logs'a yazılır.
Tenant isolation zorunludur.

17. Webhook Business Rules

Her webhook HMAC ile imzalanır.
Timestamp replay protection vardır.
Başarısız delivery retry edilir.
5 başarısız denemeden sonra DLQ.
Webhook payload event schema version içerir.

18. Enterprise Test Suite

Communication

conversation create
message send
message receive
assignment
tenant isolation
agent assigned scope

delivery log

Ticket / SLA

ticket create
ticket assign
ticket resolve
ticket close
SLA start
SLA breach
CSAT created

Dealer

dealer login
dealer sees own catalog
dealer cannot see another dealer price
dealer order create
dealer balance visible

Approval

approval request create
approve
reject
delegation
expired approval
quote discount blocked until approval

Public API

api key auth
scope allowed
scope denied
rate limit
tenant isolation
usage log

Webhook

event subscribed
signature valid
signature invalid rejected
retry

DLQ
delivery log

19. Enterprise Non-Functional Requirements

Performance

Inbox list p95 < 500ms
Chat thread p95 < 500ms
Ticket list p95 < 500ms
Dealer catalog p95 < 600ms
Public API p95 < 400ms
Webhook dispatch p95 < 2s

Reliability

Webhook retry guaranteed
Message delivery log durable
Approval state transition atomic
API key revocation immediate
SLA timers recover after restart

Security

Dealer isolation mandatory
Branch scope mandatory
Public API rate limit mandatory
Webhook signature mandatory
Audit export permission required

20. Enterprise Acceptance Criteria

Enterprise sürüm kabul için:

1. Conversation inbox çalışır.
2. Mesaj gönderme/alma akışı çalışır.
3. Conversation'dan ticket açılır.
4. Ticket SLA takip eder.
5. SLA breach event üretir.
6. Dealer portal giriş yapar.

7. Dealer katalog/fiyat/sipariş görür.
8. Dealer başka bayi verisini göremez.
9. Approval engine quote discount akışında çalışır.
10. API key oluşturulur ve scope uygulanır.
11. Public API çalışır.
12. Webhook HMAC imzalı gönderilir.
13. Retry/DLQ çalışır.
14. Branch/dealer scope testleri geçer.
15. Advanced audit filtrelenebilir.
16. Enterprise security ekranları çalışır.

21. Enterprise Deliverables

Omnichannel inbox
Ticket system
SLA engine
Dealer portal
Approval engine
Public API
API key management
Webhook engine
Integration logs
Branch/dealer scope
Enterprise audit
SSO preparation
Security events
Enterprise reports
Permission updates
E2E tests
Security tests
Updated OpenAPI
Updated docs
Release notes

22. Enterprise Go-Live Checklist

Communication flow tested
Ticket/SLA tested
Dealer isolation tested

Approval engine tested
Public API tested
Webhook signature tested
Webhook retry tested
API rate limit active
Advanced audit active
Branch scope tested
Dealer scope tested
Security regression passed
Performance smoke passed
UAT approved

23. Enterprise Final Definition

Enterprise sürüm şu zincirleri canlıda çalıştırır:

Inbox
→ Ticket
→ SLA
Dealer
→ Catalog
→ Order
→ Balance
Approval
→ Decision
→ Audit
Public API
→ Event
→ Webhook

Sonraki paket:

CRM OS AI & Analytics Build Package v1

CRM OS AI & Analytics Build Package v1

1. Paket Amacı

Bu paket CRM OS'i klasik operasyonel CRM'den **akıllı karar, tahmin, raporlama ve otomasyon platformuna** dönüştürür.

Enterprise sürümde çalışan zincir:

CRM

- Sales
- Order
- Inventory
- Finance
- Support
- Workflow

AI & Analytics sürümde şu yetenekler eklenir:

Ask CRM
AI Assistant
AI Summaries
Lead Scoring
Opportunity Prediction
Quote Assistant
Customer Risk
Advanced Dashboards
Custom Reports
Metric Snapshots
Data Quality
AI Usage Governance

2. Dahil Modüller

AI GATEWAY
AI ASSISTANT
AI AGENTS
AI SUMMARIZATION
AI CLASSIFICATION
AI PREDICTION
AI RECOMMENDATION
AI DOCUMENT READING
ANALYTICS
REPORTING
CUSTOM DASHBOARDS
METRIC SNAPSHOTS
DATA QUALITY

SEARCH / ASK CRM

Dahil değil:

Tam otonom agent işlemleri
Muhasebe kararlarının otomatik verilmesi
Otomatik ödeme iptali
Otomatik stok düzeltme
Yetki değiştiren AI aksiyonları

Bunlar insan onayı gerektirir.

3. Ana AI Akışları

Ask CRM

User question

- Permission check
- Tenant context
- Semantic / SQL intent parse
- Query planning
- Safe query execution
- Answer generation
- Citation to records
- Audit + AI usage log

Örnek:

"Bu ay en çok kâr ettiren ürünler hangileri?"

Cevap:

1. Flora Gold – ₺420.000 kâr
2. Zen Pattern – ₺310.000 kâr
3. Aurora Premium – ₺255.000 kâr

Conversation Summary

MessageReceived

- Conversation context loaded

- PII masking
- AI summary
- Sentiment
- Next action
- Timeline event

Lead Scoring

LeadCreated

- Source + profile + activity data
- AI score
- Temperature
- Assignment recommendation
- Sales rep notification

Opportunity Prediction

OpportunityStageChanged

- Customer history
- Deal value
- Stage velocity
- Activity count
- Win probability
- Risk reason
- Next best action

Quote Assistant

QuoteCreated

- Customer segment
- Product items
- Margin
- Past deals
- AI text generation
- Sales rep review

4. AI Database Scope

Yeni Migration Set

031_ai_gateway
032_ai_agents
033_ai_prompts
034_ai_conversations
035_ai_documents_embeddings
036_ai_predictions
037_ai_recommendations
038_ai_usage_governance
039_analytics_core
040_reporting_builder
041_data_quality

Yeni Tablolar

ai_providers
ai_provider_models
ai_provider_settings

ai_agents
ai_agent_tools
ai_agent_permissions
ai_agent_runs
ai_agent_run_steps

ai_prompts
ai_prompt_versions

ai_conversations
ai_conversation_messages

ai_documents
ai_document_chunks
ai_embeddings

ai_summaries
ai_classifications
ai_predictions
ai_recommendations

ai_usage_logs

ai_usage_quotas

ai_feedback

analytics_datasets

analytics_dataset_fields

analytics_queries

analytics_query_runs

dashboards

dashboard_widgets

dashboard_permissions

dashboard_filters

reports

report_columns

report_filters

report_schedules

report_exports

metrics

metric_snapshots

kpi_targets

data_quality_rules

data_quality_issues

data_quality_runs

saved_searches

semantic_search_logs

5. AI API Scope

AI Assistant

POST /api/v1/ai/ask

GET /api/v1/ai/conversations

POST /api/v1/ai/conversations

GET /api/v1/ai/conversations/{id}

POST /api/v1/ai/conversations/{id}/messages

AI Summaries

POST /api/v1/ai/summarize/conversation/{conversationId}
POST /api/v1/ai/summarize/customer/{customerId}
POST /api/v1/ai/summarize/opportunity/{opportunityId}
POST /api/v1/ai/summarize/ticket/{ticketId}

AI Predictions

POST /api/v1/ai/predict/lead-score/{leadId}
POST /api/v1/ai/predict/opportunity-win/{opportunityId}
POST /api/v1/ai/predict/customer-risk/{customerId}
POST /api/v1/ai/predict/payment-risk/{customerId}

AI Recommendations

GET /api/v1/ai/recommendations
GET /api/v1/ai/recommendations/customer/{customerId}
GET /api/v1/ai/recommendations/opportunity/{opportunityId}
POST /api/v1/ai/recommendations/{id}/dismiss
POST /api/v1/ai/recommendations/{id}/accept

AI Admin

GET /api/v1/ai/providers
POST /api/v1/ai/providers
PATCH /api/v1/ai/providers/{id}

GET /api/v1/ai/agents
POST /api/v1/ai/agents
PATCH /api/v1/ai/agents/{id}

GET /api/v1/ai/usage
GET /api/v1/ai/usage/tenant
GET /api/v1/ai/feedback

6. Analytics / Reporting API Scope

Dashboards

GET /api/v1/dashboards
POST /api/v1/dashboards

GET /api/v1/dashboards/{id}
PATCH /api/v1/dashboards/{id}
DELETE /api/v1/dashboards/{id}

POST /api/v1/dashboards/{id}/widgets
PATCH /api/v1/dashboard-widgets/{id}
DELETE /api/v1/dashboard-widgets/{id}

Reports

GET /api/v1/reports
POST /api/v1/reports
GET /api/v1/reports/{id}
PATCH /api/v1/reports/{id}
POST /api/v1/reports/{id}/run
POST /api/v1/reports/{id}/export

Metrics

GET /api/v1/metrics/sales
GET /api/v1/metrics/pipeline
GET /api/v1/metrics/finance
GET /api/v1/metrics/inventory
GET /api/v1/metrics/support
GET /api/v1/metrics/ai

Data Quality

GET /api/v1/data-quality/issues
POST /api/v1/data-quality/rules
POST /api/v1/data-quality/runs
POST /api/v1/data-quality/issues/{id}/resolve

7. Frontend Scope

Yeni Ekranlar

/ai
/ai/ask
/ai/conversations
/ai/agents
/ai/usage

/ai/settings

/reports

/reports/new

/reports/[id]

/reports/exports

/dashboards

/dashboards/new

/dashboards/[id]

/analytics/executive

/analytics/sales

/analytics/finance

/analytics/inventory

/analytics/support

/analytics/ai

/data-quality

/data-quality/issues

/data-quality/rules

8. Yeni UI Bileşenleri

AskCrmInput

AiAnswerPanel

AiCitationList

AiConfidenceBadge

AiRecommendationCard

AiSummaryBox

AiUsageMeter

AiAgentCard

AiAgentBuilderShell

DashboardBuilder

WidgetLibrary

MetricWidget

ChartWidget

TableWidget

KpiTargetWidget

ReportBuilder
ReportFilterBuilder
ReportColumnSelector
ReportPreviewTable
ReportExportDrawer

DataQualityIssueTable
DuplicateRecordResolver
DataQualityScoreCard

9. Backend Module Scope

ai module

providers
models
gateway
agents
prompts
conversations
summaries
classifications
predictions
recommendations
usage
feedback

analytics module

datasets
query_engine
metrics
snapshots
dashboards
reports
exports

data-quality module

rules

issues

deduplication

validation

cleanup_suggestions

search module

global_search

semantic_search

saved_searches

natural_language_search

10. AI Gateway Design

AI Gateway görevi:

Provider abstraction

Prompt template management

Tenant-aware execution

Permission-aware context loading

PII masking

Token/cost tracking

Rate limiting

Audit logging

Fallback handling

Provider adapter:

OpenAIAdapter

AzureOpenAIAdapter

AnthropicAdapter

LocalModelAdapter

İç arayüz:

```
interface AiProvider {  
    complete(input: AiCompletionInput): Promise<AiCompletionResult>;  
    embed(input: AiEmbeddingInput): Promise<AiEmbeddingResult>;  
}
```

}

11. AI Safety Rules

AI asla doğrudan şu işlemleri yapamaz:

payment.cancel
stock.adjust
quote.approve
order.cancel
customer.delete
permission.update
api_key.create
workflow.activate

Bu aksiyonlar için:

AI önerir

- Kullanıcı inceler
- Approval gerekirse açılır
- İnsan onayı sonrası işlem yapılır

12. Permission Model

AI Permissions

ai.assistant.use
ai.ask_crm.use
ai.summary.generate
ai.recommendation.read
ai.prediction.read
ai.lead_score.read
ai.quote.generate
ai.agent.read
ai.agent.manage
ai.usage.read
ai.settings.manage
ai.debug.read

Analytics Permissions

analytics.dashboard.read
analytics.dashboard.create
analytics.dashboard.update
analytics.dashboard.delete
analytics.report.read
analytics.report.create
analytics.report.update
analytics.report.export
analytics.metric.read
analytics.executive.read
analytics.custom_query.run

Data Quality Permissions

data_quality.issue.read
data_quality.issue.resolve
data_quality.rule.read
data_quality.rule.manage
data_quality.merge.suggest
data_quality.merge.execute

13. AI Events

AIActionRequested
AIActionCompleted
AIActionFailed
AIUsageLimitExceeded

AISummaryGenerated
AIClassificationCompleted
AILeadScoreCalculated
AIOpportunityWinProbabilityCalculated
AICustomerRiskCalculated
AIPaymentRiskCalculated
AIRecommendationCreated
AIRecommendationAccepted
AIRecommendationDismissed

AIAgentCreated

AIAgentRunStarted
AIAgentRunCompleted
AIAgentRunFailed

ReportGenerated
ReportExportRequested
ReportExportCompleted
DashboardWidgetUpdated
MetricSnapshotCreated

DataQualityIssueDetected
DataQualityIssueResolved
DuplicateRecordDetected
DuplicateRecordMerged

14. Ask CRM Business Rules

Kullanıcının görme yetkisi olmayan veri cevaba dahil edilemez.
Cevapta kullanılan kayıtlar citation olarak gösterilir.
Finansal veriler field-level permission'a göre maskelenir.
AI SQL üretirse sadece read-only query çalışır.
DELETE/UPDATE/INSERT çalıştırılmaz.
Query timeout zorunludur.
Sonuç sayısı limitlidir.

Safe SQL policy:

SELECT only
tenant_id enforced
permission filter enforced
limit mandatory
timeout mandatory
no raw arbitrary SQL from user

15. AI Context Loading Rules

Context şu sırayla yüklenir:

Tenant scope
↓
User permission scope
↓
Entity scope
↓
Field-level masking
↓
PII minimization
↓
Prompt context

Context kaynakları:

customers
leads
opportunities
quotes
orders
payments
tickets
messages
activities
timeline_events
products
stocks

16. RAG / Embedding Scope

İlk aşama:

pgvector

Embedding yapılacak içerikler:

product descriptions
knowledge base articles
quote templates
support articles
customer notes
ticket resolutions

uploaded documents

Embedding metadata:

tenant_id
entity_type
entity_id
permission_scope
created_at
source

17. Analytics Model

Analytics iki katmanlı olacak:

Operational Metrics

Canlı uygulama DB'den veya replica'dan:

lead count
pipeline value
quote count
order count
payment total
stock alerts
ticket SLA

Snapshot Metrics

Zaman serisi snapshot:

daily_sales
daily_pipeline_value
daily_collections
daily_stock_value
daily_ticket_sla
daily_ai_usage

Tablo:

metric_snapshots

Alanlar:

tenant_id
metric_key
dimensions
value
snapshot_date
created_at

18. Dashboard Builder Scope

Widget tipleri:

metric
line_chart
bar_chart
pie_chart
table
funnel
kanban_summary
ai_insight
target_progress

Widget config:

```
{  
  "metricKey": "sales.revenue",  
  "dimensions": ["branch", "sales_rep"],  
  "filters": {  
    "dateRange": "this_month"  
  },  
  "visualization": "line_chart"  
}
```

19. Report Builder Scope

Report kaynakları:

customers
leads
opportunities
quotes

orders
payments
products
stocks
tickets
activities

Özellikler:

column select
filters
sorting
grouping
aggregations
scheduled export
CSV/XLSX/PDF export
permission-aware fields

20. Data Quality Scope

Kurallar:

missing email
invalid phone
duplicate customer
duplicate tax number
missing assigned user
stale opportunity
inactive customer with open balance
quote expired but still open

Data Quality Score:

Customer Data Quality
Lead Quality
Sales Data Quality
Finance Data Quality

21. AI Usage Governance

Her AI çağrısı loglanır:

tenant_id
user_id
agent_id
action_type
model
input_tokens
output_tokens
cost_amount
latency_ms
status

Quota:

monthly_token_limit
monthly_cost_limit
per_user_limit
per_agent_limit

Aşımda:

AIUsageLimitExceeded

event'i üretilir.

22. Sprint Plan

Sprint 33 – AI Gateway

AI-001 ai_providers migration
AI-002 ai_models migration
AI-003 AI gateway interface
AI-004 Provider adapter v1
AI-005 Token/cost logging
AI-006 AI usage quota
AI-007 AI settings UI

Sprint 34 – Ask CRM v1

- AI-008 Ask CRM API
- AI-009 Permission-aware context loader
- AI-010 Safe query planner
- AI-011 AI answer generator
- AI-012 citation builder
- AI-013 Ask CRM UI
- AI-014 AI audit logs

Sprint 35 – Summaries / Recommendations

- AI-015 conversation summary
- AI-016 customer summary
- AI-017 opportunity summary
- AI-018 ticket summary
- AI-019 recommendation table
- AI-020 recommendation cards UI

Sprint 36 – Predictions

- AI-021 lead scoring
- AI-022 opportunity win probability
- AI-023 customer risk
- AI-024 payment risk
- AI-025 prediction widgets
- AI-026 prediction tests

Sprint 37 – Analytics Core

- AN-001 analytics datasets
- AN-002 metric definitions
- AN-003 metric snapshots
- AN-004 sales metrics API
- AN-005 finance metrics API
- AN-006 inventory metrics API
- AN-007 support metrics API

Sprint 38 – Dashboard Builder

- AN-008 dashboards migration
- AN-009 dashboard widgets
- AN-010 widget config schema
- AN-011 dashboard builder UI

AN-012 metric widgets
AN-013 chart widgets
AN-014 dashboard permissions

Sprint 39 – Report Builder

AN-015 reports migration
AN-016 report filters
AN-017 report columns
AN-018 report run API
AN-019 export jobs
AN-020 report builder UI
AN-021 report export UI

Sprint 40 – Data Quality

DQ-001 data quality rules
DQ-002 data quality issues
DQ-003 duplicate detection v1
DQ-004 data quality score
DQ-005 issue resolver UI
DQ-006 duplicate merge flow

23. Acceptance Criteria

AI & Analytics paketi kabul için:

1. AI Gateway en az bir provider ile çalışır.
2. Her AI çağrısı usage log yazar.
3. Quota aşımı engellenir.
4. Ask CRM kullanıcı yetkisine göre cevap verir.
5. Ask CRM cevaplarında citation gösterilir.
6. AI summary customer/conversation/ticket için çalışır.
7. Lead scoring sonucu lead üzerinde görünür.
8. Opportunity win probability görünür.
9. Dashboard builder widget ekleyebilir.
10. Report builder filtreli rapor çalıştırır.
11. Export job CSV/XLSX üretir.
12. Data quality issue listesi çalışır.
13. Duplicate customer detection v1 çalışır.
14. AI tenant isolation testleri geçer.

15. AI sensitive action'ları doğrudan yapamaz.

24. Test Suite

AI Tests

provider adapter test
prompt template test
tenant context test
permission context test
PII masking test
quota test
usage log test
AI action blocked test
citation test

Analytics Tests

metric calculation test
snapshot generation test
dashboard widget test
report filter test
report export test
permission-aware report test

Data Quality Tests

duplicate detection
missing field detection
invalid email
invalid phone
merge permission
resolve issue

25. Non-Functional Requirements

AI answer p95 < 8s
Summary p95 < 10s
Metric dashboard p95 < 1s
Report run p95 < 5s for standard reports

Export async for large data
AI usage log durability 100%

26. Final Definition

Bu paket tamamlandığında CRM OS şu seviyeye gelir:

Kullanıcı CRM'e soru sorabilir.
AI müşteri ve görüşmeleri özetleyebilir.
Lead/fırsat risk ve skor üretilebilir.
Yönetici özel dashboard oluşturabilir.
Raporlar filtrelenip dışa aktarılabilir.
Veri kalitesi izlenebilir.
AI kullanımı güvenli, limitli ve audit'li çalışır.

Sonraki çıktı:

CRM OS Final Architecture Pack v1

CRM OS Final Architecture Pack v1

Executive Architecture Summary

CRM OS artık aşağıdaki katmanları kapsayan tam kapsamlı bir Enterprise SaaS Platform'dur:

CRM
Sales
Quote
Order
Inventory
Finance Lite
Customer Portal

Omnichannel
Support
SLA

Dealer Portal

Workflow Engine

Approval Engine

Public API

Webhook Platform

AI Platform

Analytics Platform

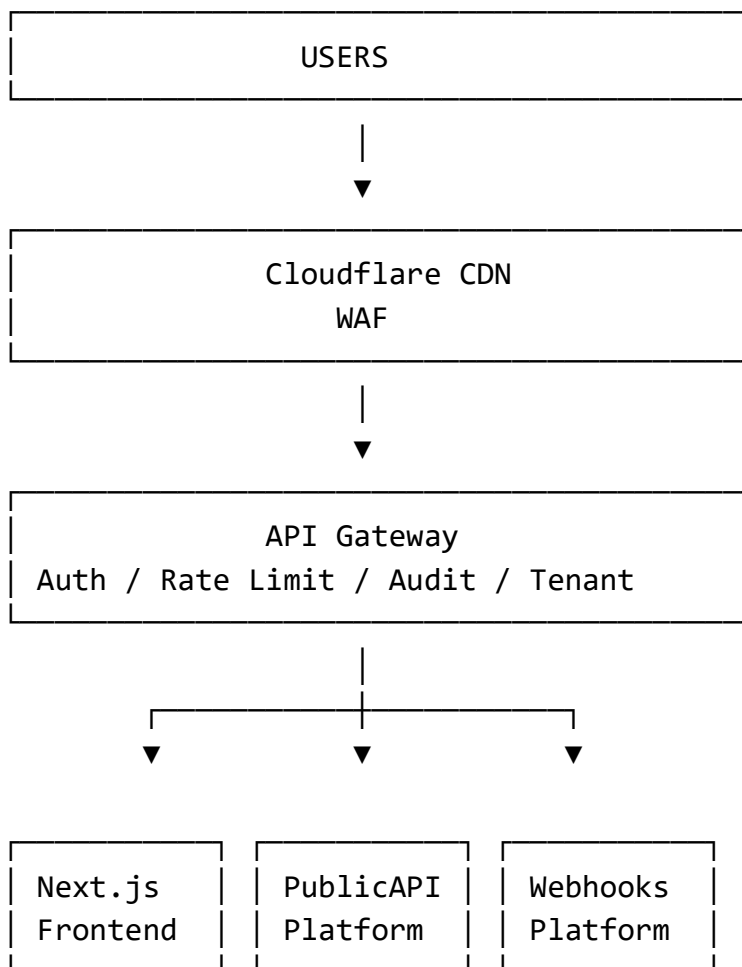
Multi-Tenant Security

Enterprise IAM

DevOps Platform

Observability Platform

1. High-Level Architecture



|

NestJS Application Layer



Domain Modules

- IAM
- CRM
- LEAD
- SALES
- QUOTE
- ORDER
- PRODUCT
- INVENTORY
- FINANCE
- SUPPORT
- COMMUNICATION
- WORKFLOW
- APPROVAL
- AI
- ANALYTICS



Domain Event Bus
RabbitMQ + Outbox Pattern



PostgreSQL + RLS + Read Replica

2. Final Deployment Architecture

Production

Region A

Kubernetes Cluster

api-deployment
worker-deployment
scheduler-deployment
frontend-deployment

postgres-primary

redis-cluster

rabbitmq-cluster

minio/s3

grafana
prometheus
loki
tempo

vault

Enterprise Scale

Region A

Region B

Active / Passive

PostgreSQL PITR

Cross Region Backups

Disaster Recovery

RT0:

< 2 hours

RPO:

< 15 min

3. Final Repository Structure

crm-os/

apps/

- |— api
- |— worker
- |— scheduler
- |— web
- |— customer-portal
- |— dealer-portal

packages/

- |— ui
- |— design-system
- |— api-client
- |— auth
- |— permissions
- |— analytics
- |— ai
- |— shared

infra/

- |— terraform
- |— helm
- |— k8s
- |— monitoring
- |— security

```
docs/  
├─ architecture  
├─ api  
├─ security  
└─ operations
```

scripts/

tests/

4. Final Backend Architecture

apps/api

src/

core/

modules/

```
iam/  
crm/  
lead/  
sales/  
quote/  
order/  
inventory/  
finance/  
support/  
communication/  
workflow/  
approval/  
integration/  
analytics/  
ai/
```

common/

Her modül:

module/

controllers/

dto/

services/

repositories/

entities/

events/

listeners/

guards/

policies/

tests/

5. Frontend Architecture

Stack

Next.js

React

TypeScript

TanStack Query

TanStack Table

Zustand

React Hook Form

Zod

Tailwind

Shadcn

Framer Motion

App Layout

AppShell

Sidebar

Topbar

Command Palette

Workspace

Pages

Drawers
Modals
Notifications

6. Design System

Theme

Dark First

Graphite Black
Obsidian
Glass Layers
Neon Accent

Style

Linear
Stripe
Vercel
Raycast
Warp
Attio

karışımı.

UI Rules

8px grid

16 radius

Glass surfaces

Soft shadows

Minimal borders

High density

Keyboard first

7. Database Architecture

PostgreSQL

Primary

Read Replica

Partitioned Tables

RLS

Heavy Tables

audit_logs

domain_events

timeline_events

messages

notifications

ai_usage_logs

metric_snapshots

Partition:

monthly

8. Caching Architecture

Redis:

sessions

permissions

dashboard

metrics

feature flags

workflow cache

Key pattern:

tenant:{tenantId}:...

9. Event Architecture

Outbox Pattern

Business Transaction

↓

Insert Event

↓

Commit

↓

Outbox Worker

↓

RabbitMQ

↓

Consumers

Main Event Topics

customer.*

lead.*

opportunity.*

quote.*

order.*

payment.*

ticket.*

workflow.*

ai.*

analytics.*

10. Security Architecture

Authentication

JWT

Refresh Token Rotation

2FA

SSO

SAML

OIDC

Authorization

RBAC

Permission

Scope

Field Security

RLS

Data Protection

TLS

Encryption At Rest

Secrets Vault

Audit Trail

Rate Limiting

11. AI Architecture

AI Gateway

altında:

OpenAI

Azure OpenAI

Anthropic

Local LLM

providerları.

AI Services

Ask CRM

Lead Scoring

Win Prediction

Quote Assistant

Conversation Summary

Ticket Summary

Recommendation Engine

AI Safety Layer

Permission Check

Tenant Isolation

PII Masking

Cost Tracking

Quota

Approval Required Actions

12. Analytics Architecture

Operational Metrics

Sales

Pipeline

Inventory

Finance

Support

Snapshot Metrics

Daily

Weekly

Monthly

Dashboards

Executive

Sales

Finance

Inventory

Support

AI

13. Integration Architecture

Public API

REST

OpenAPI

API Keys

Scopes

Webhooks

HMAC

Retry

DLQ

Logs

Connectors

Hazır connector katmanı:

SAP

Logo

Mikro

Netsis

Nebim

Shopify

WooCommerce

Meta

WhatsApp

Google

Microsoft

14. Observability Architecture

Metrics

Prometheus

api_latency

error_rate

queue_depth

db_connections

cache_hit_rate

webhook_failures

ai_cost

Logs

Loki

structured logs

tenant id

user id
request id

Tracing

Tempo

frontend

api

worker

db

tam izlenebilir.

15. Scalability Targets

MVP

10 tenant

100 user

Commercial

100 tenant

1000 user

Enterprise

1000 tenant

10000 user

Scale Target

5000 tenant

100000+ user

16. Final Team Structure (50 Kişi)

Platform Squad

6 kişi

IAM + Security

4 kişi

CRM Squad

6 kişi

Sales Squad

6 kişi

Order + Inventory

6 kişi

Finance

4 kişi

Support + Omnichannel

5 kişi

AI + Analytics

5 kişi

Frontend Platform

4 kişi

DevOps + SRE

2 kişi

QA Automation

2 kişi

17. Release Strategy

Dev

↓

Integration

↓

Staging

↓

UAT

↓

Production

Deployment:

Blue/Green

veya

Canary

18. Commercial Launch Package

İlk satılacak paket:

CRM

Lead

Pipeline

Quote

Task

Dashboard

Professional

CRM

Sales

Quote

Order

Inventory

Finance Lite

Portal

Workflow

Enterprise

Omnichannel

Support

Dealer Portal

Approval

Public API

Webhooks

SSO

Advanced Audit

AI Enterprise

Ask CRM

AI Assistant

Predictions

Analytics

Custom Dashboards

Data Quality

19. Final Go-Live Gates

Canlıya çıkmadan önce:

Tenant Isolation PASS

Permission PASS

RLS PASS

Performance PASS

Security PASS

Backup PASS

Restore PASS

Audit PASS

E2E PASS

UAT PASS

20. Final Architecture Verdict

Bu blueprint ile ortaya çıkan ürün:

Attio

+

HubSpot

+
Pipedrive
+
Zoho CRM
+
Freshworks
+
Monday CRM
+
Salesforce Essentials
+
Dealer Portal
+
AI Copilot

seviyesinde modüler bir SaaS CRM platformudur.

Bu noktadan sonra mimari fazı tamamlanmıştır. Sonraki adım artık belge üretmek değil, doğrudan:

Sprint-1 Repository Bootstrap Pack
Sprint-1 PostgreSQL Migration Pack
Sprint-1 NestJS Source Tree
Sprint-1 Next.js Source Tree
Sprint-1 Docker/Kubernetes Pack

çıktılarının üretilmesidir.

Eğer hedefiniz 50 kişilik ekibin 12 ayda yapacağı işi Cursor + AI Agent + CI/CD fabrikasına dönüştürmekse, amaç tek tek kod yazdırmak değil, üretim hattı kurmaktır.

Mimariyi çıkarmışsınız. Bundan sonra asıl verim şurada gelir:

Blueprint
→ Spec
→ Task
→ Code Generation
→ Review
→ Test

- Deploy
- Documentation

insansız veya yarı-insansız olacak.

Faz 1 – AI Factory Kurulumu

Önce CRM değil üretim fabrikası yapılır.

Repo içine:

.ai/

agents/
prompts/
templates/
playbooks/

specs/
modules/

pipelines/

Agent Roller

Her agent tek iş yapar.

Architect Agent
Backend Agent
Frontend Agent
Database Agent
DevOps Agent
QA Agent
Security Agent
Reviewer Agent
Documentation Agent

Architect Agent

Input:

module: customer

features:

- customer crud
- contacts
- addresses

Output:

ERD

DTO

API Contract

Events

Permissions

Acceptance Criteria

Backend Agent

Input:

customer-module-spec.yaml

Output:

NestJS module

controller

service

repository

dto

events

tests

Frontend Agent

Input:

customer-ui-spec.yaml

Output:

pages

forms

tables

drawers

modals

queries

hooks

QA Agent

Input:

customer module

Output:

unit tests

integration tests

playwright tests

permission tests

tenant isolation tests

Faz 2 – Spec Driven Development

Kod değil spec üret.

Her modül:

modules/

customer/

lead/

opportunity/

quote/

inventory/

finance/

Örnek:

module: customer

entities:

- customer
- contact
- address

permissions:

- customer.read
- customer.create

events:

- CustomerCreated
- CustomerUpdated

screens:

- customer-list
- customer-detail

Bu dosya tek kaynak olur.

Faz 3 – Code Generation

Cursor'a:

Kod yaz

demezsınız.

Şunu dersiniz:

Generate customer module from spec.

Agent:

migration

entity

dto

controller

service

tests

swagger

üretir.

Faz 4 – Monorepo Generator

Tek komut:

pnpm generate customer

çıktı:

backend module

frontend pages

tests

docs

Generator stack:

plop

hygen

nx generators

custom scripts

Faz 5 – Vertical Slice Strategy

En kritik nokta.

Modül modül git.

Yanlış:

Önce tüm backend

sonra frontend

Doğru:

Customer

DB

API

UI

Test

biter

sonra

Lead

DB

API

UI

Test

Faz 6 – Cursor Swarm

Tek Cursor yerine çok agent.

Örnek:

Agent-1

Customer Backend

Agent-2

Customer Frontend

Agent-3

Customer Tests

Agent-4

Customer Documentation

aynı anda.

Faz 7 – CI Factory

PR açılınca:

Lint

Typecheck

Unit Test

Integration Test

Security Scan

Architecture Rules

E2E Smoke

çalışır.

Faz 8 – Architecture Guardrails

AI'ın kafasına göre kod yazmasını engellersiniz.

Kurallar:

No direct SQL

Repository Pattern

Event Driven

DTO Required

Swagger Required

Tests Required

Permission Required

Audit Required

Cursor her üretimde bu checklistten geçer.

Faz 9 – Auto Review Agents

Kod yazan agent ayrı.

Kod inceleyen agent ayrı.

Pipeline:

Code Agent

↓

Review Agent

↓

Security Agent

↓

Performance Agent

↓

Merge

Faz 10 – Documentation Factory

Her merge sonrası:

OpenAPI

ERD

Permissions

Events

Release Notes

otomatik güncellenir.

Faz 11 – Enterprise Ölçeği

50 kişi yerine:

10 senior engineer

+

Cursor Agents

+

CI/CD Factory

yeterli hale gelir.

En Hızlı Üretim Modeli

Sizin mevcut blueprint için ideal üretim hattı:

Blueprint

↓

Module Spec

↓

Generator

↓

Cursor Backend Agent

↓

Cursor Frontend Agent

↓

Cursor Test Agent

↓

Review Agent

↓

CI Validation

↓

Auto Documentation

↓

Merge

Bu model kurulduğunda Sprint-1'den Sprint-40'a kadar olan planın büyük kısmı **spec-first ve generator-first** ilerler. İnsanlar mimari karar, iş kuralları ve kalite kontrolüne odaklanır; tekrar eden CRUD, DTO, form, tablo, test ve dokümantasyon üretimi büyük ölçüde otomatikleşir.

Evet, yöntem doğru; ama daha gelişmiş hâli şu olmalı:

Spec-Driven AI Factory

+ Cursor Rules

+ Generators

+ Background Agents

+ MCP

+ CI Quality Gates

+ Auto Review

Yani sadece Cursor'a "kod yaz" demek değil; Cursor'u **kontrollü üretim hattına** bağlamak.

En iyi kurgu

Blueprint

↓

Module Spec

↓

Code Generator

↓

Cursor Agent

↓

Tests

↓

BugBot / Review Agent

↓

CI/CD

↓

Staging

Cursor'un resmi dokümanlarında Agent, Rules, MCP, Skills ve CLI gibi özellikler bu tarz agentic geliştirme için temel araçlar olarak geçiyor. ([Cursor](#))

Daha gelişmiş kullanım:

1. `.cursor/rules` ile mimari kuralları sabitle
2. Her modül için YAML spec yaz
3. Generator ile dosya iskeletlerini üret
4. Cursor Agent sadece spec'e göre kod yazsın
5. MCP ile Jira, GitHub, DB, Swagger, docs bağla
6. Background Agent ile paralel PR üret
7. CI her PR'da lint/test/security/e2e çalıştırsın
8. BugBot veya review agent PR'ı incelesin
9. İnsan sadece mimari ve kritik iş kurallarını onaylasın

Büyük ekiplerde Cursor'un değeri en çok burada çıkıyor: kalıcı kurallar, agent iş akışı, test üretimi ve review süreci. Nvidia'nın binlerce mühendiste Cursor benzeri özelleştirilmiş iş akışlarıyla kod üretimini ciddi artırdığı da raporlanmış; burada kritik nokta insan denetimi ve custom rules kullanımı. ([Tom's Hardware](#))

En gelişmiş model:

Cursor = Kod yazan kişi değil

Cursor = Yazılım üretim fabrikasının işçisi

Senin CRM için ideal sıra:

1. .cursor/rules mimari paketi
2. module-spec şablonları
3. code generator
4. Sprint-1 bootstrap
5. Customer vertical slice
6. CI kalite kapıları
7. Background agent paralel üretim

Bu şekilde kullanırsan Cursor'dan maksimum verim alınır.

Hayır, sadece bu kadar değil. O 7 madde **temel kurgu**. Daha güçlü versiyon şöyle olur:

1. Blueprint → module-spec otomasyonu
2. .cursor/rules mimari kuralları
3. AGENTS.md / task playbook
4. Code generator
5. Cursor Agent task queue
6. Background Agents ile paralel PR
7. MCP bağlantıları
8. Auto test generation
9. Auto code review
10. Security review
11. Performance review
12. Migration validator
13. OpenAPI validator
14. Visual regression
15. Auto documentation
16. Release bot
17. Staging deploy bot
18. Bug triage bot
19. Regression bot
20. Human approval gates

En gelişmiş kurgu:

Product Spec
↓
Module Spec
↓
Generator
↓
Cursor Agent
↓
Background Agent PR
↓
BugBot / Review Agent
↓
CI Quality Gates
↓
Staging Deploy
↓
E2E + Security + Performance
↓
Human Approval
↓
Production

Cursor tarafında Agent, Rules, MCP, Skills, CLI ve Teams/Enterprise gibi araçlar bu tarz üretim hattını destekleyen ana parçalar olarak geçiyor. ([Cursor](#))

Senin CRM için gerçek “AI factory” seviyesi şudur:

Cursor kod yazmaz.
Cursor kontrollü task alır.
Spec dışına çıkamaz.
Test geçmeden PR açamaz.
Review geçmeden merge olmaz.
Security gate geçmeden release olmaz.

Yani evet, önceki plan iyi başlangıçtı; ama profesyonel seviye **20** adımlı AI üretim hattı olmalı.